CROSS SECTIONAL ANATOMY OF THE BRAIN



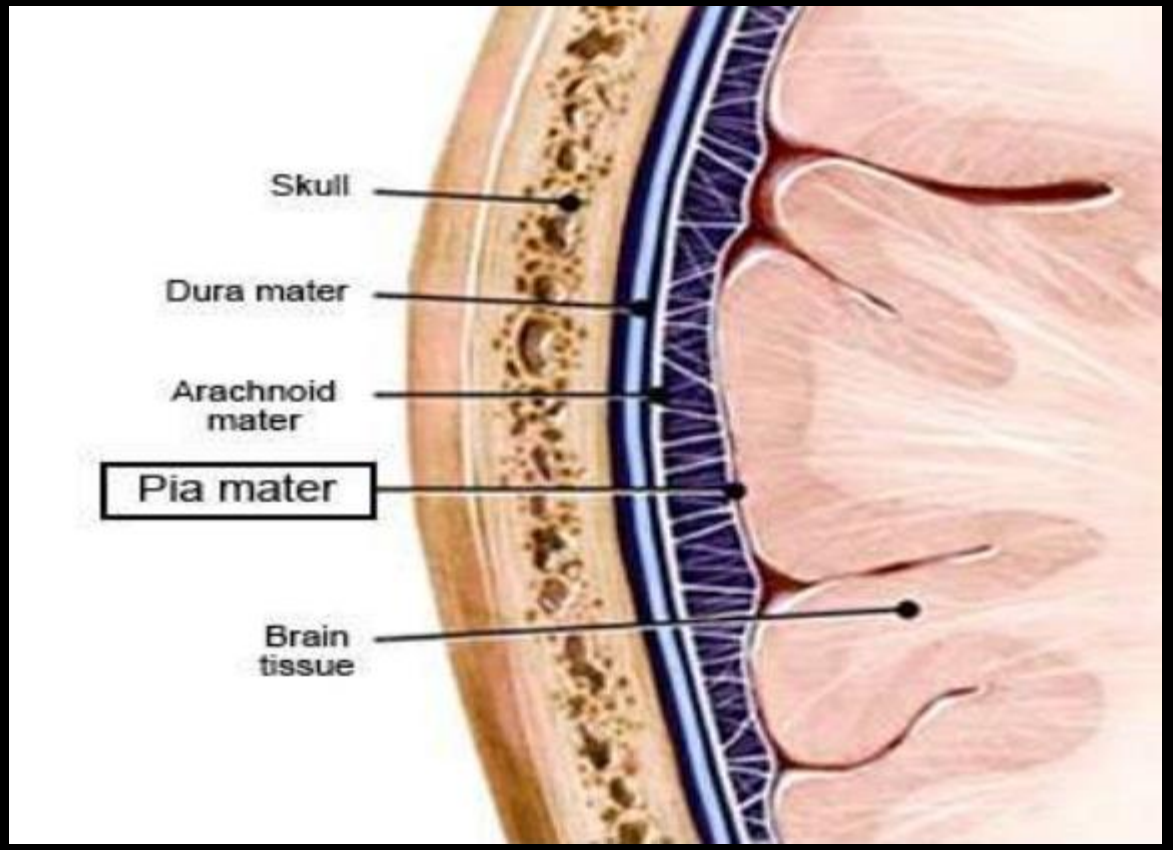
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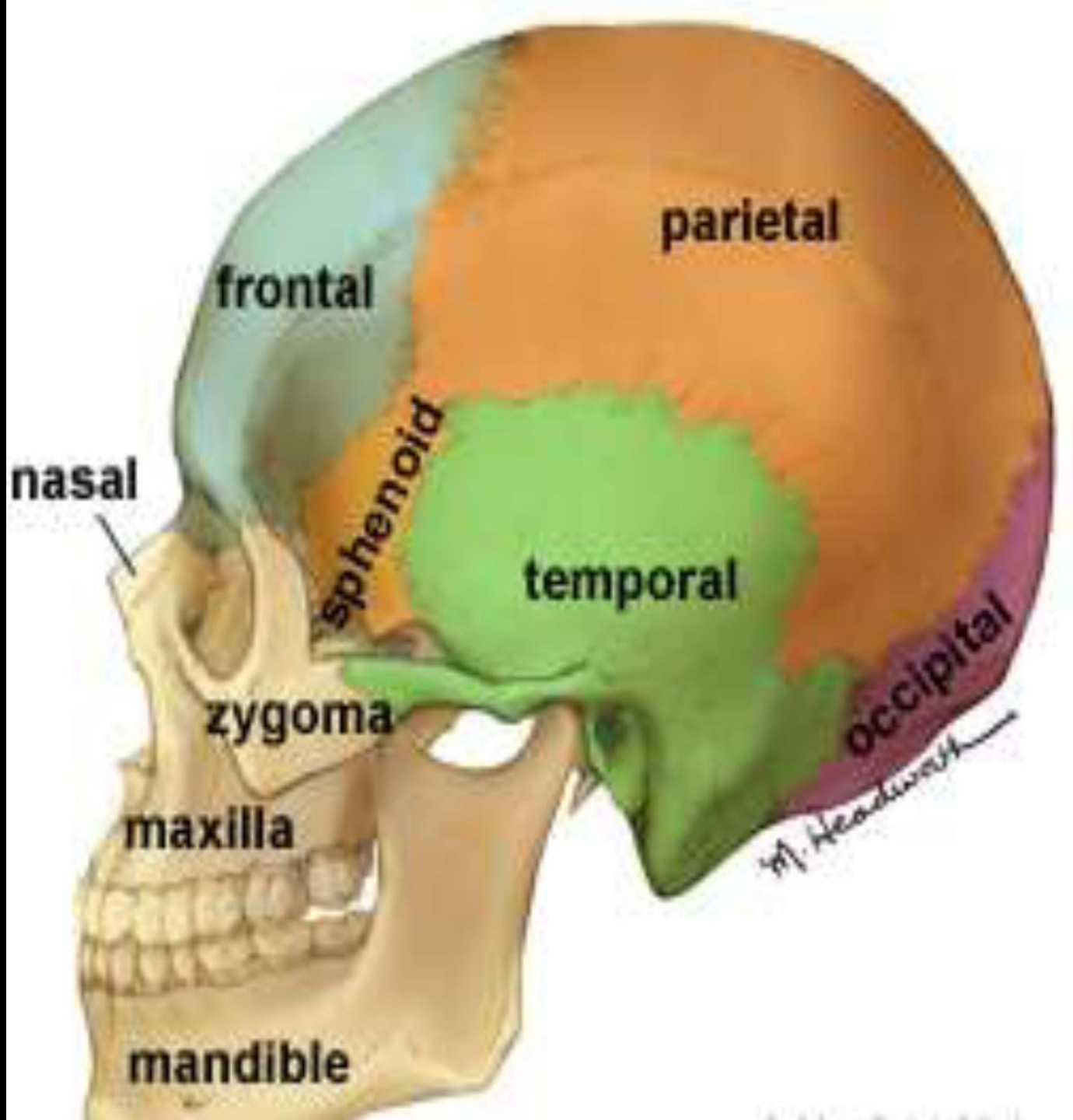
Brain protective layers

- Skull .
- Meninges.
- Cerebrospinsal fluid (CSF).



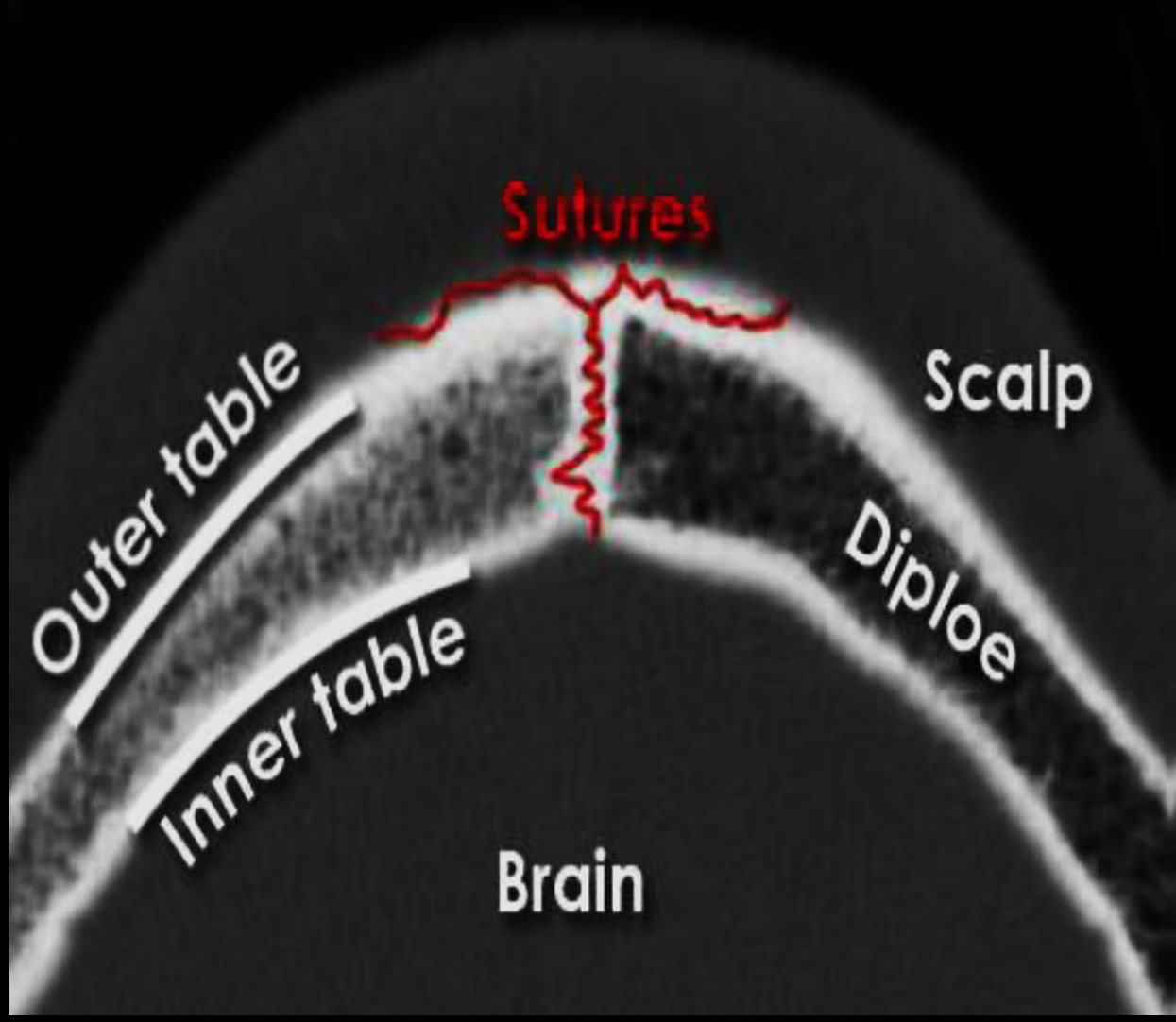
SKULL

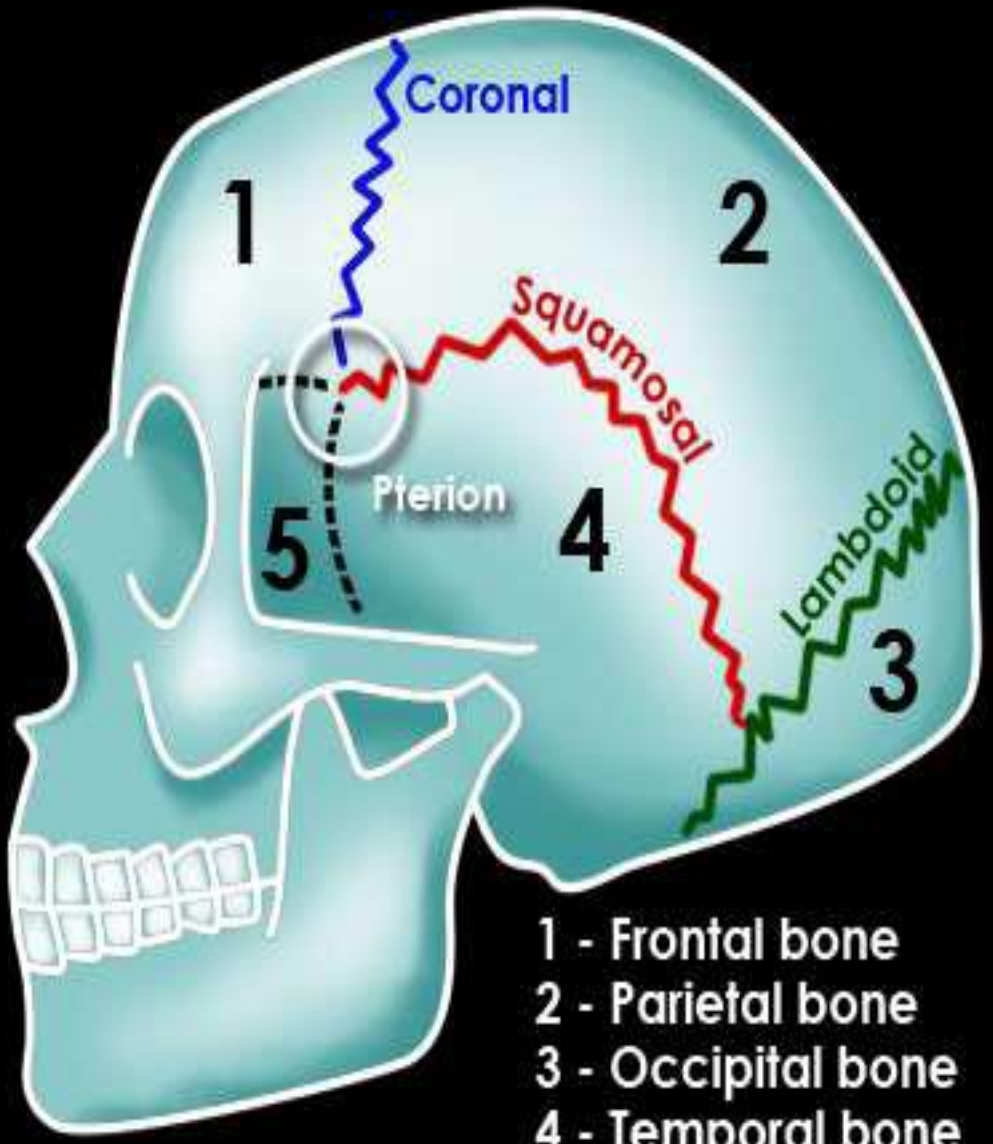
- *The skull is formed from 8 bones ; frontal, parietal (2), temporal (2), sphenoid, occipital and ethmoid.*
- *They ossify separately and gradually become united at the skull sutures.*
- *The skull has inner and outer tables of cortical bone with central cancellous bone called dipolic space.*



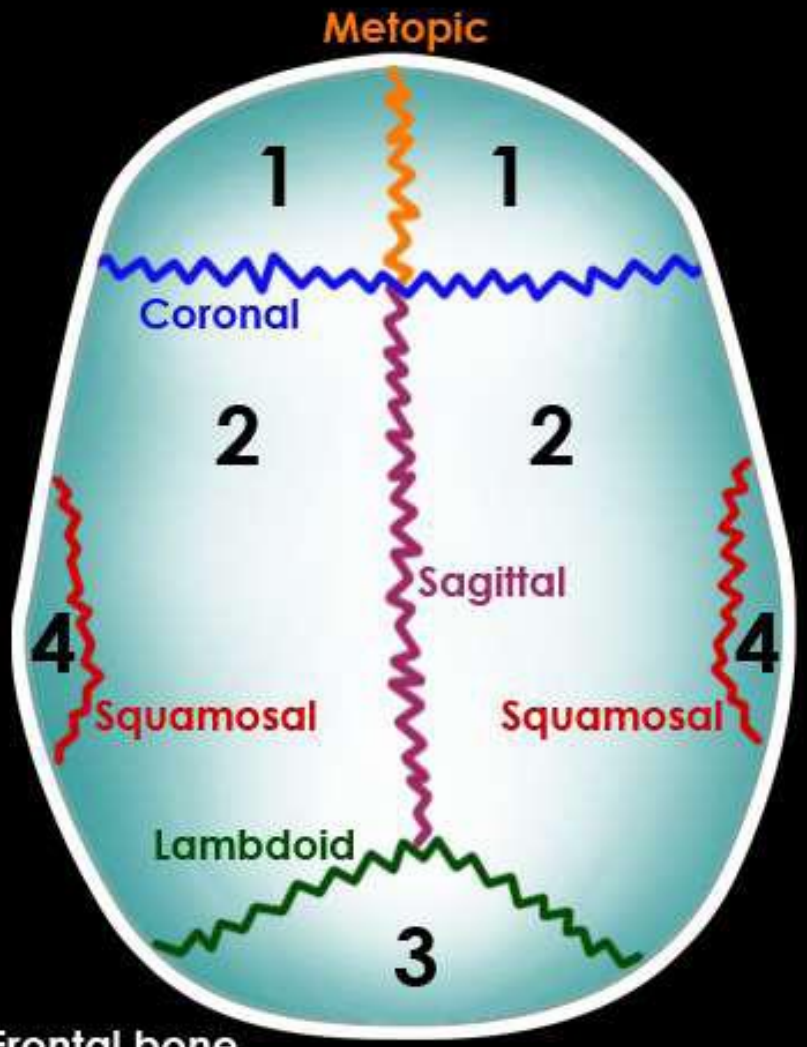
Sutures

- *The main sutures of the skull are the coronal, sagittal, lambdoid and squamosal sutures.*
- *Coronal suture - unites the frontal bone with theparietal bones*
- *Sagittal suture - unites the 2 parietal bones in the midline*
- *Lambdoid suture - unites the parietal bones with the occipital bone*
- *Squamosal suture - unites the squamous portion of the temporal bone with the parietal bones*
- *Metopic suture - (if present) unites the 2 fontal bones*





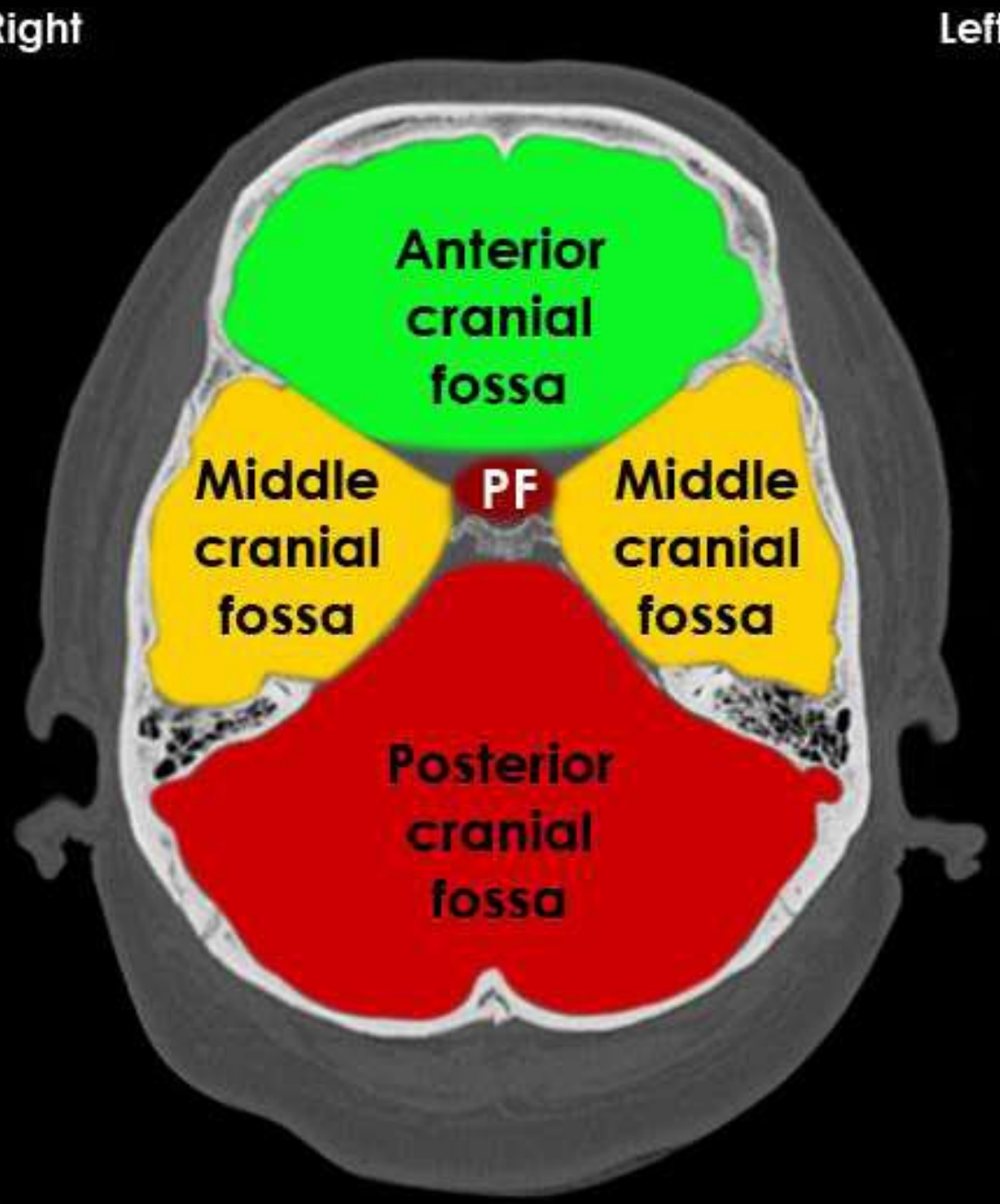
1 - Frontal bone
2 - Parietal bone
3 - Occipital bone
4 - Temporal bone
5 - Sphenoid bone



1 - Frontal bone
2 - Parietal bones
3 - Occipital bone
4 - Temporal bones

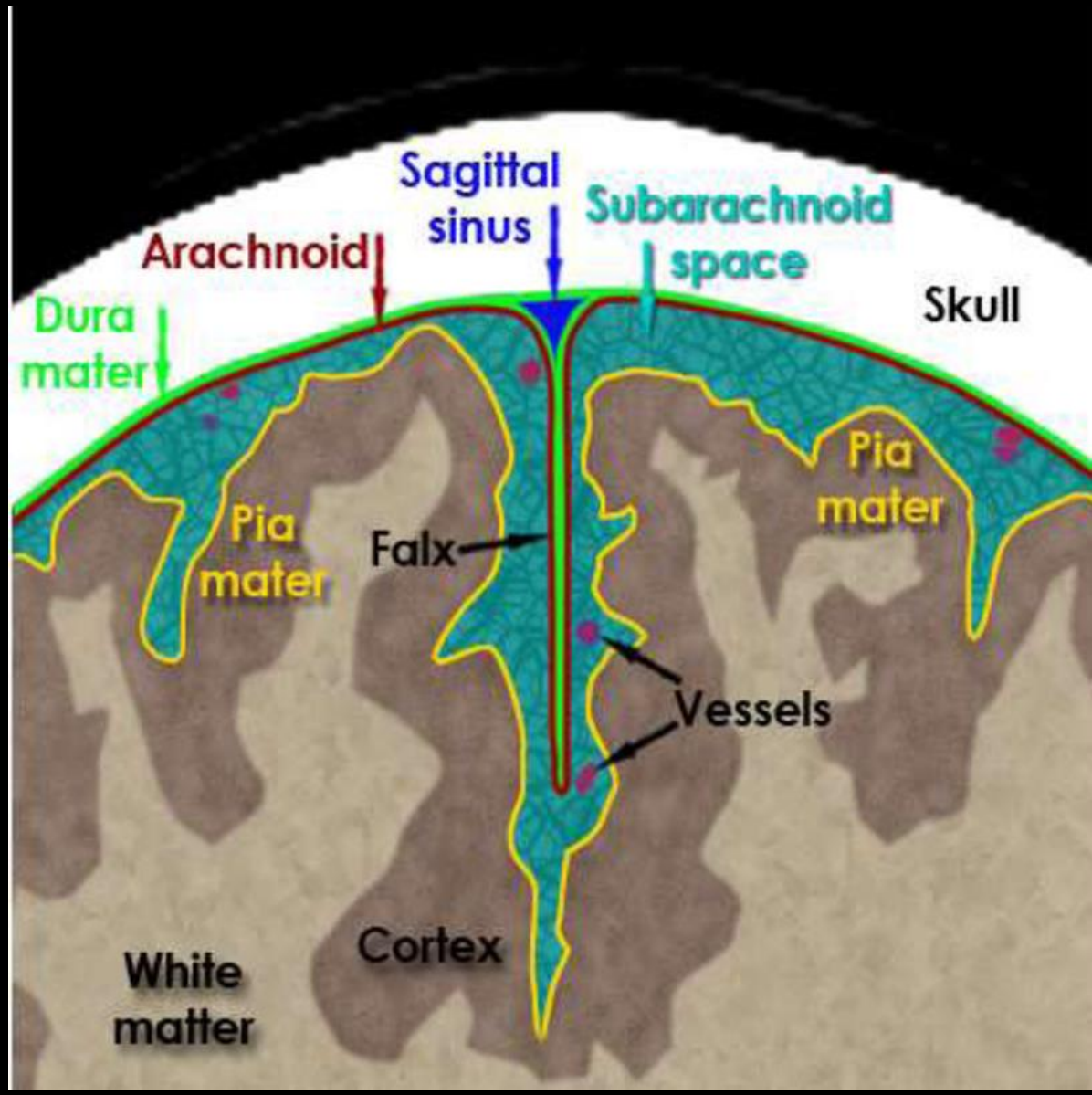
Inside the skull are three distinct areas: anterior, middle, and posterior fossae.

- *Anterior cranial fossa accommodates the anterior part of the frontal lobes*
- *Middle cranial fossa accommodate the temporal lobes*
- *Posterior cranial fossa accommodates the cerebellum and brain stem*
- *Pituitary fossa (PF) accommodates the pituitary gland*



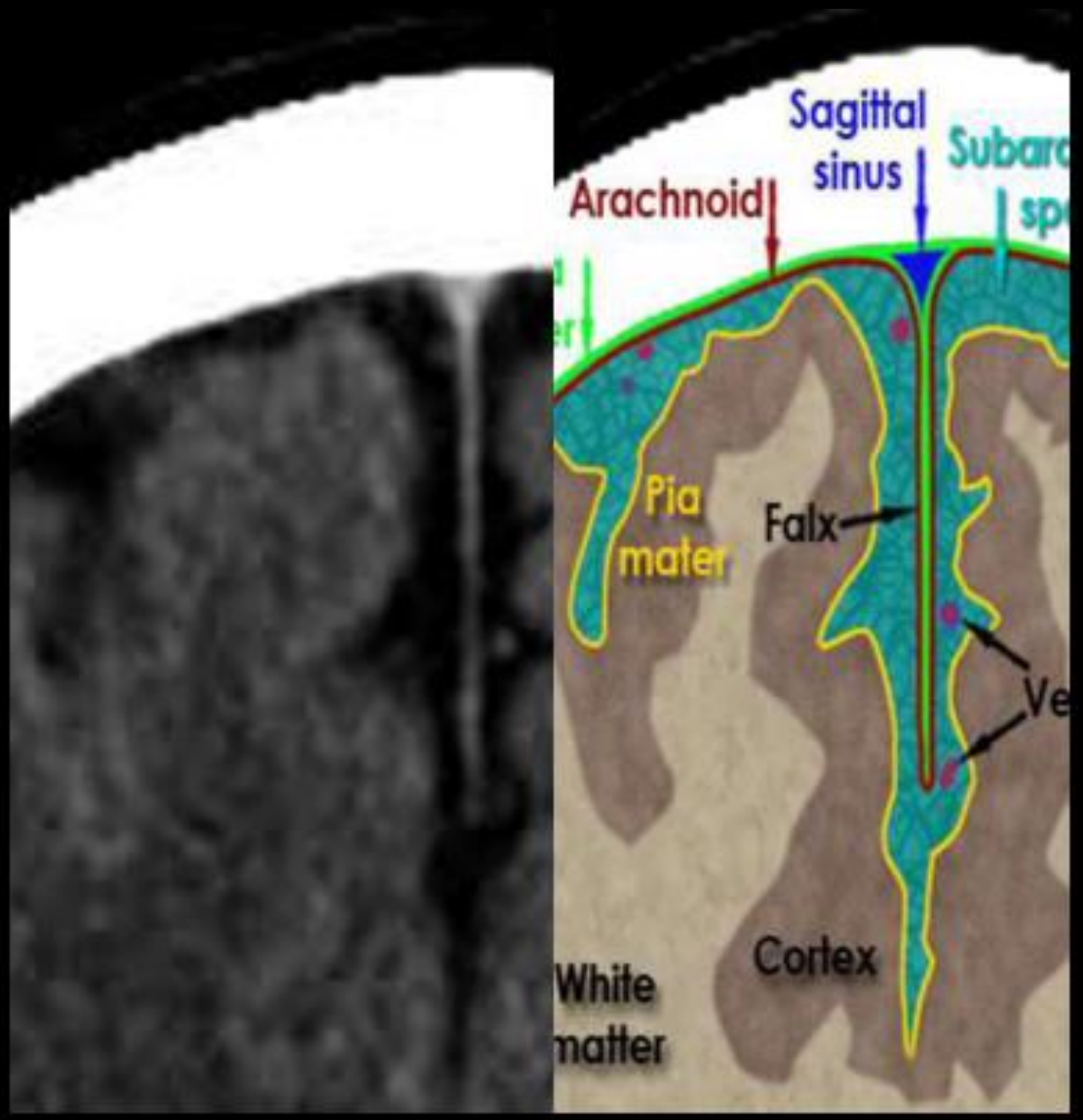
Meninges

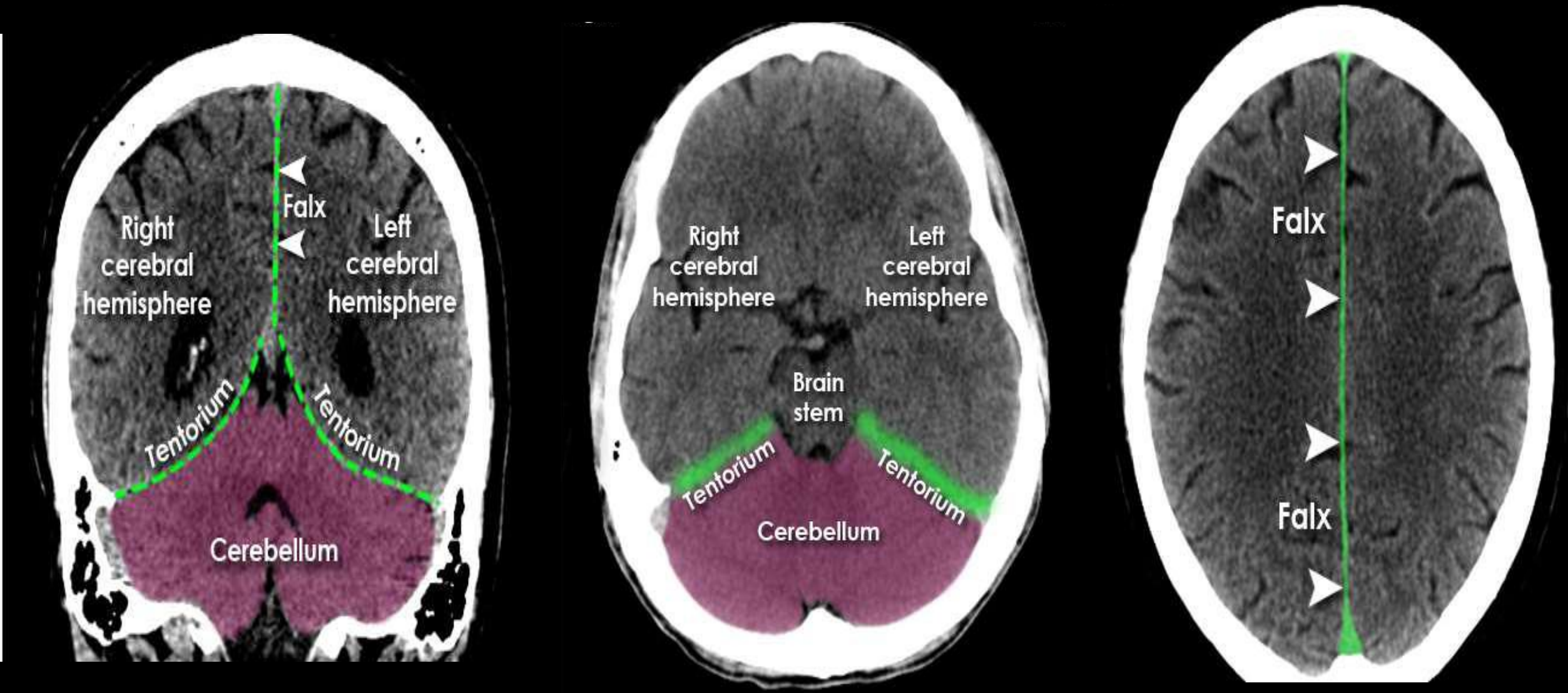
- *The brain and spinal cord are covered and protected by three layers of tissue called meninges. From the outermost layer inward they are:*
  - *Outer layer:the dura mater.*
  - *Middle layer: arachnoid mater.*
  - *Inner layer: pia mater.*



Meninges…

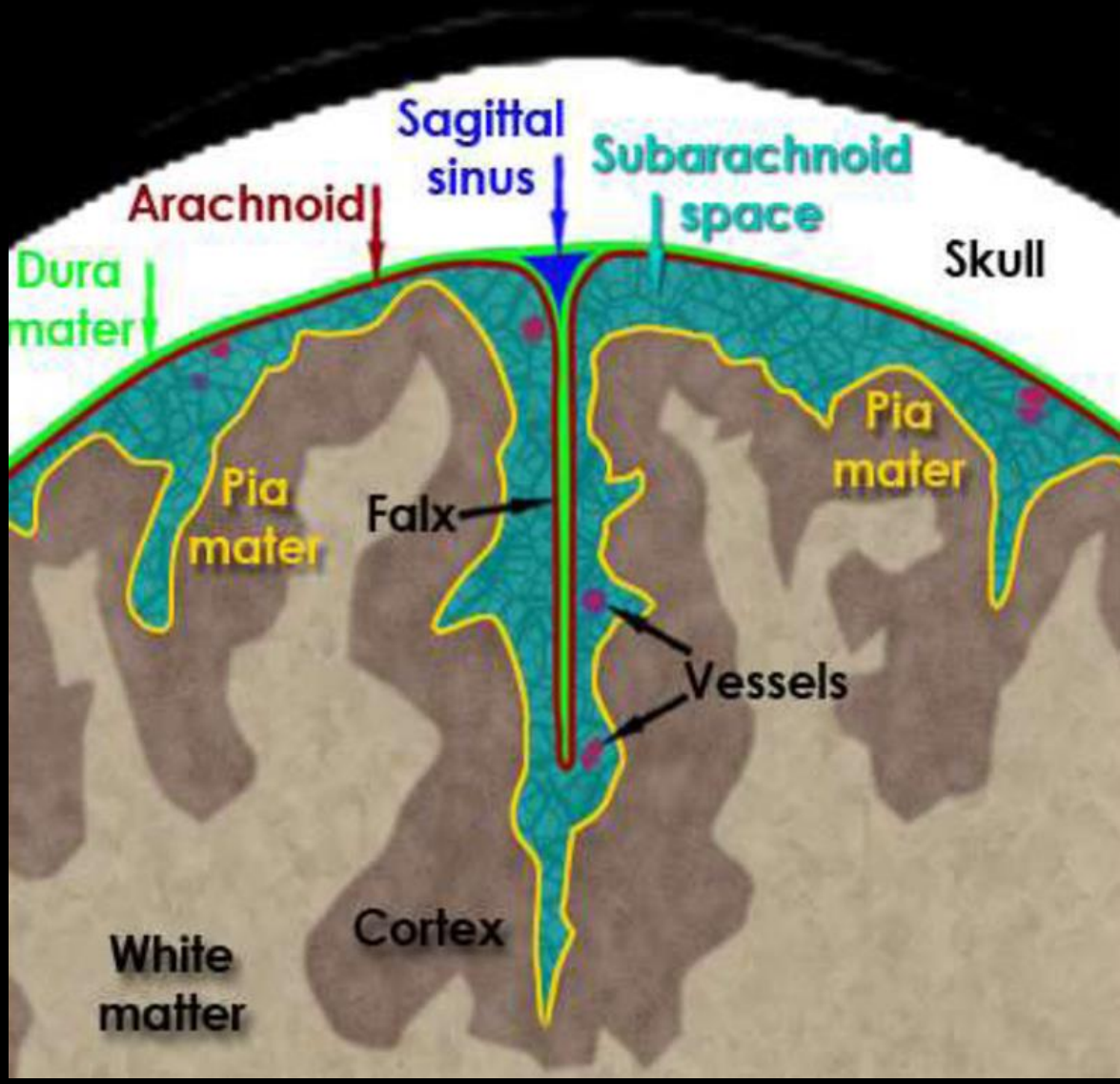
- *The dura mater is a strong, thick membrane that closely lines the inside of the skull; its two layers, the periosteal and meningeal dura, are fused and separate only to form venous sinuses.*
- *There are two special dural folds, the falx cerebri and the tentorium cerebelli. The falx separates the right and left hemispheres of the brain and the tentorium separates the cerebrum from the cerebellum. They are visible on CT imaging. Elsewhere the meningeal layers are not visible .*




Falx
Right cerebral hemisphere
Left cerebral hemisphere
Tentorium
Tentorium
Cerebellum
Right cerebral hemisphere
Left cerebral hemisphere
Brain stem
Tentorium
Tentorium
Cerebellum
Falx
Falx

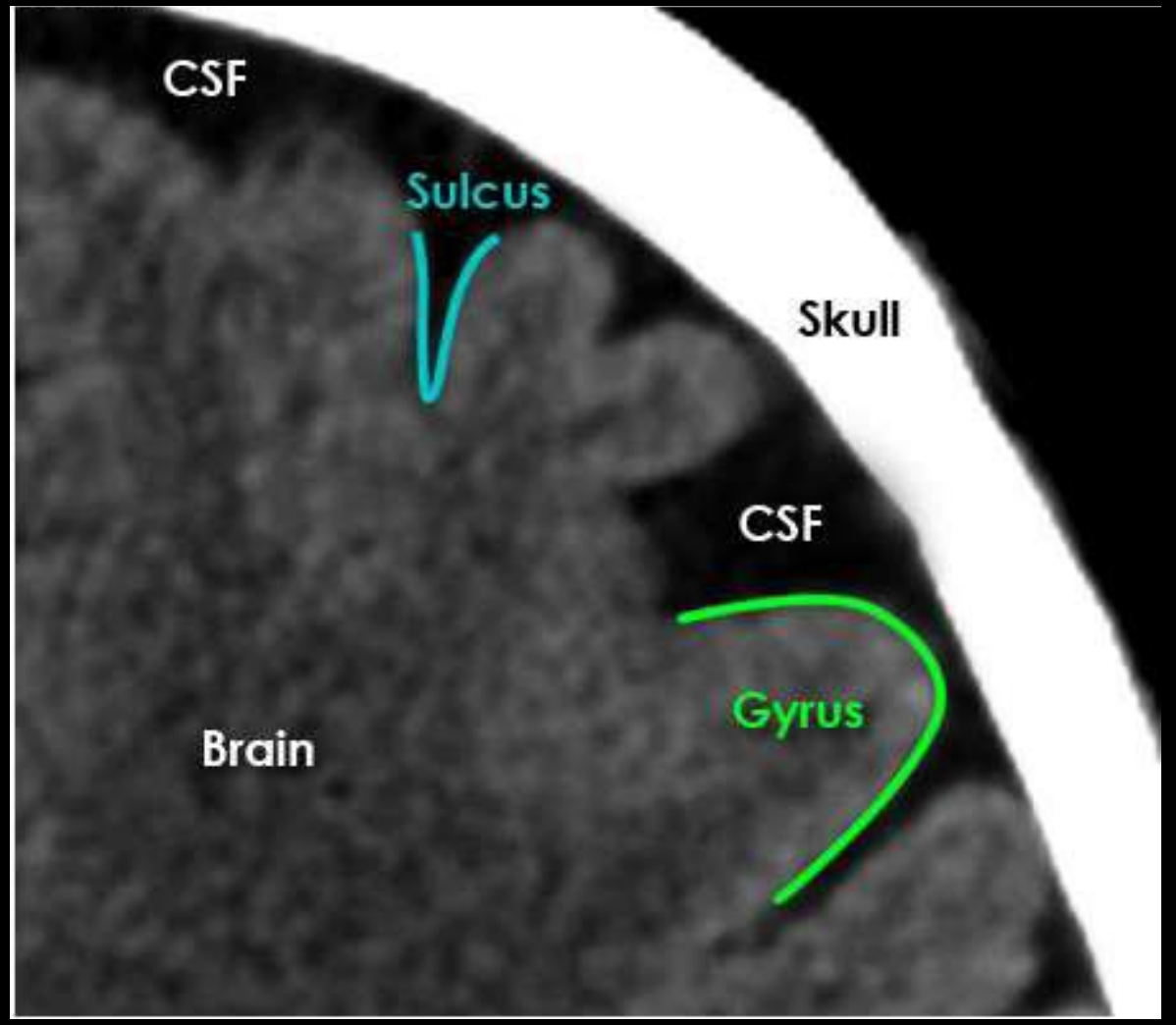
Meninges…

- *The arachnoid mater is a thin, web-like membrane that covers the entire brain. The space between the dura and arachnoid membranes is called the subdural space.*
- *The pia mater hugs the surface of the brain following its folds and grooves. The space between the arachnoid and pia is called the subarachnoid space. It is where the cerebrospinal fluid bathes and cushions the brain.*



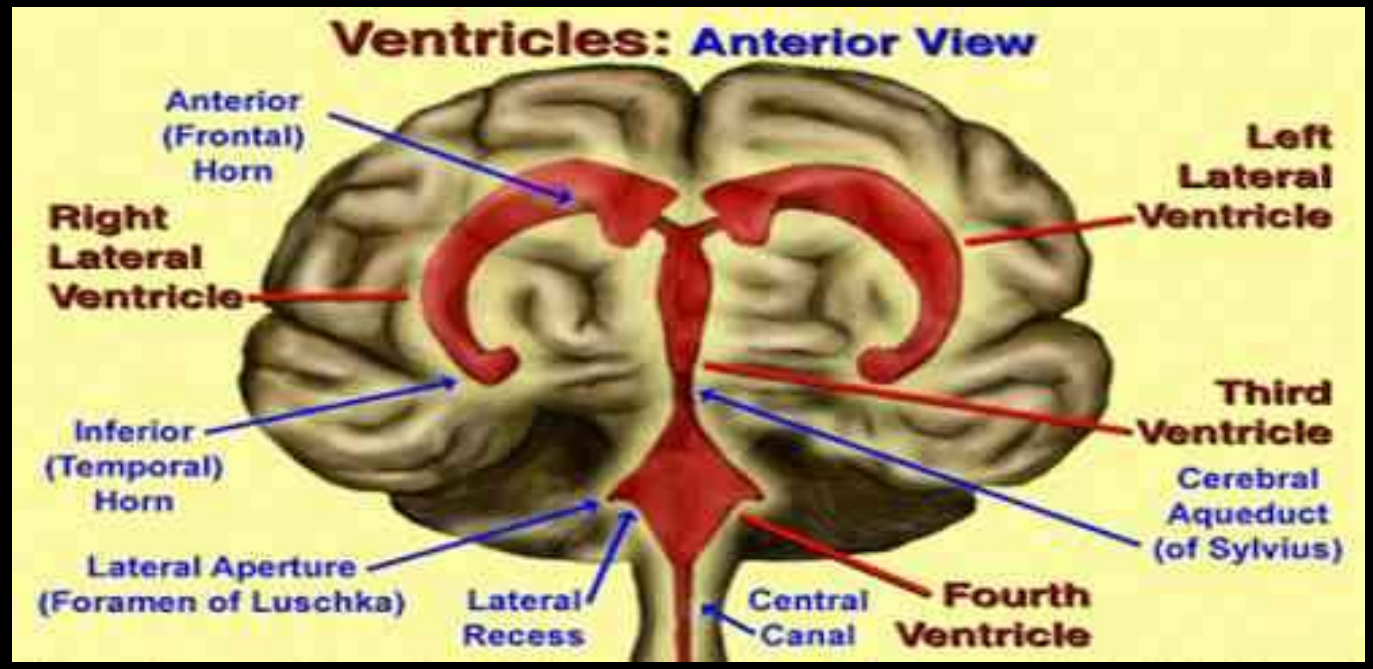
CSF spaces

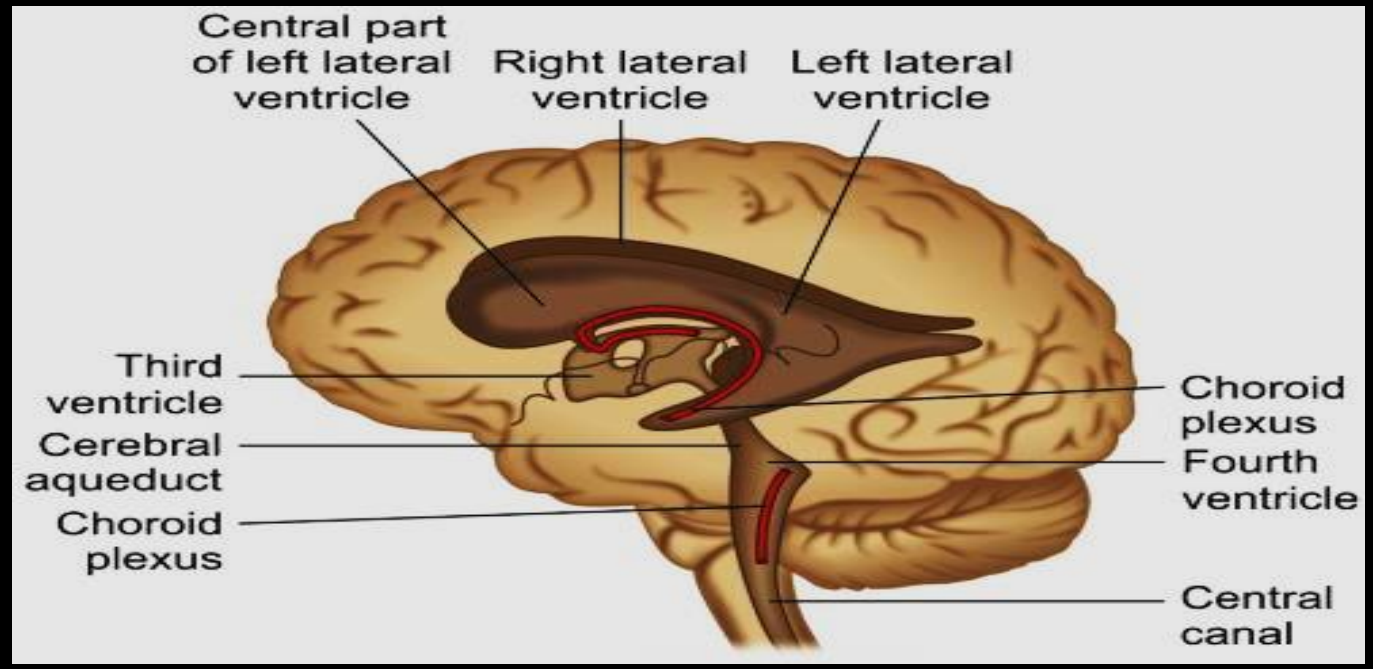
- *The brain is surrounded by cerebrospinal fluid (CSF) within the sulci, fissures and basal cisterns and ventricles together form the 'CSF spaces', also known as the 'extra-axial spaces'.*
- *The brain surface is formed by folds of the cerebral cortex known as gyri. Between these gyri there are furrows, known as sulci, which contain CSF.*



ventricles

- *Two lateral ventricles.*
- *Third ventricle.*
- *Fourth ventricle.*

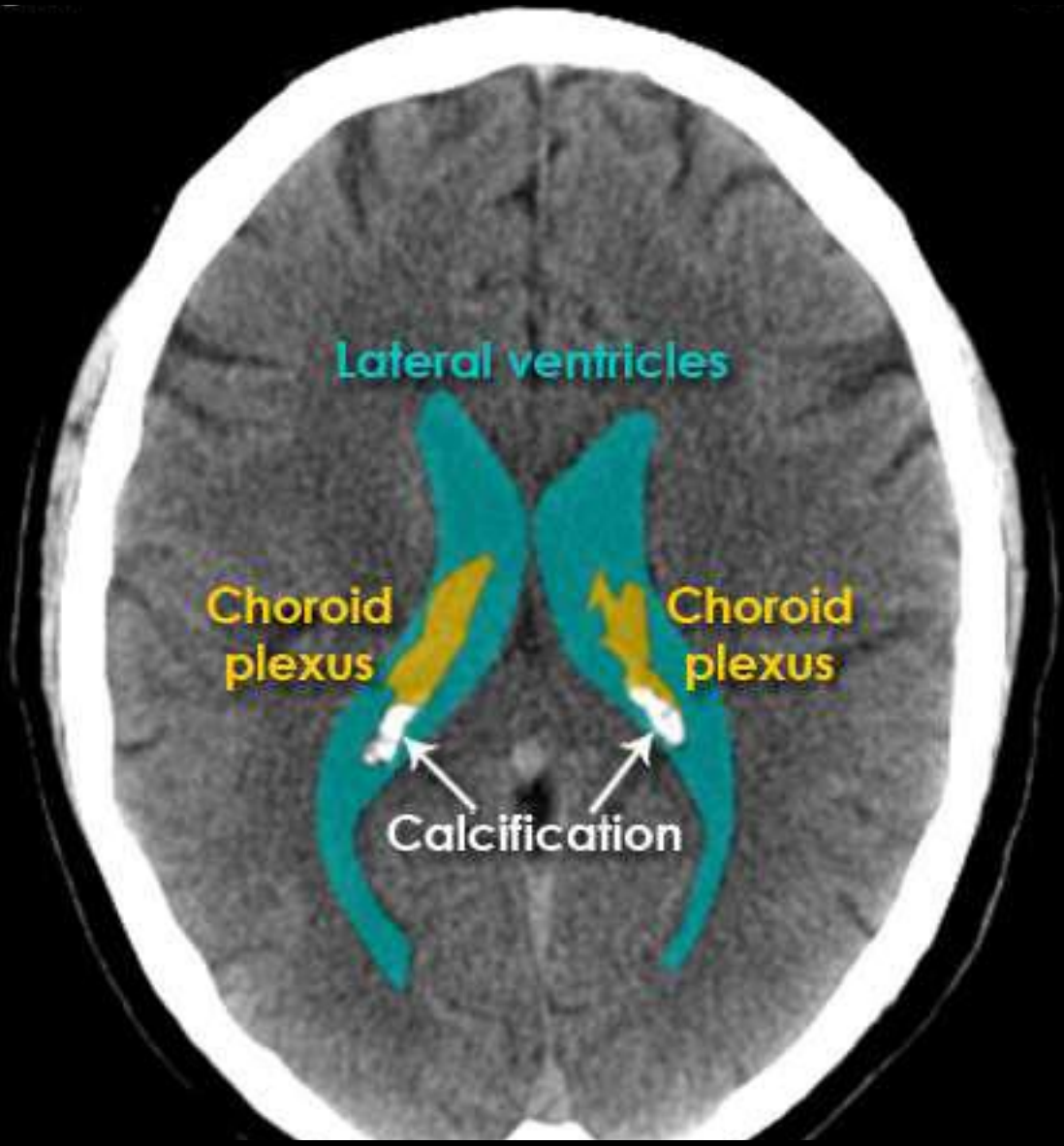




Lateral ventricles

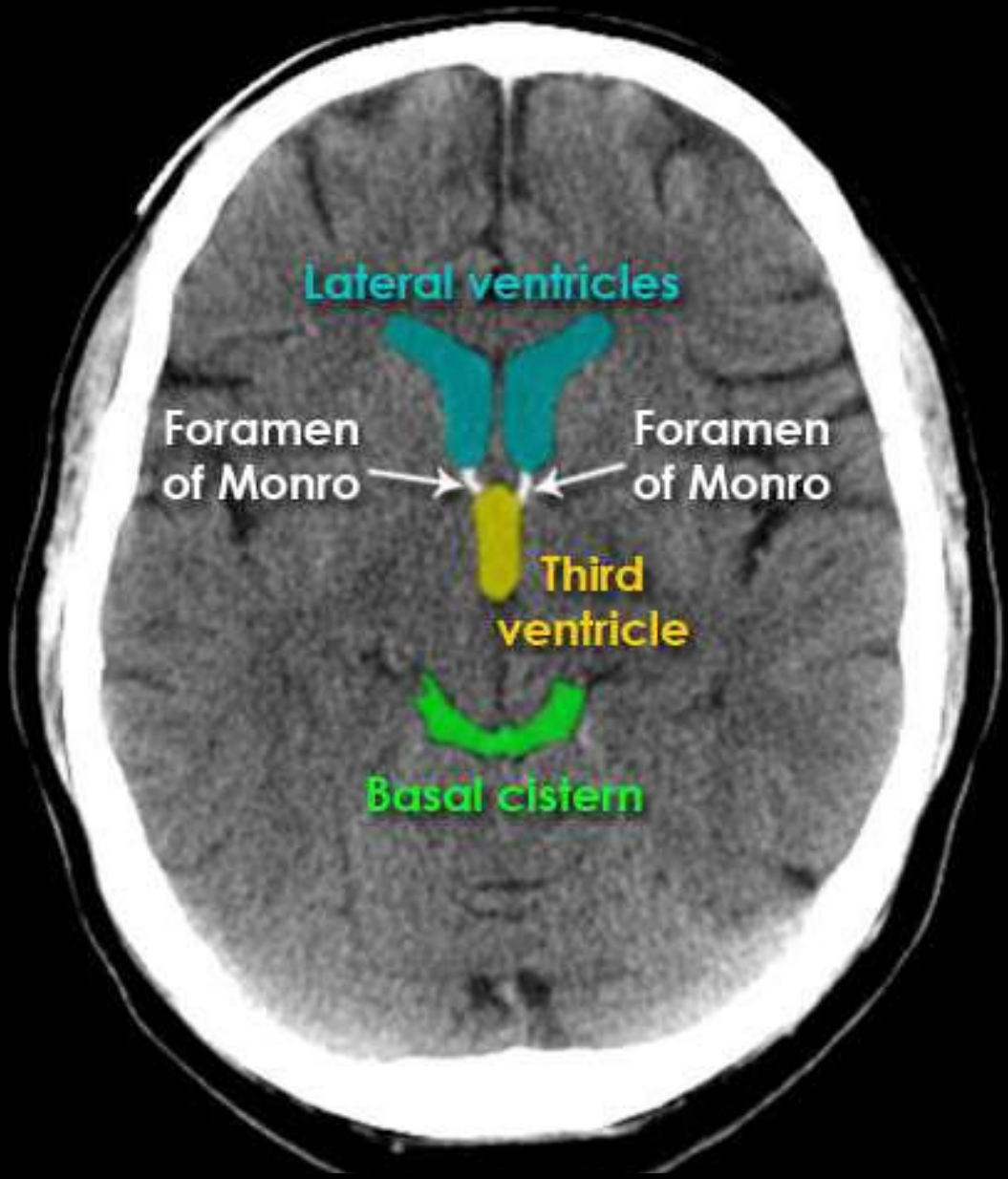
- *The paired lateral ventricles are located on either side of the brain.*
- *Each ventricle has trigone, body,frontal , temporal & occipital horns*
- *They contain the choroid plexus which produces CSF.*

Note : The choroid plexus is almost always calcified in adults.



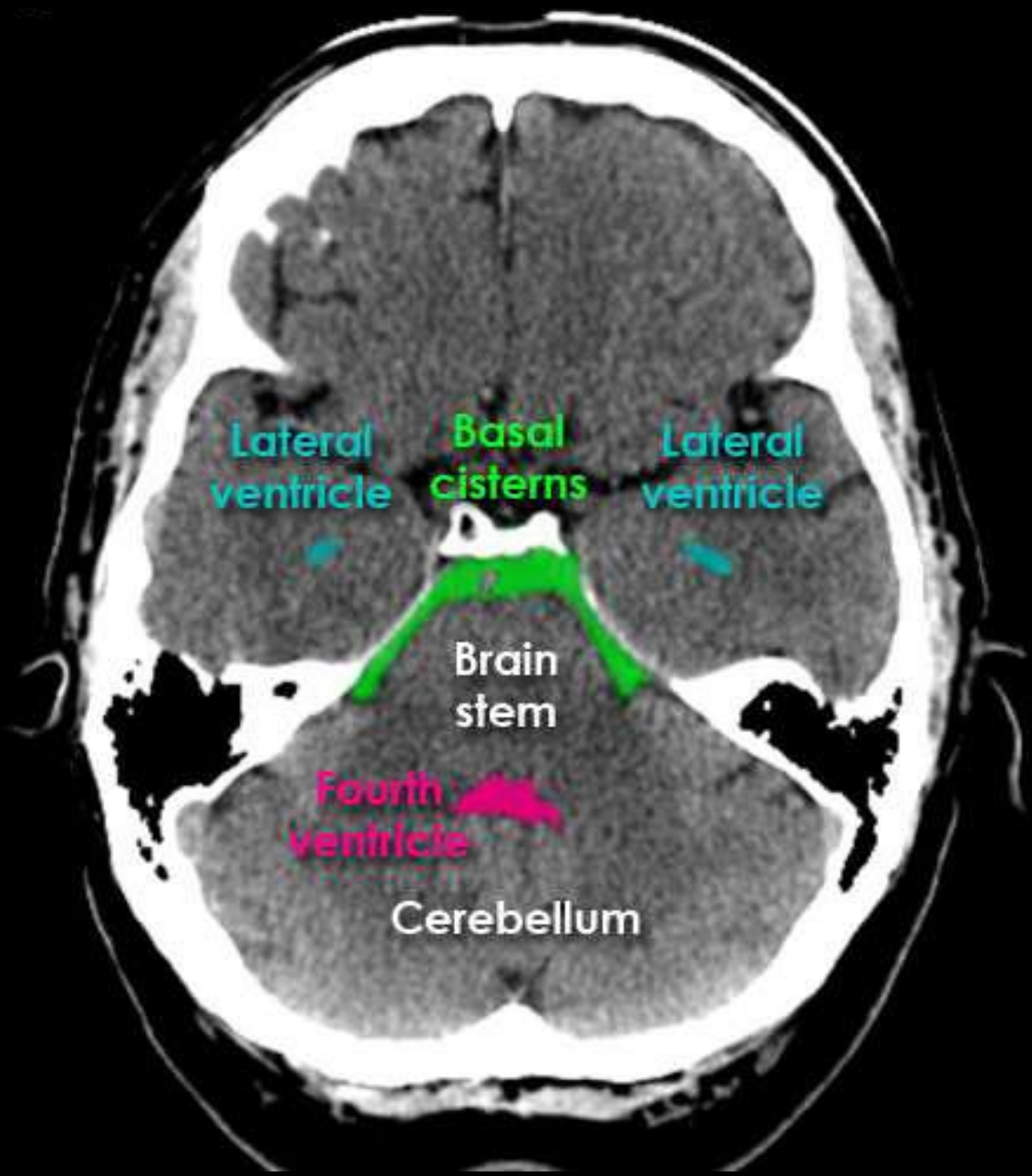
Third ventricle

- *The third ventricle is located centrally.*
- *The lateral ventricles communicate with the third ventricle via small holes (foramina of Monro).*



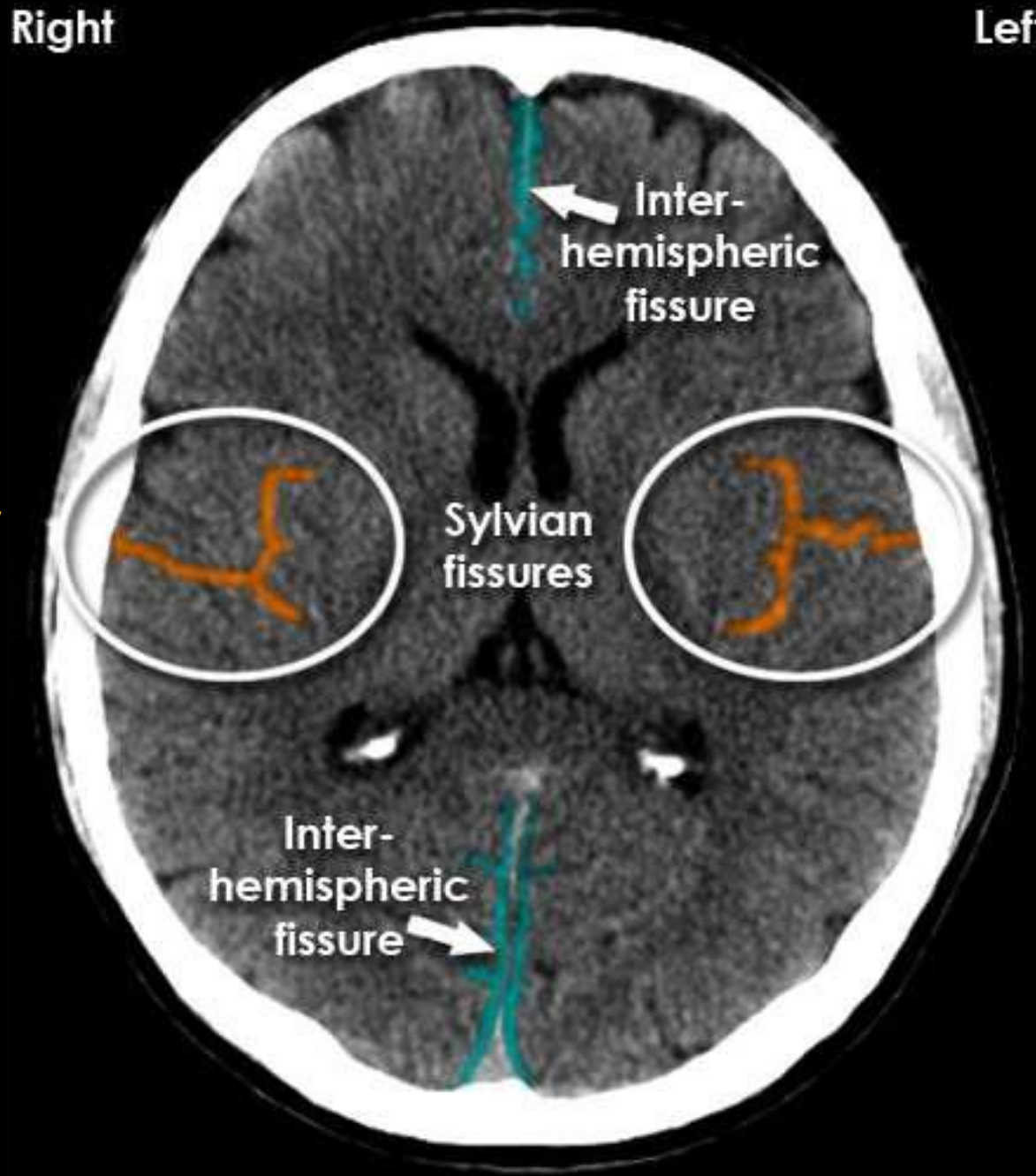
Fourth ventricle

- *Is located in the posterior fossa between the brain stem and cerebellum.*
- *It communicates with the third ventricle above via a very narrow canal, the aqueduct of Sylvius .*
- *Basal cisterns CSF in the basal cisterns surrounds the brain stem structures.*

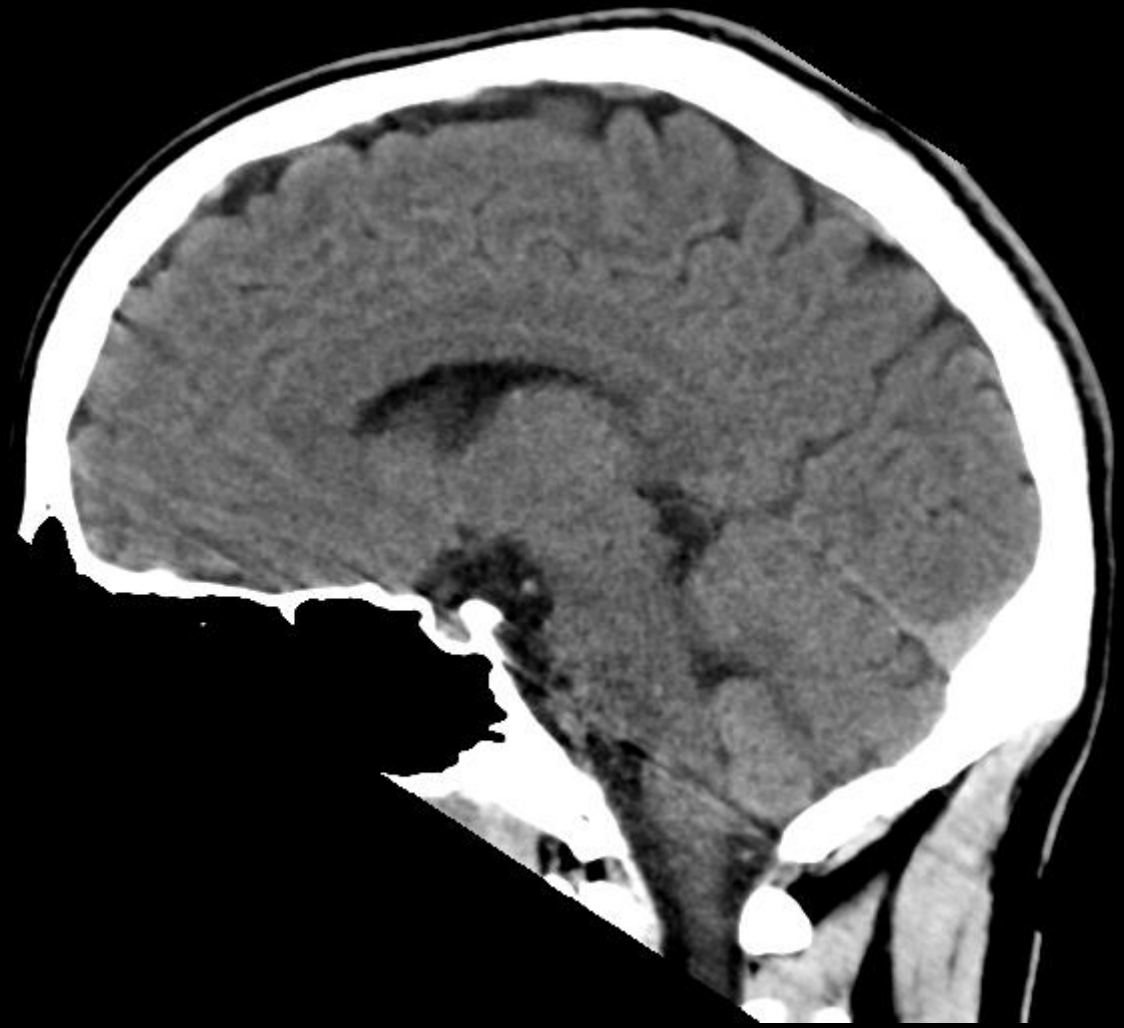


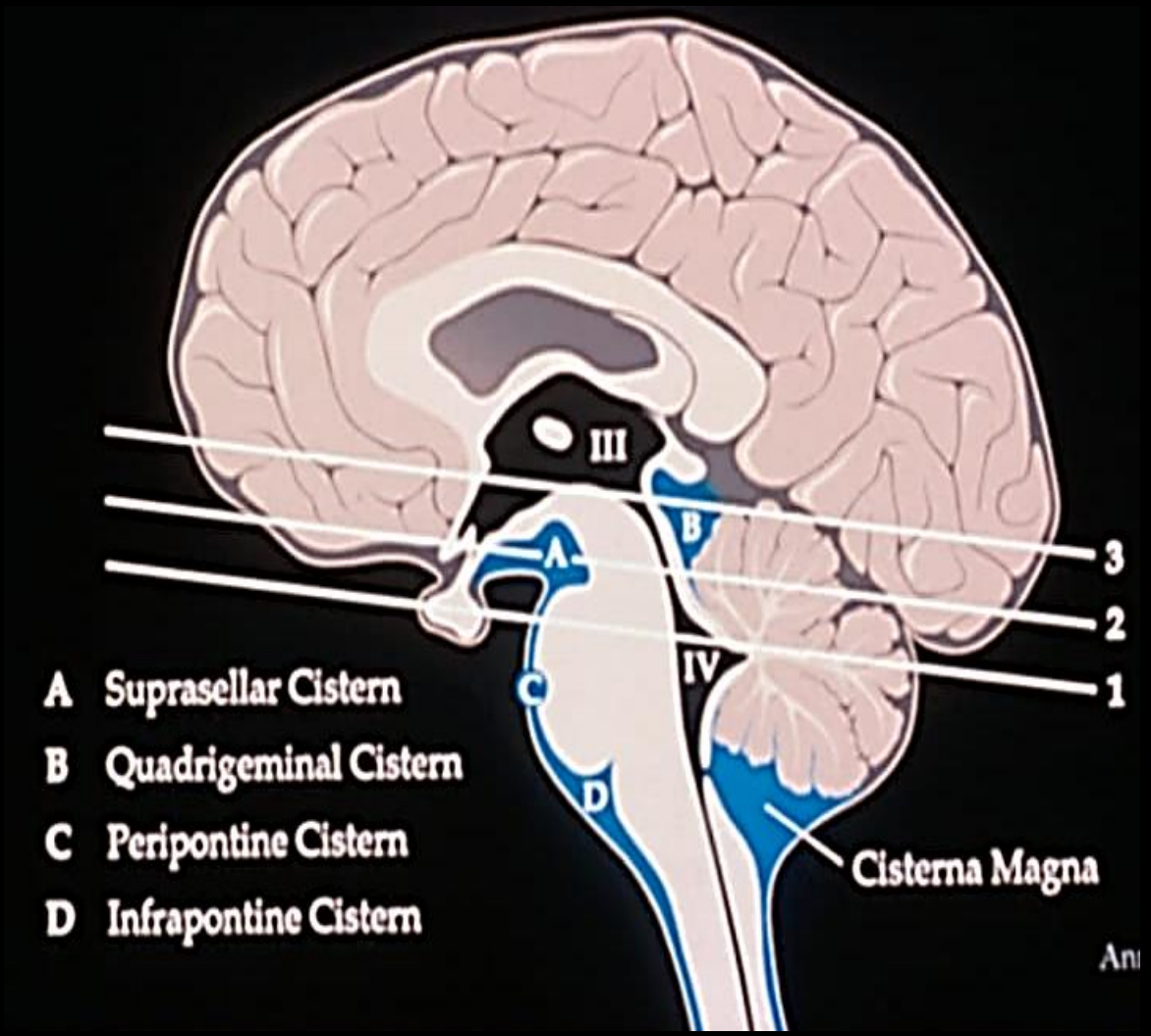
Fissures

- *The interhemispheric fissure separates both cerebral hemispheres.*
- *The Sylvian fissures separate the frontal and temporal lobes.*

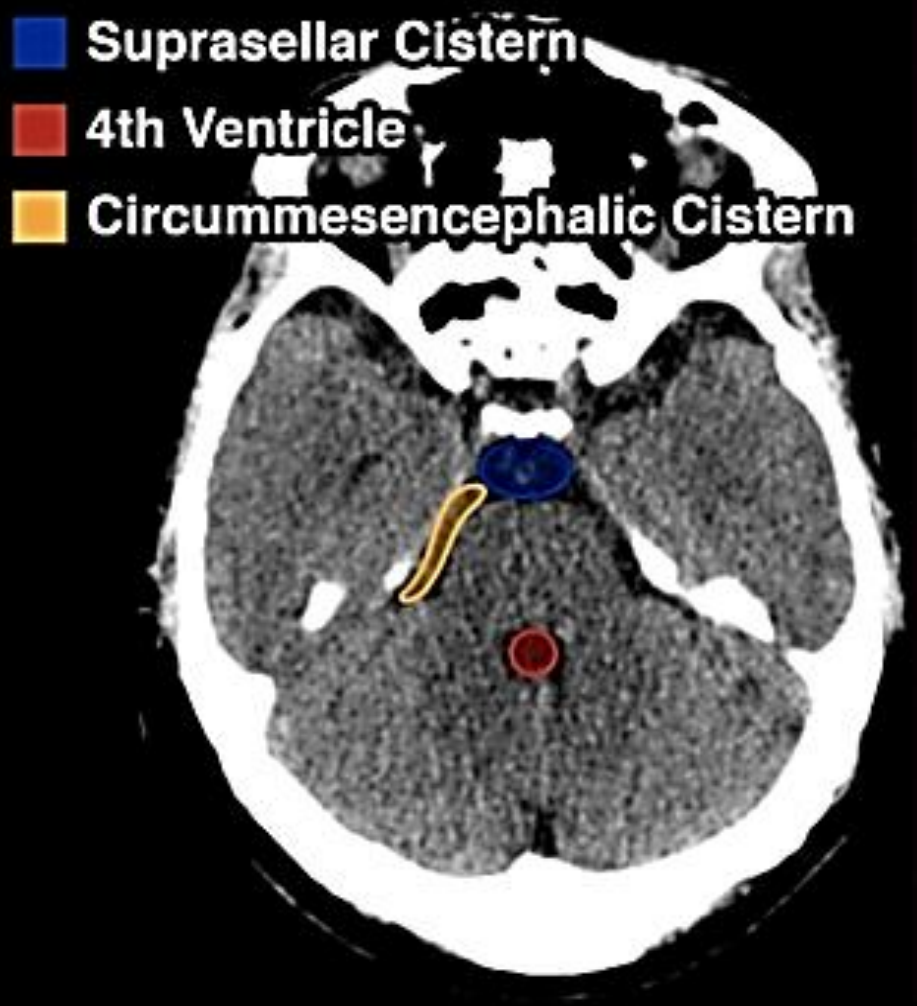


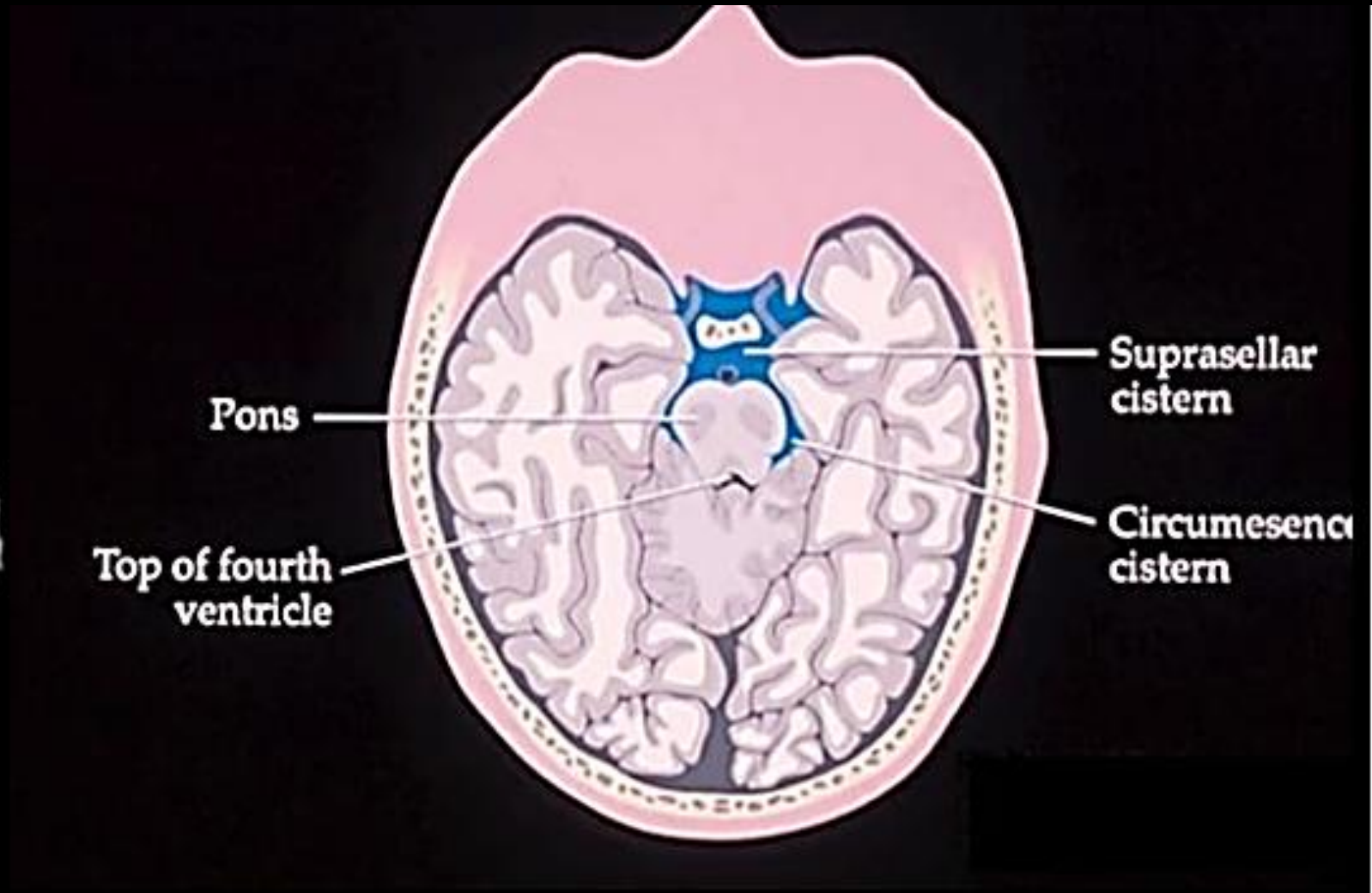
Main CSF CISTERNS:



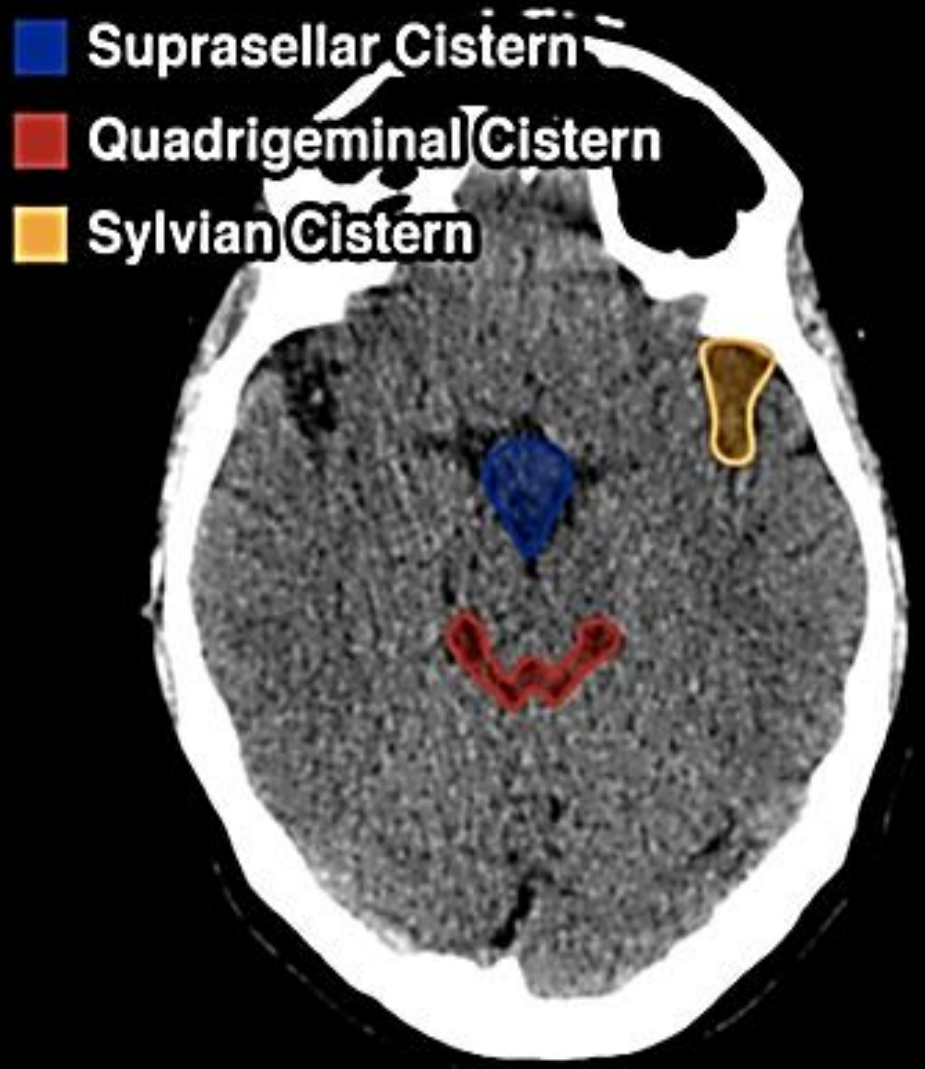


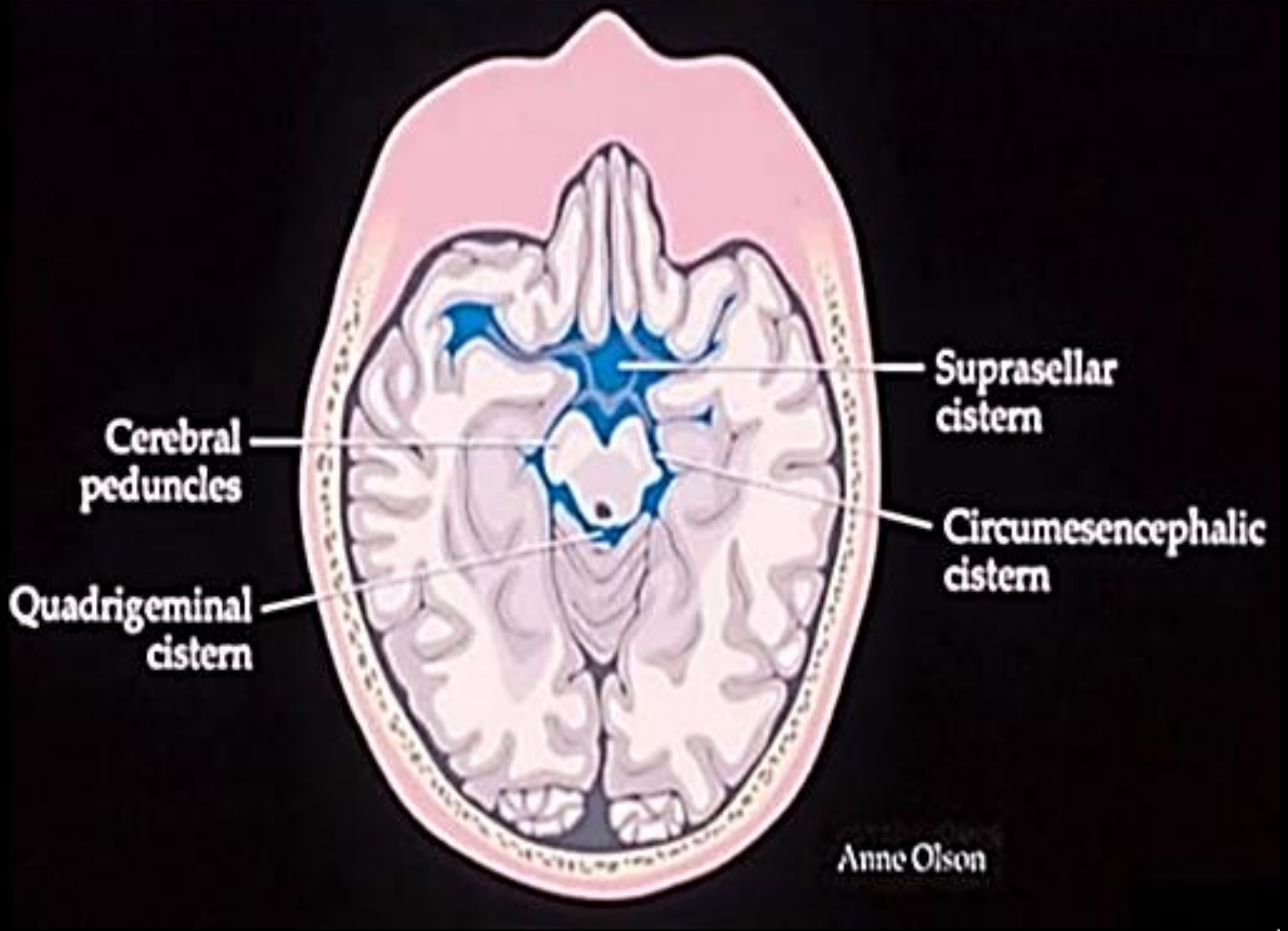
1. Cisterns Viewed at High Pontine Level





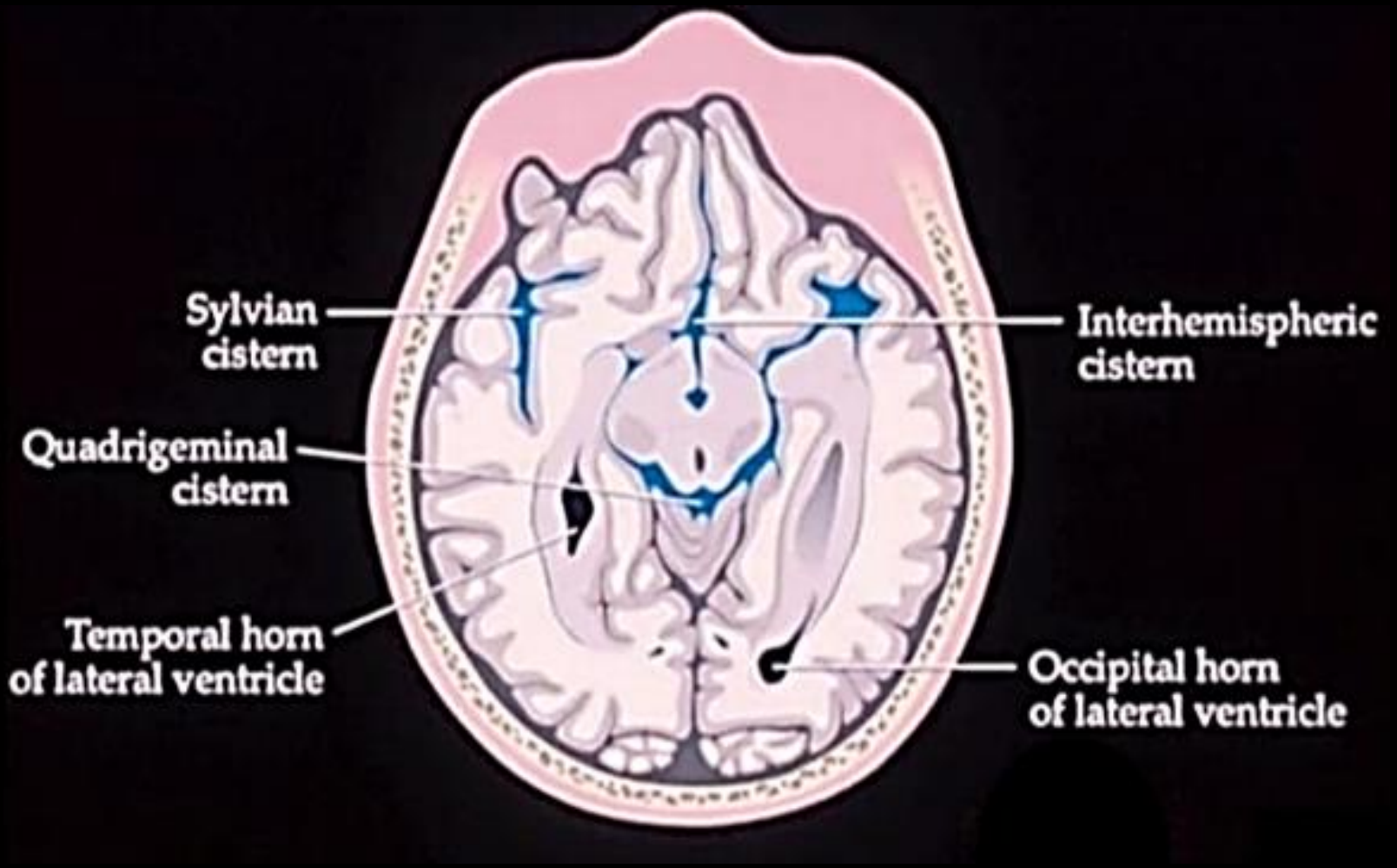
2. Cisterns Viewed at Level of Cerebral Peduncles



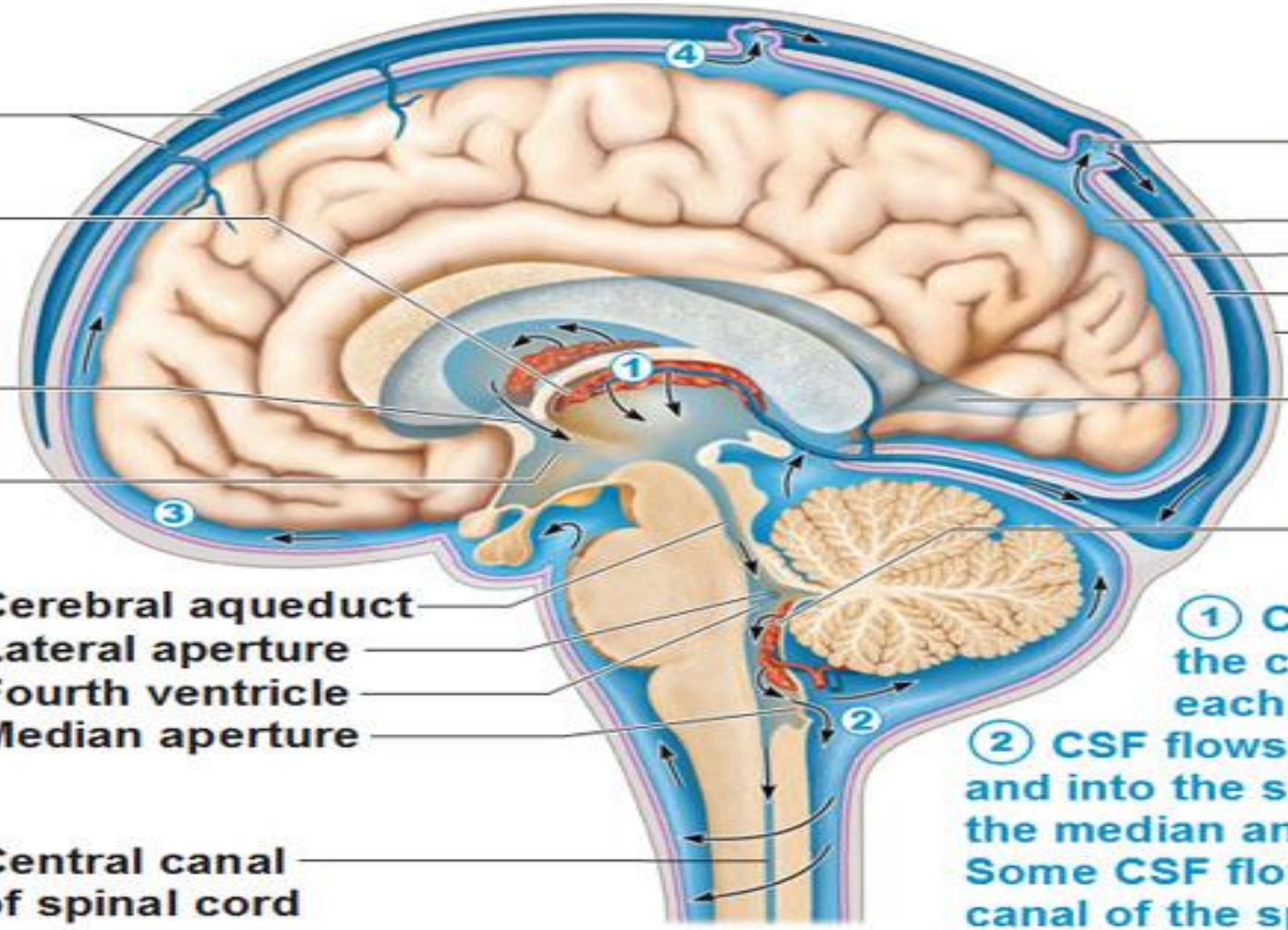


3. Cisterns Viewed at High Mid-Brain Level





Circulation of Cerebrospinal Fluid (CSF)



(1) CSF is produced by the choroid plexus of each ventricle.

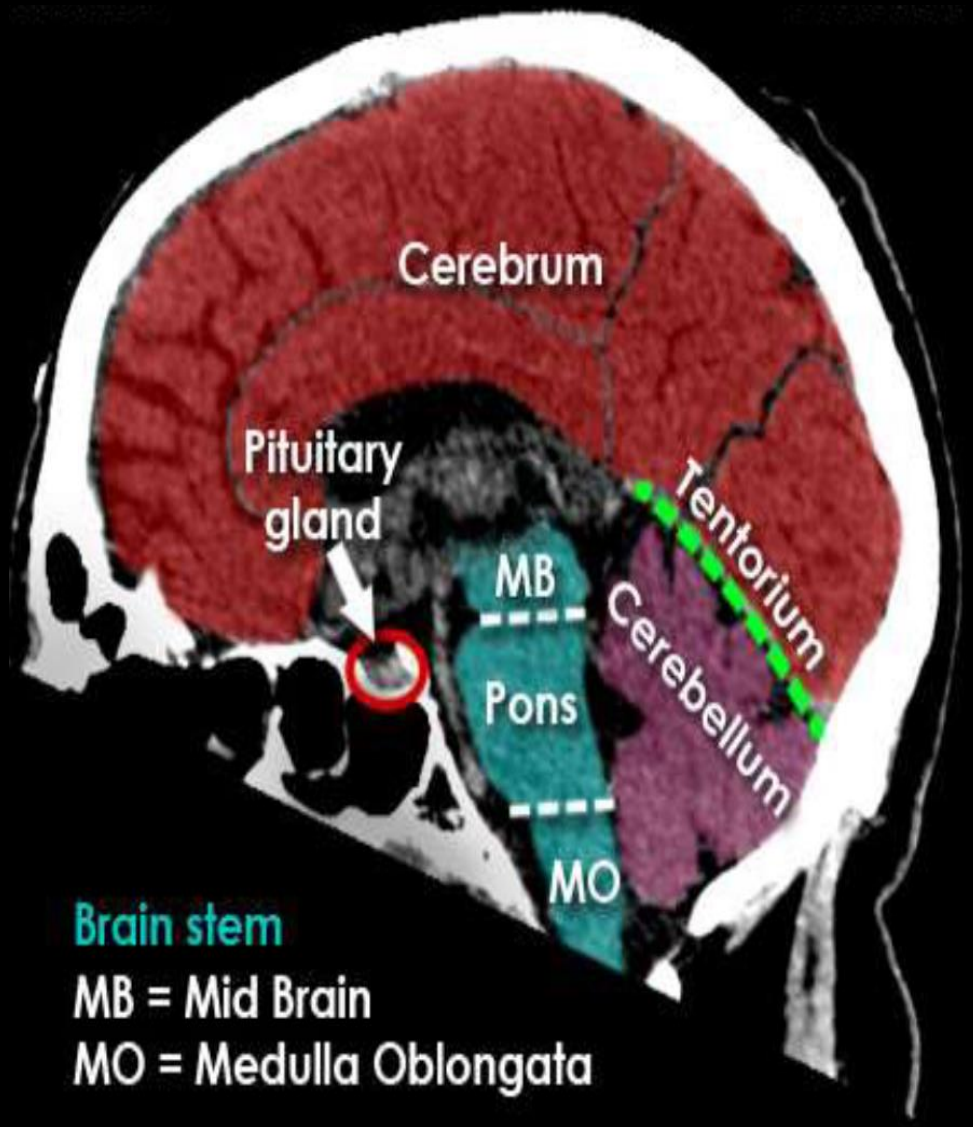
(2) CSF flows through the ventricles and into the subarachnoid space via the median and lateral apertures. Some CSF flows through the central canal of the spinal cord.

(3) CSF flows through the subarachnoid space.

(4) CSF is absorbed into the dural venous sinuses via the arachnoid villi.

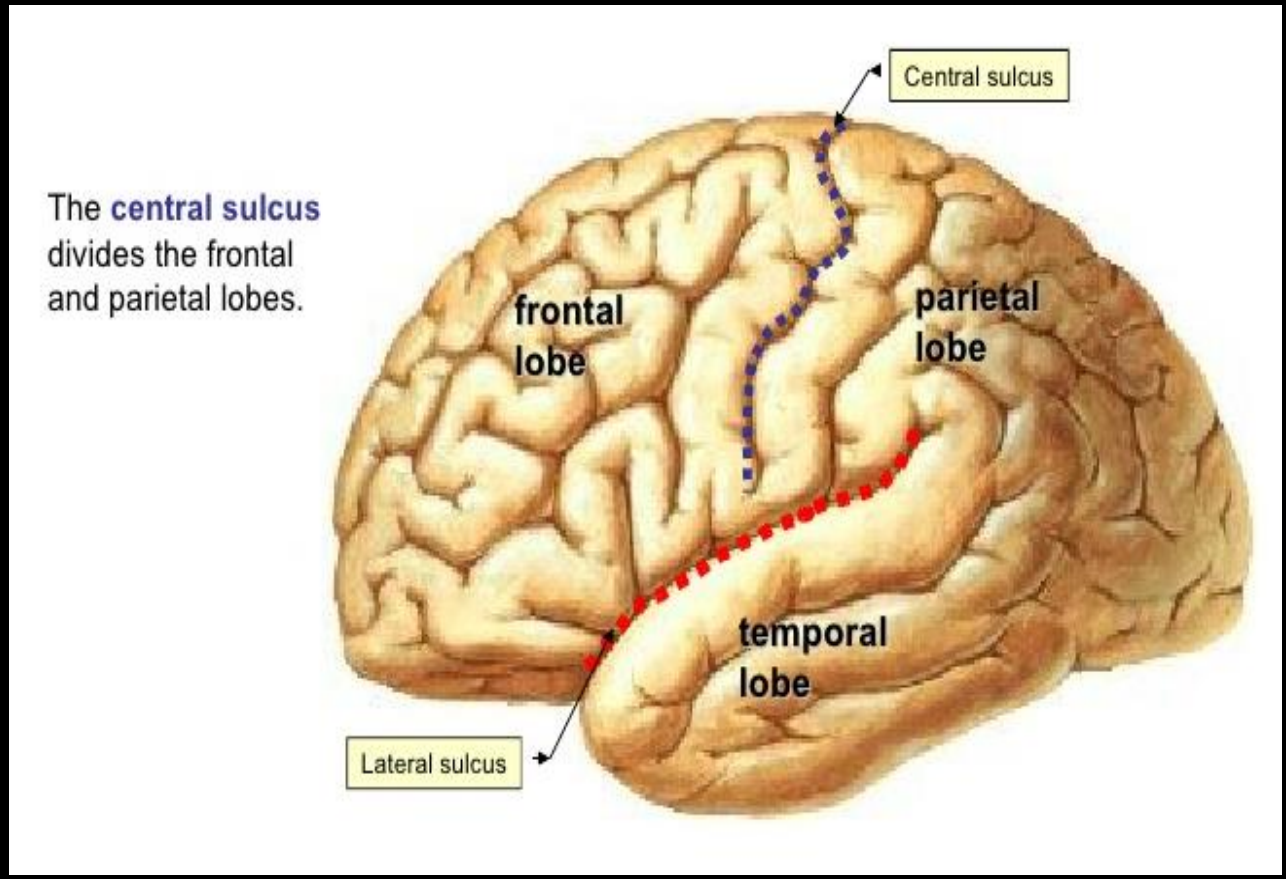
Brain.

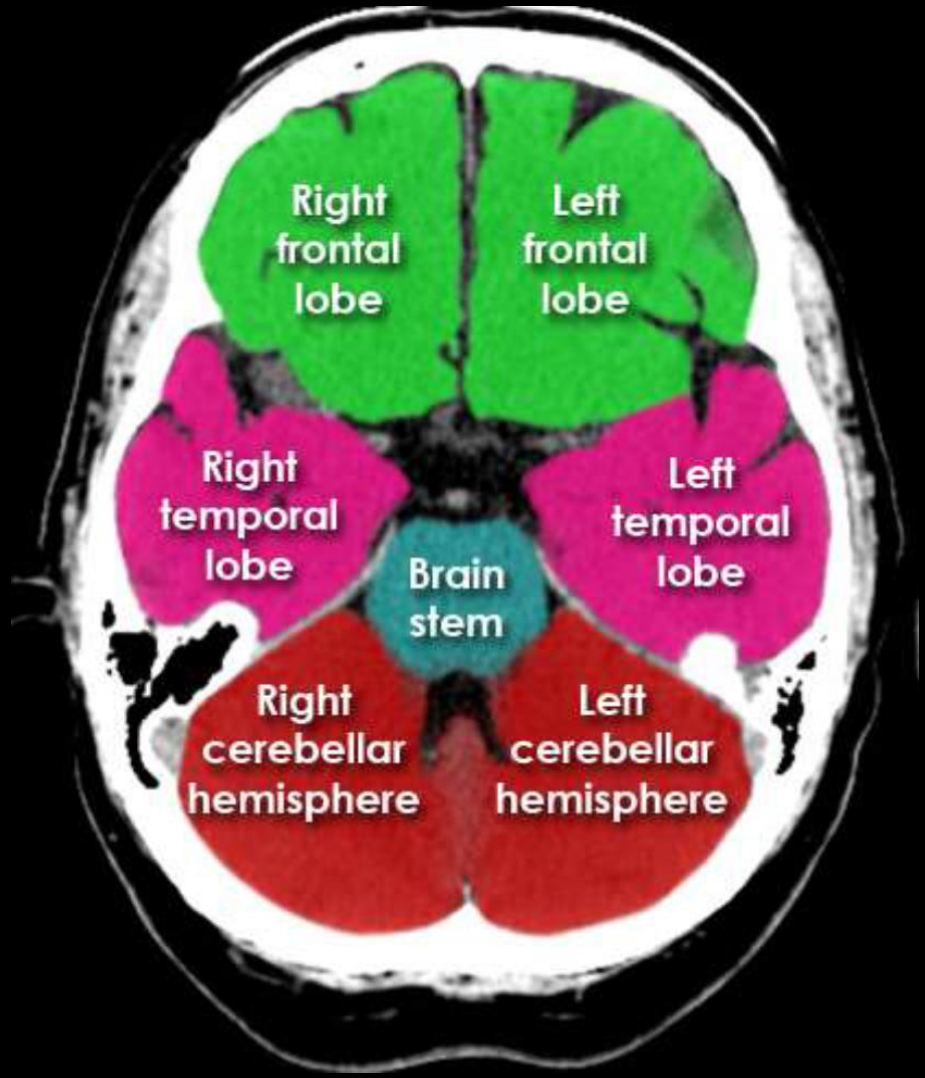
- *The brain is composed of the:*
  - *Cerebrum.*
  - *Cerebellum.*
  - *Brain stem.*



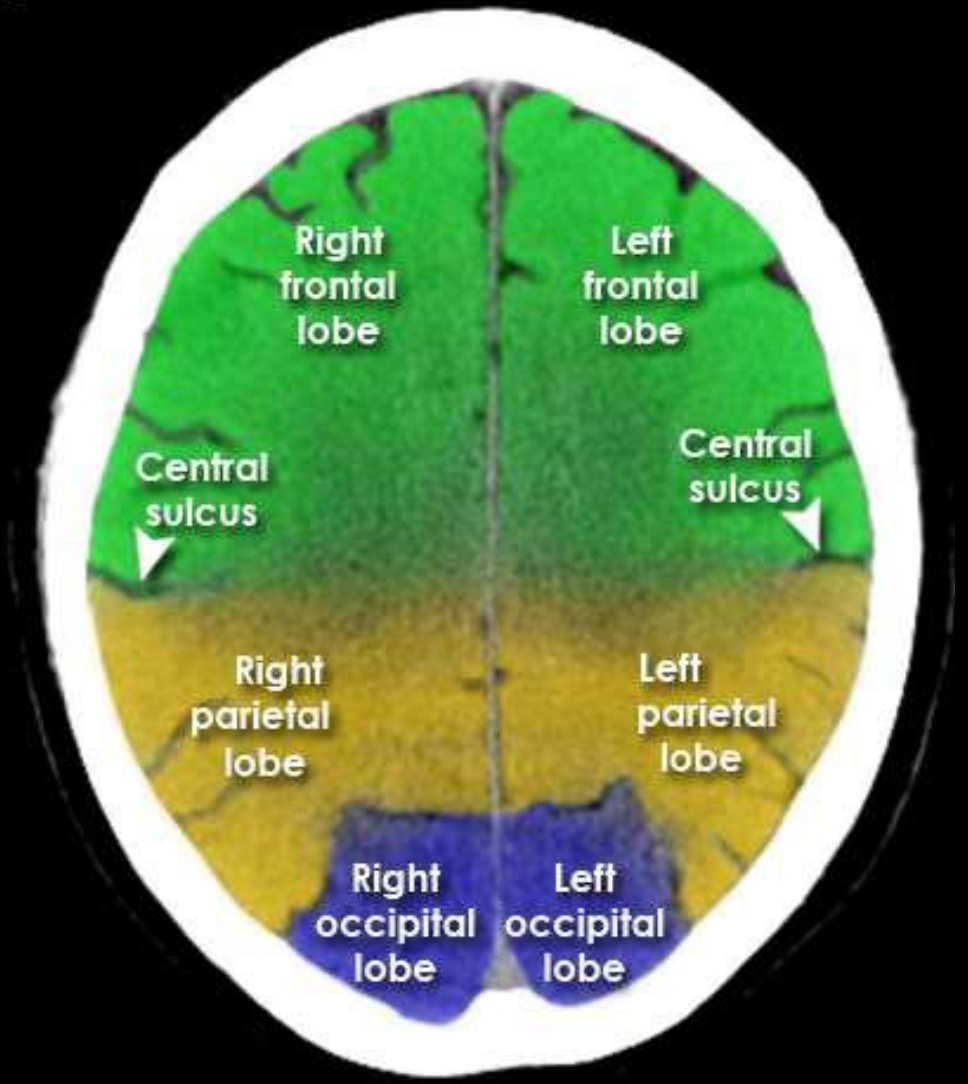
Brain.

- *The cerebrum is the largest part of the brain and is composed of right and left hemispheres. It performs higher functions like interpreting, touch, vision and hearing, as well as speech, reasoning, emotions, learning, and fine control of movement.*
- *The cerebral hemispheres have distinct fissures, which divide the brain into lobes. Each hemisphere has 4 lobes: frontal, temporal, parietal, and occipital.*





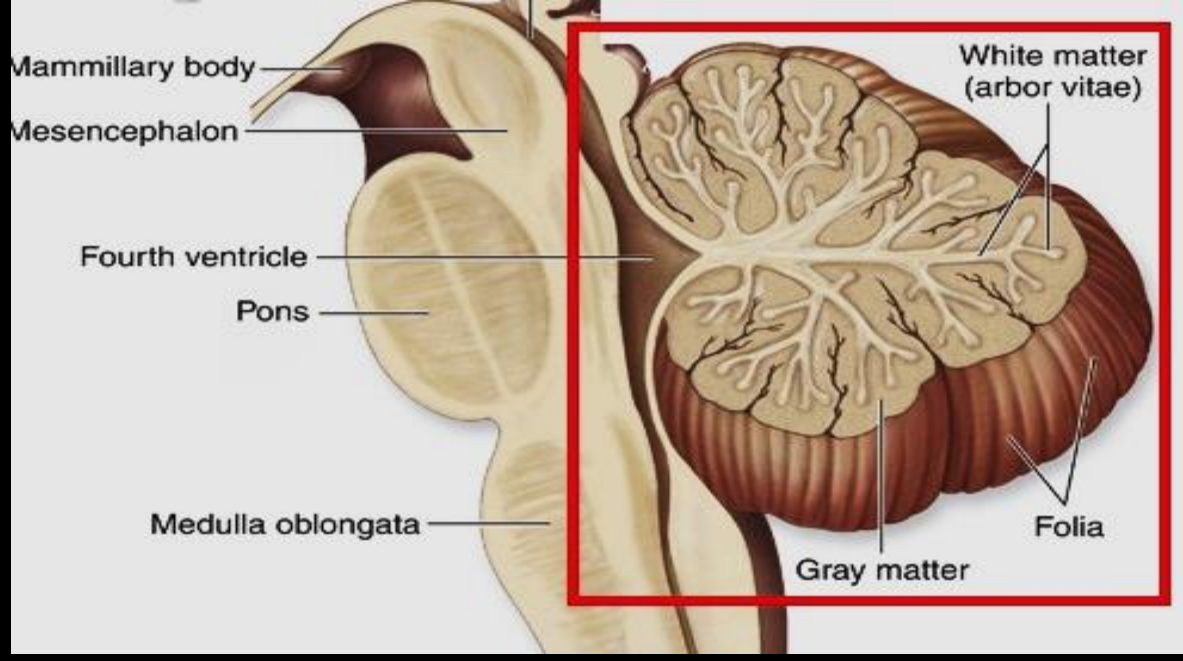
Brain lobes - CT brain (inferior slice) the frontal lobes at anterior cranial fossae .The temporal lobes at middle cranial fossae .The cerebellum and brain stem at the posterior fossa

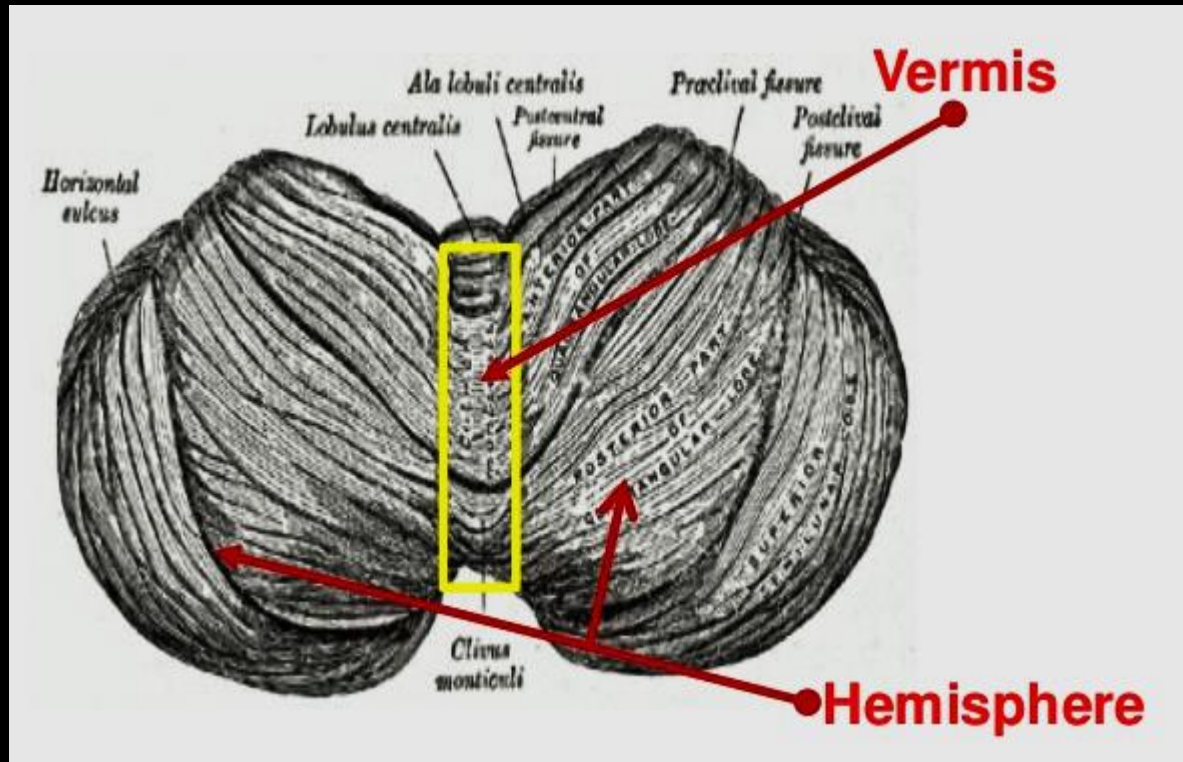


Brain lobes - CT brain (superior slice) On both sides the frontal lobes are separated from the parietal lobes by the central sulcus
Note: The large frontal lobes as compared to parietal and occipital lobes

Brain...

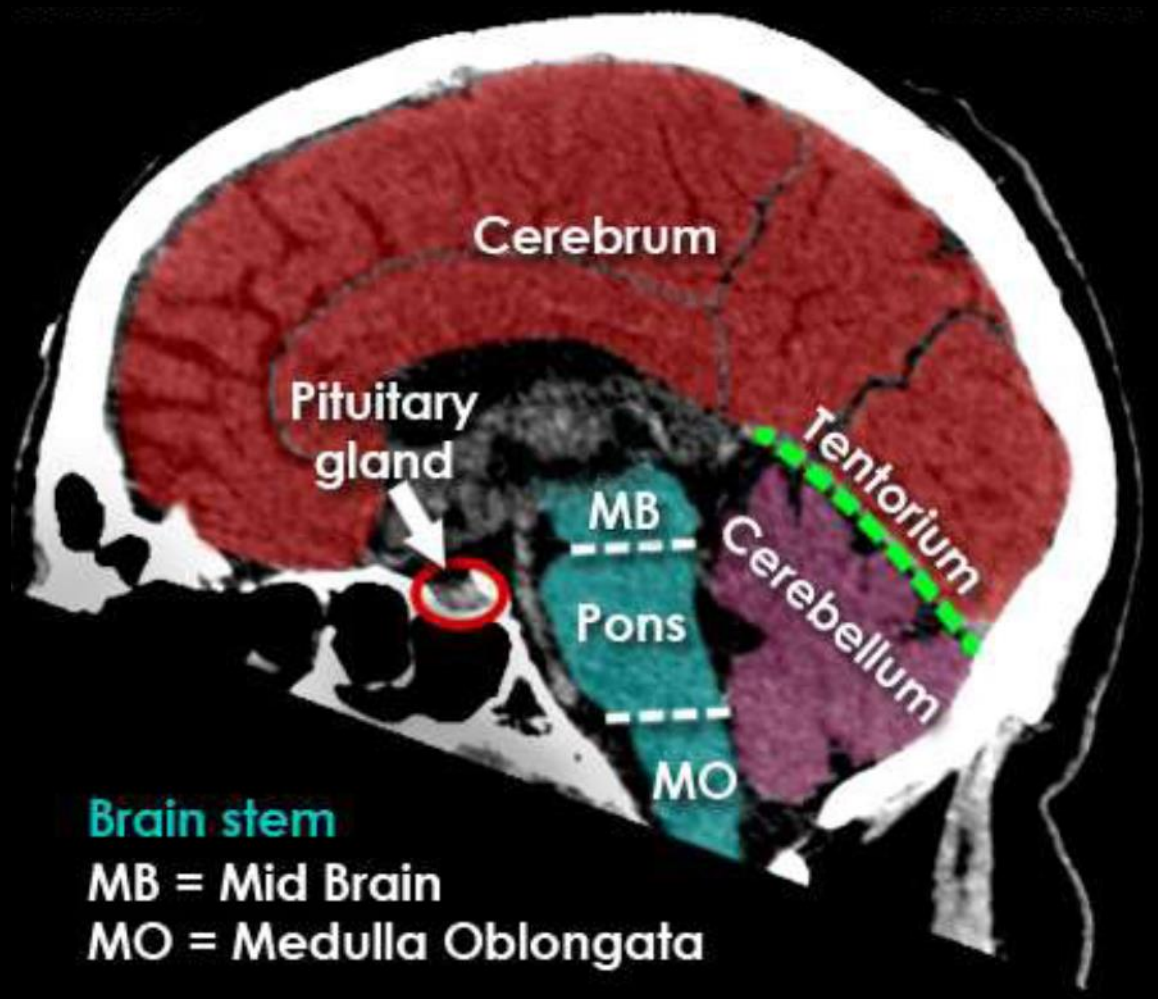
- *The cerebellum is located under the cerebrum.*
- *it's two hemishperes with midline structure named vermis.*
- *Its function is to coordinate muscle movements, maintain posture, and balance.*
- *Note the gyri in cerebellum is named foli*





Brain...

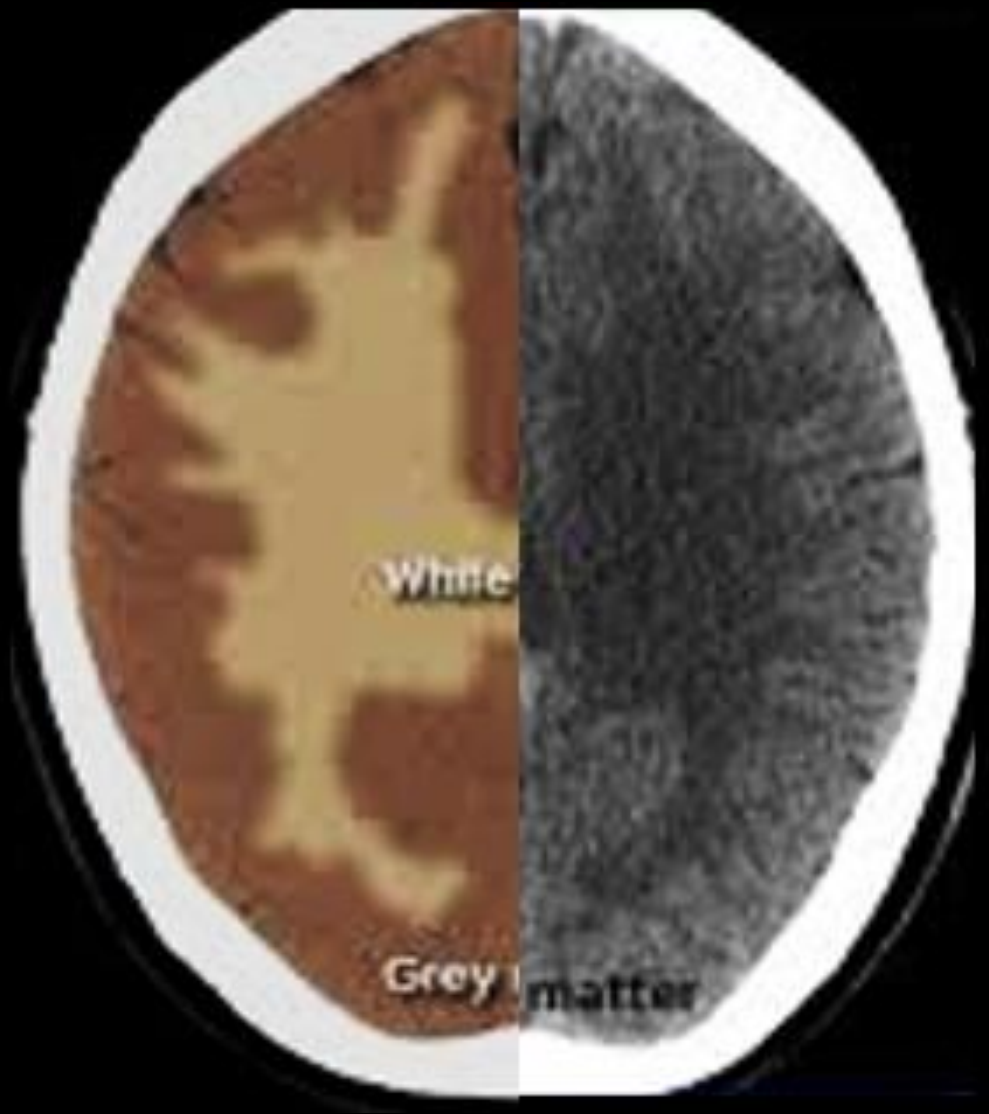
- *The brainstem is the lower extension of the brain, located in front of the cerebellum and connected to the spinal cord.*
- *It consists of three structures; the midbrain, pons and medulla oblongata.*
- *It performs many automatic functions such as breathing, heart rate, body temperature, wake and sleep cycles, digestion, sneezing, coughing, vomiting, and swallowing. Ten of the twelve cranial nerves originate in the brainstem.*



Destruction of these regions of the brain will cause "brain death." Without these key functions, humans cannot survive.

Grey / white matter

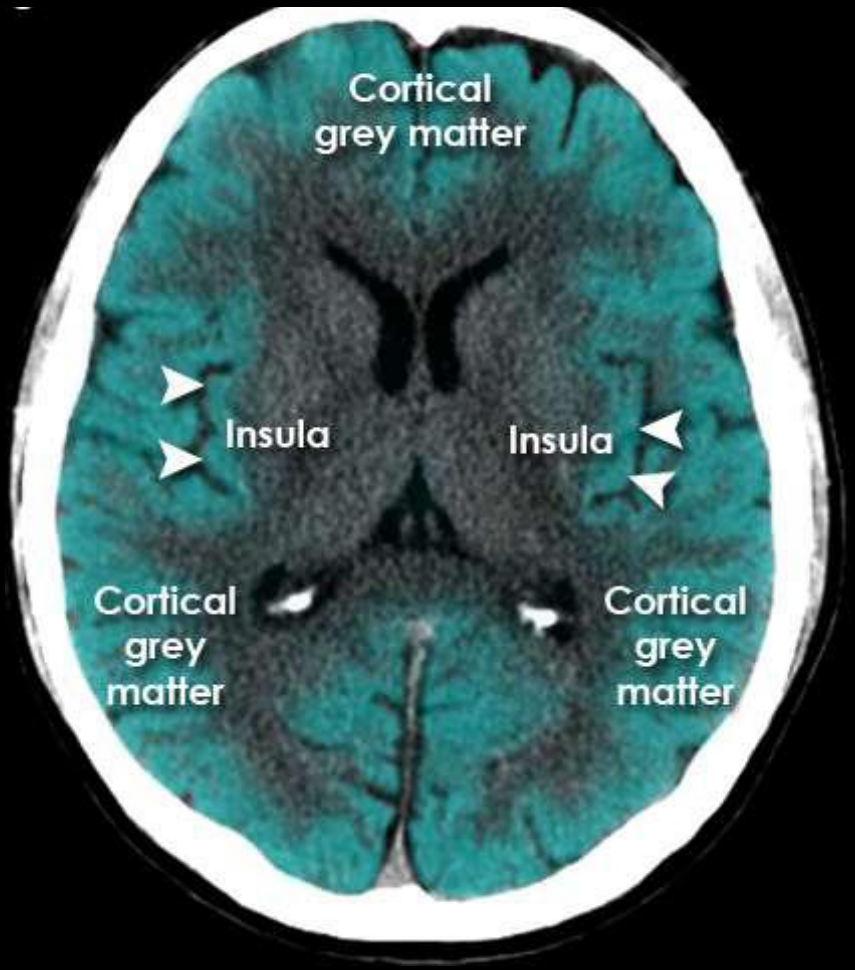
- *The brain consists of grey and white matter structures which are differentiated on CT by differences in density.*
- *White matter has a high content of myelinated axons.*
- *Grey matter contains relatively few axons and a higher number of cell bodies.*
- *As myelin is a fatty substance it is of relatively low density compared to the cellular grey matter. White matter, therefore, appears blacker than grey matter.*

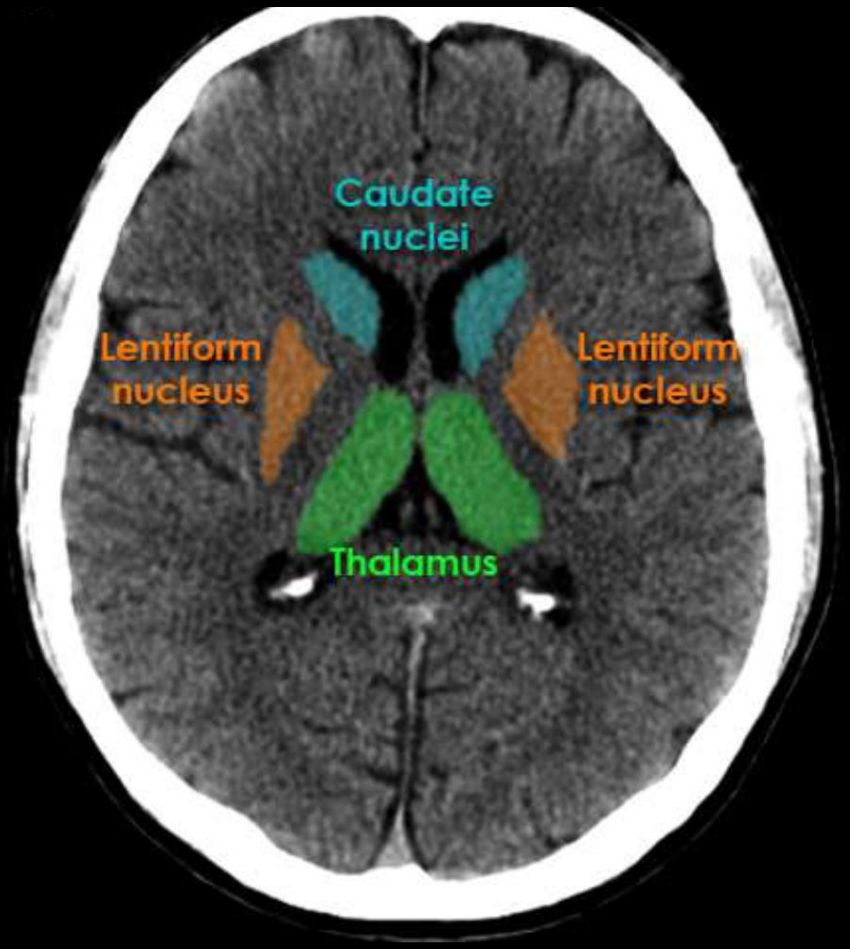


Grey matter structures

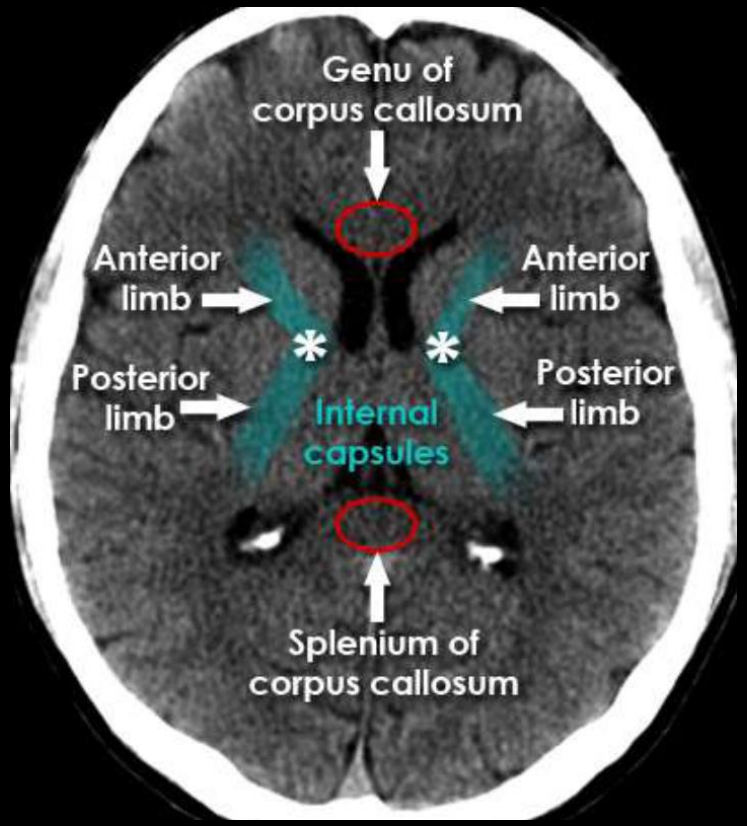
Important grey matter structures visible on CT :

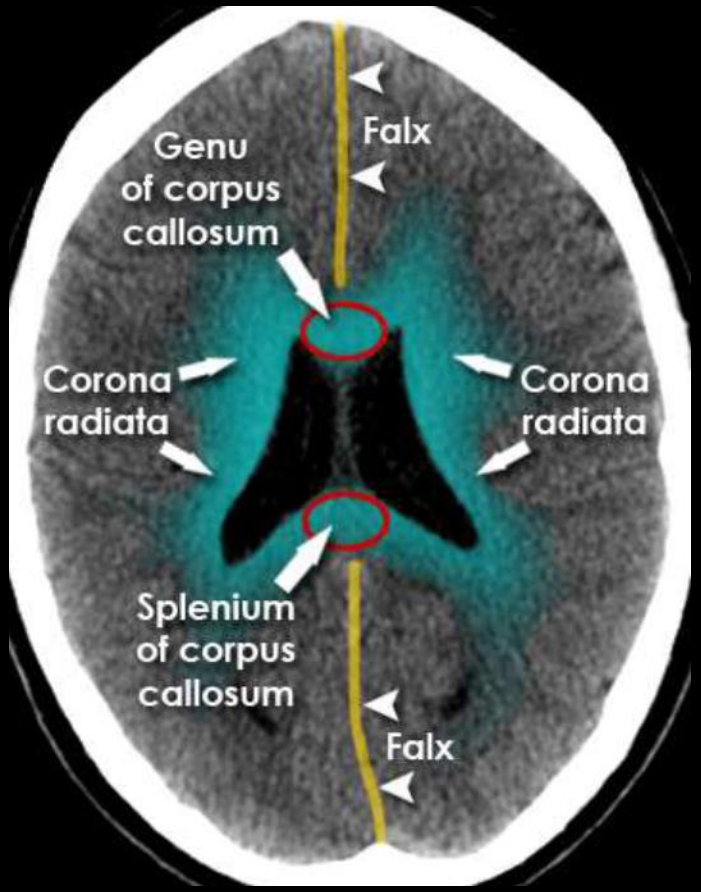
- *Cortex, insula.*
- *Basal ganglia(lentiform nucleus + caudate nucleus).*
- *Thalamus .*

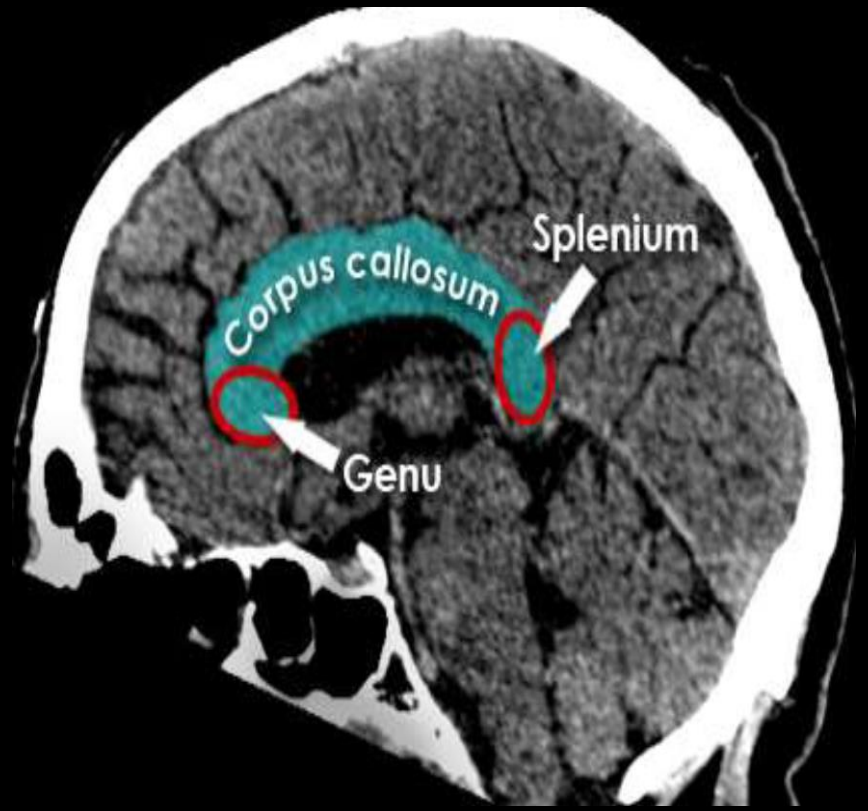




White matter structures

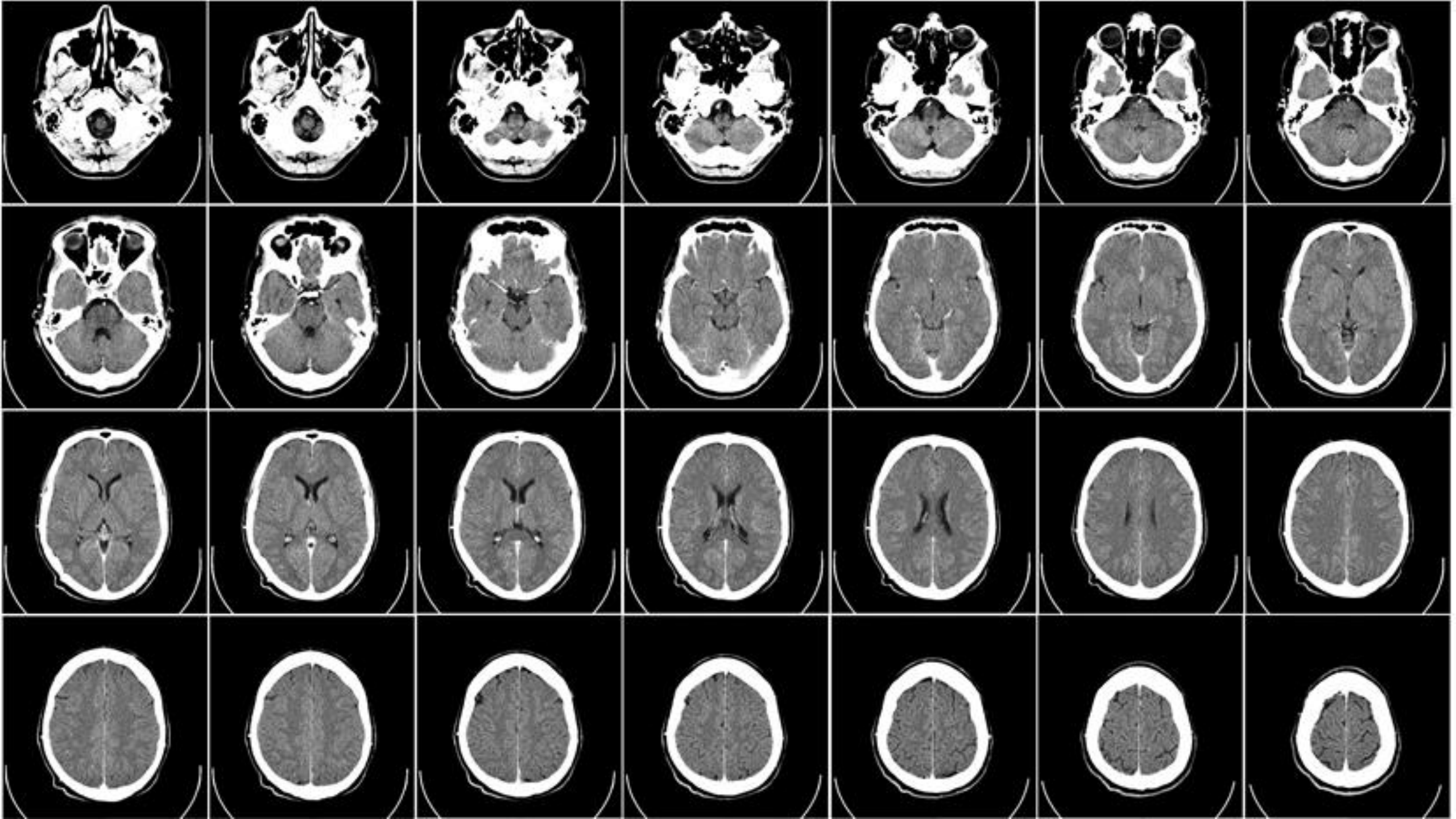






The internal capsules *are white matter tracts which connect with the* ***corona radiata*** *superiorly with the brain stem inferiorly.*

The corpus callosum *is a white matter tract located in the midline. It arches over the lateral ventricles and connects white matter of the left and right cerebral hemispheres*



AXIAL SECTIONS OF CT HEAD

POSTERIOR FOSSA CUTS

-ABOVE THE FORAMEN MAGNUM LEVEL

-LEVEL OF THE FOURTH VENTRICLE

-ABOVE THE FOURTH VENTRICULAR LEVEL

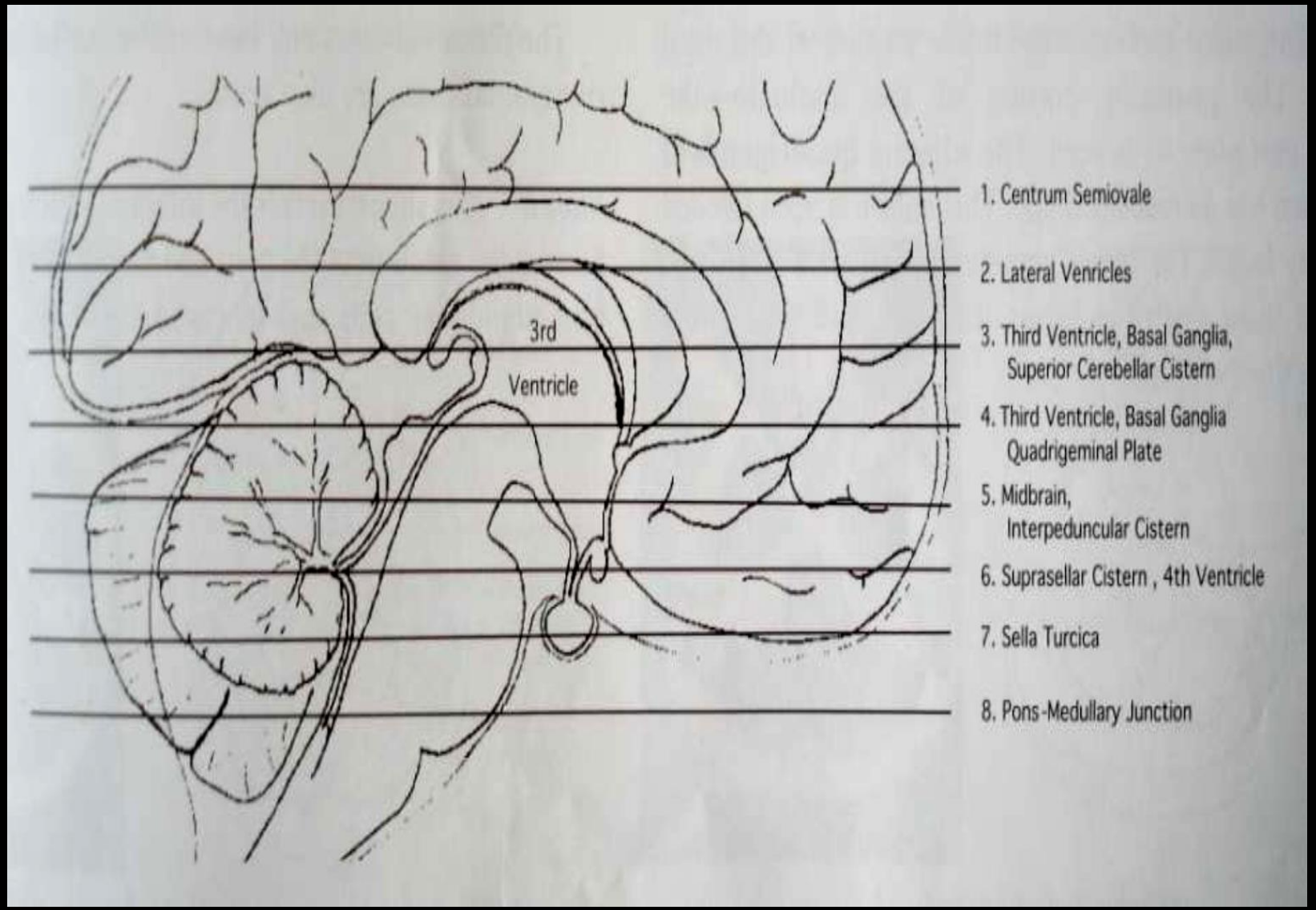
-TENTORIAL

SUPRATENTORIAL CUTS

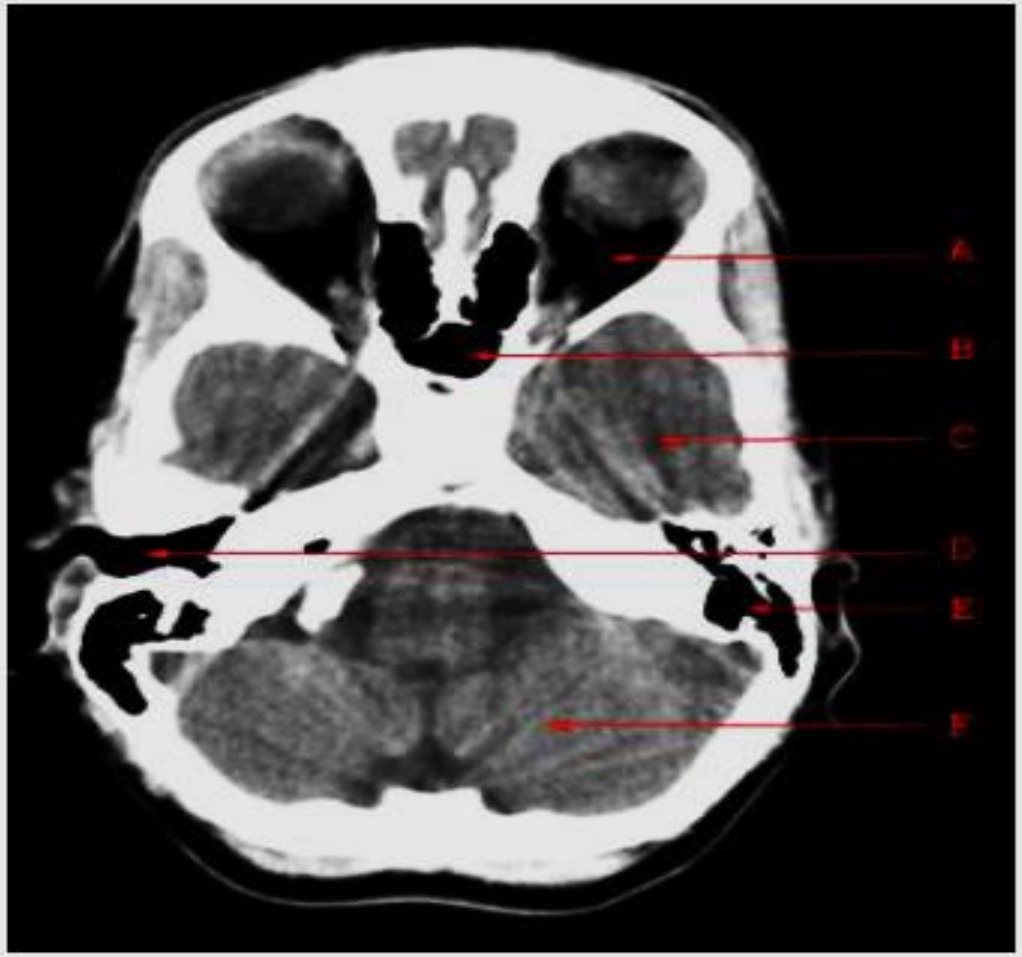
-THIRD VENTRICULAR LEVEL

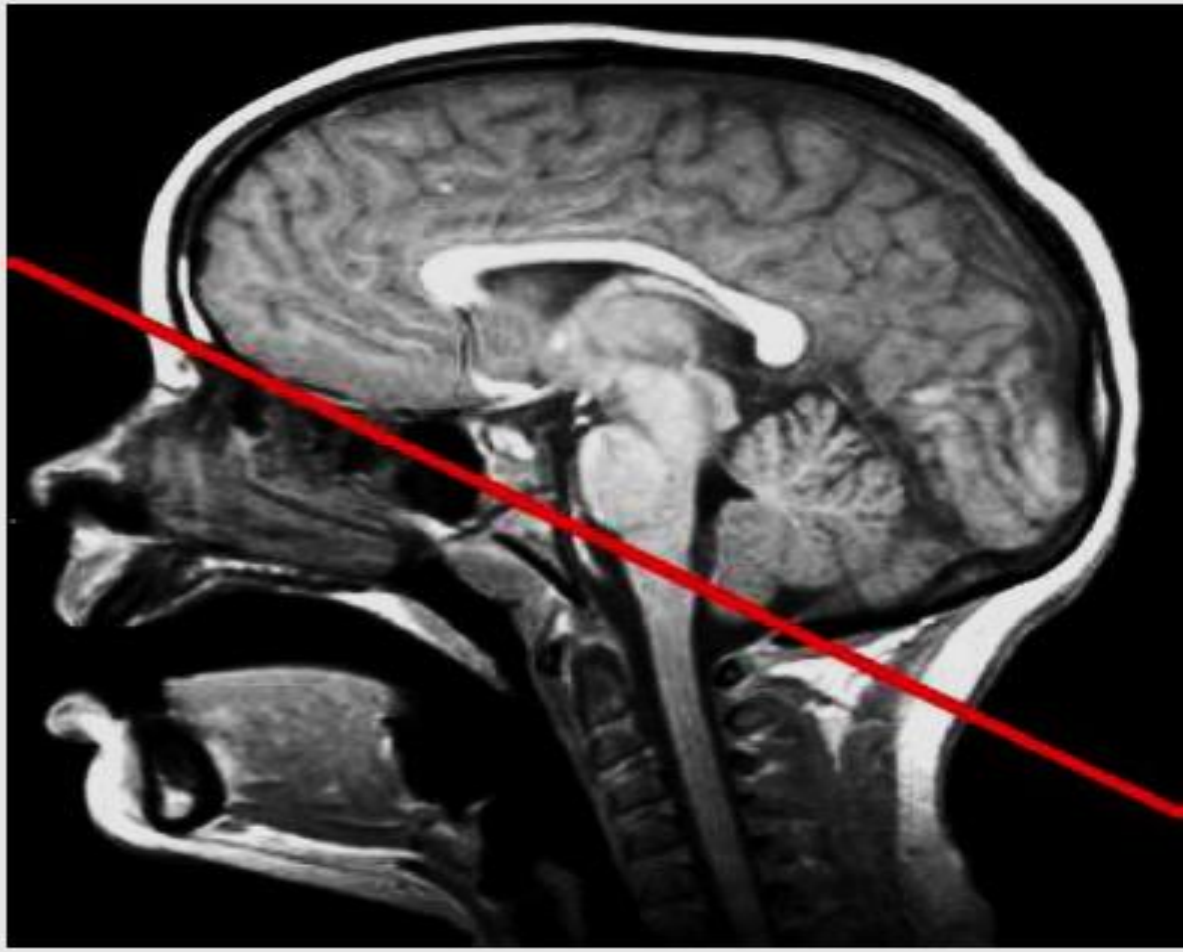
-LATERAL VENTRICULAR LEVEL

-ABOVE THE VENTRICULAR



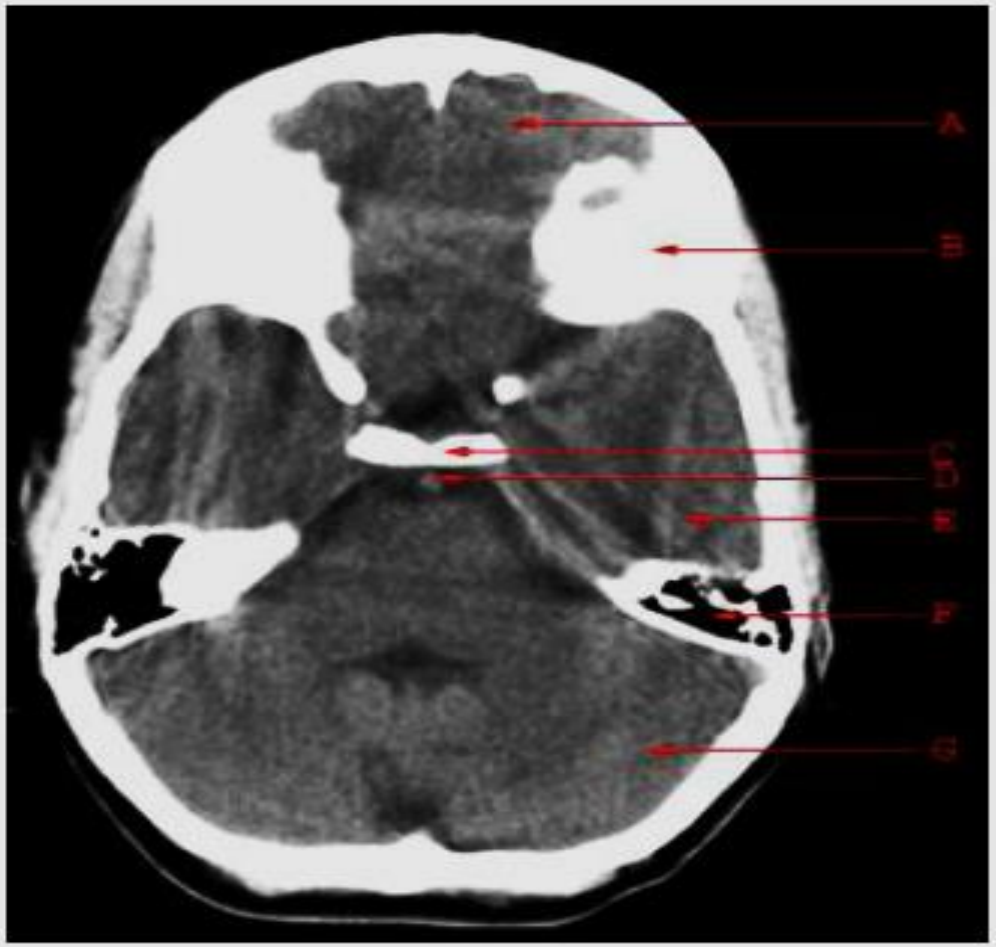
NORMAL ANATOMY.....

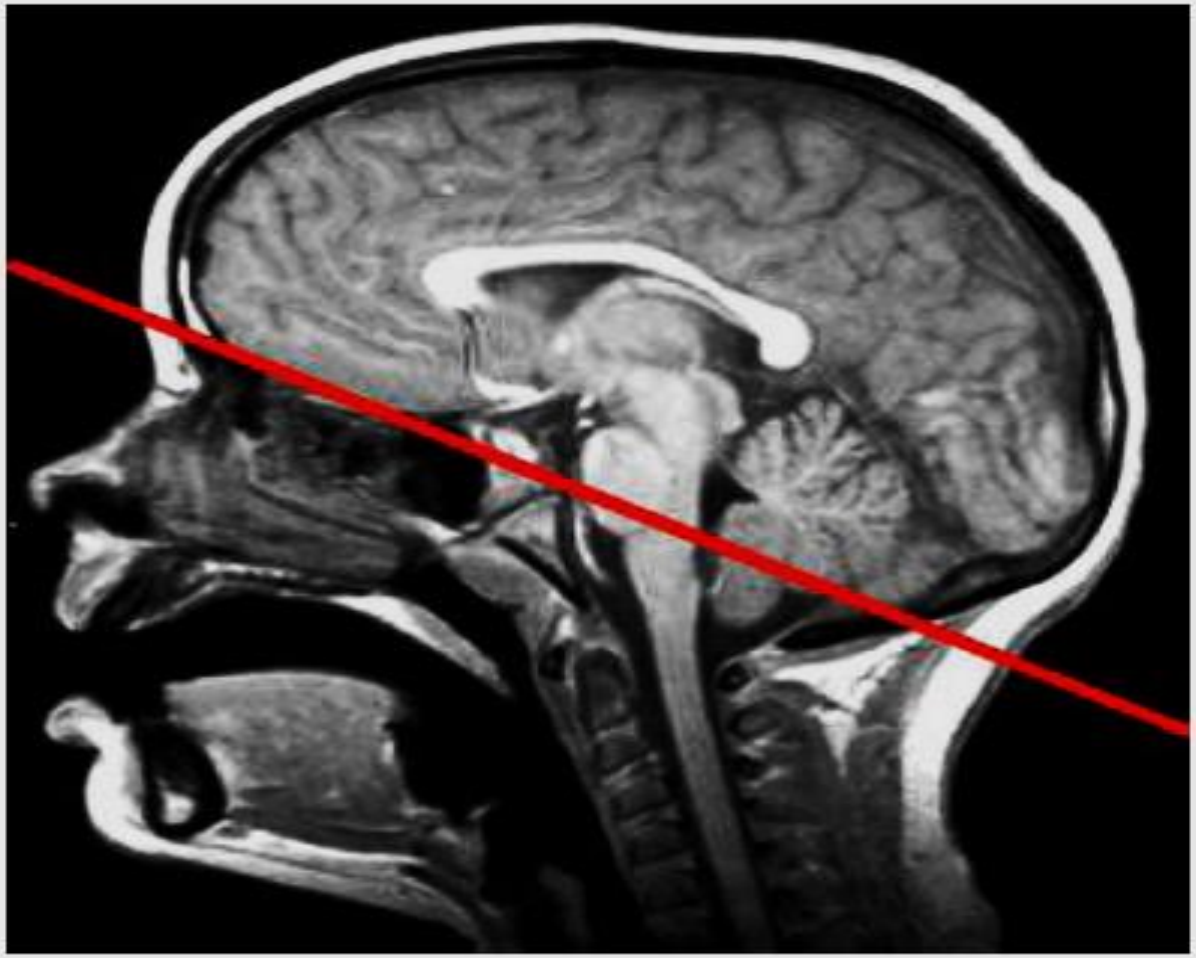




A= ORBIT , *B= SPHENOID SINUS* , C= TEMPORAL LOBE, *D=EXTERNAL AUDITORY CANAL*
E= MASTOID AIR CELLS *F= CEREBELLAR HEMISPHERES*

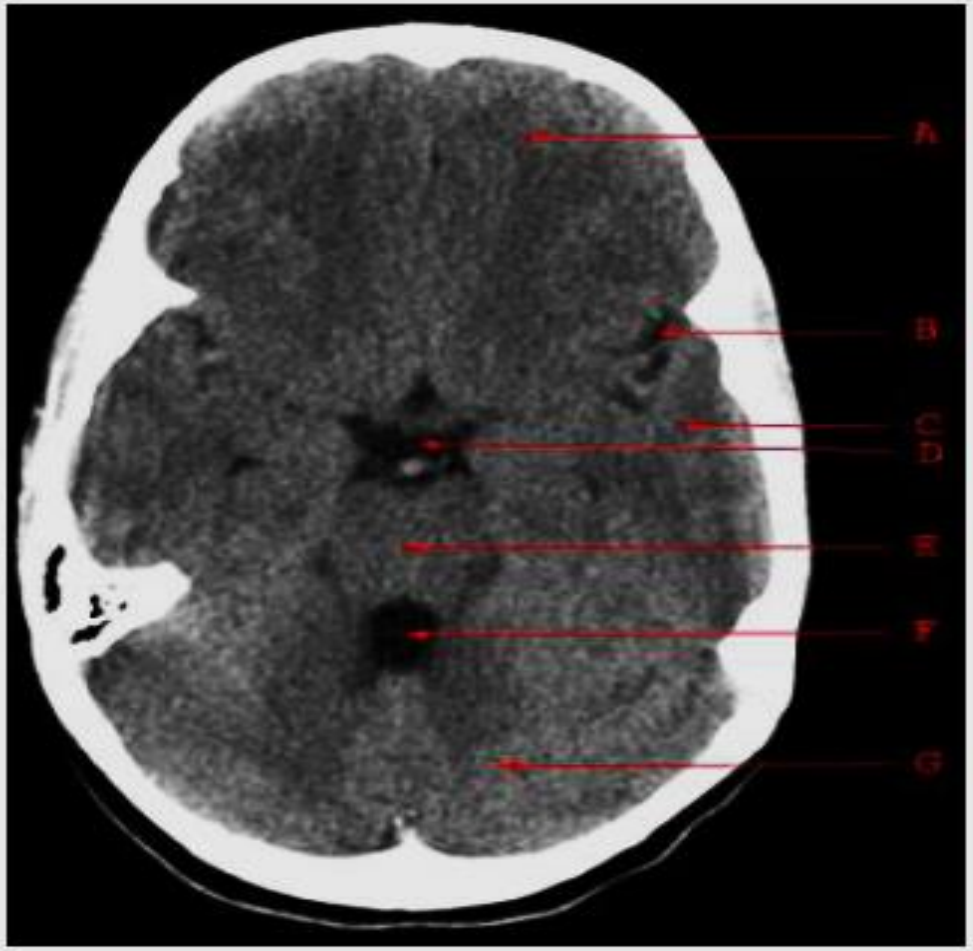
NORMAL ANATOMY.....

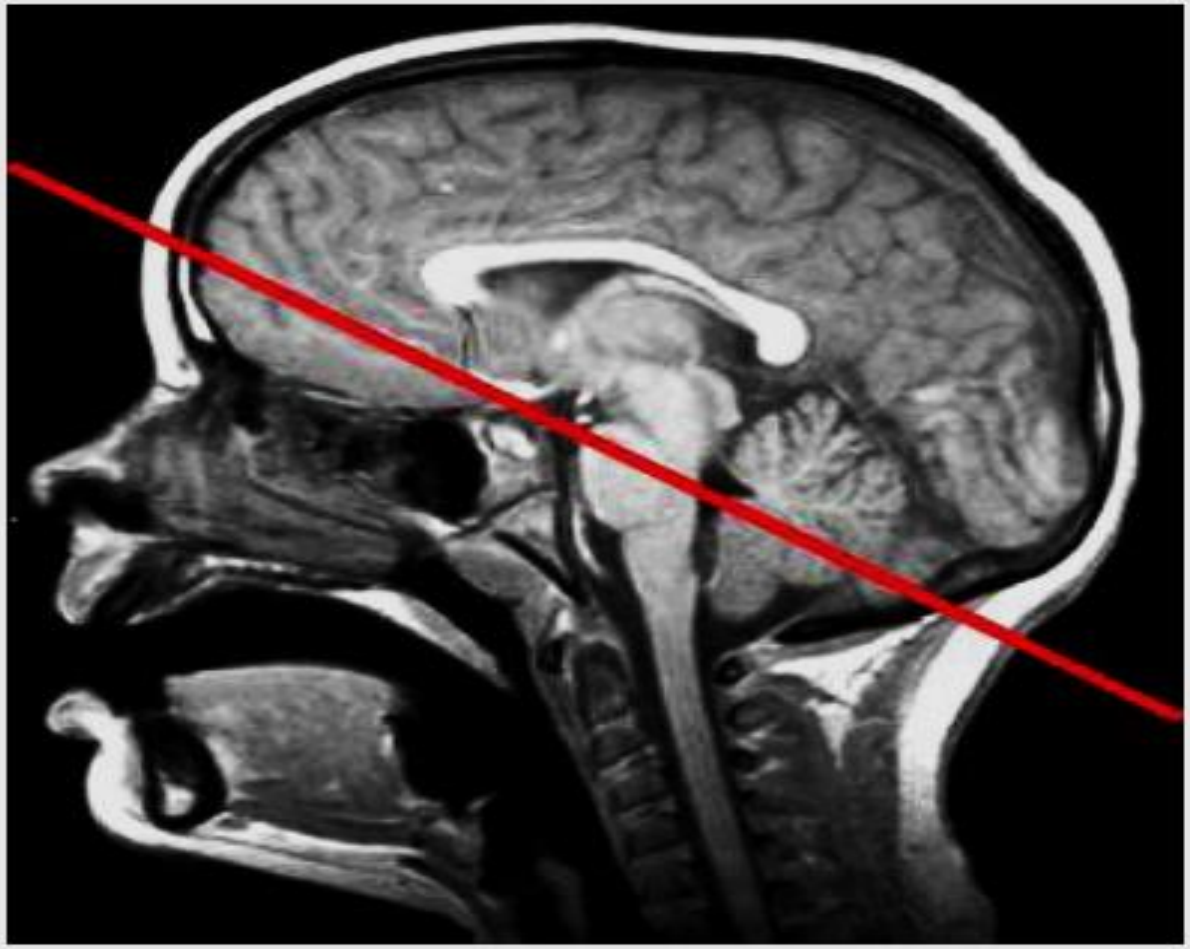




A=Frontal Lobe, *B= Frontal Bone (Superior Surface of Orbital Part)*, C= Dorsum Sellae, *D=Basilar Artery* E= Temporal Lobe *F= Mastoid Air Cells* G=Cerebellar Hemisphere

NORMAL ANATOMY.....



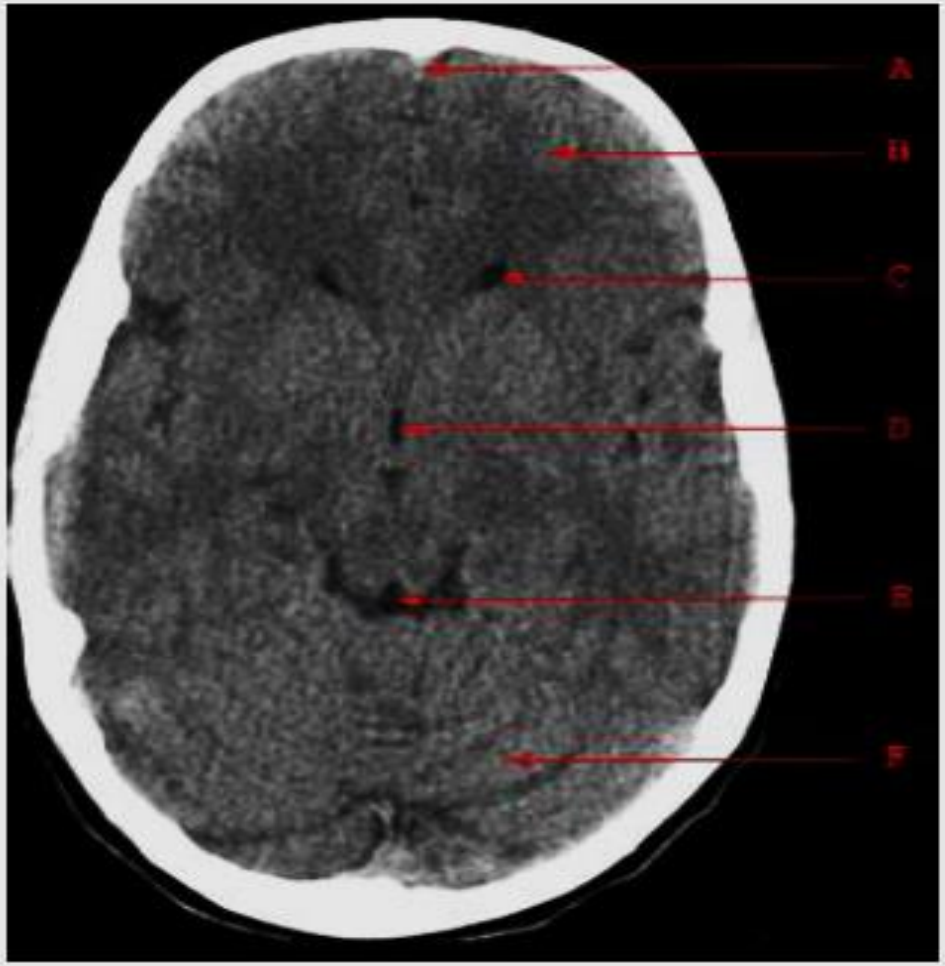


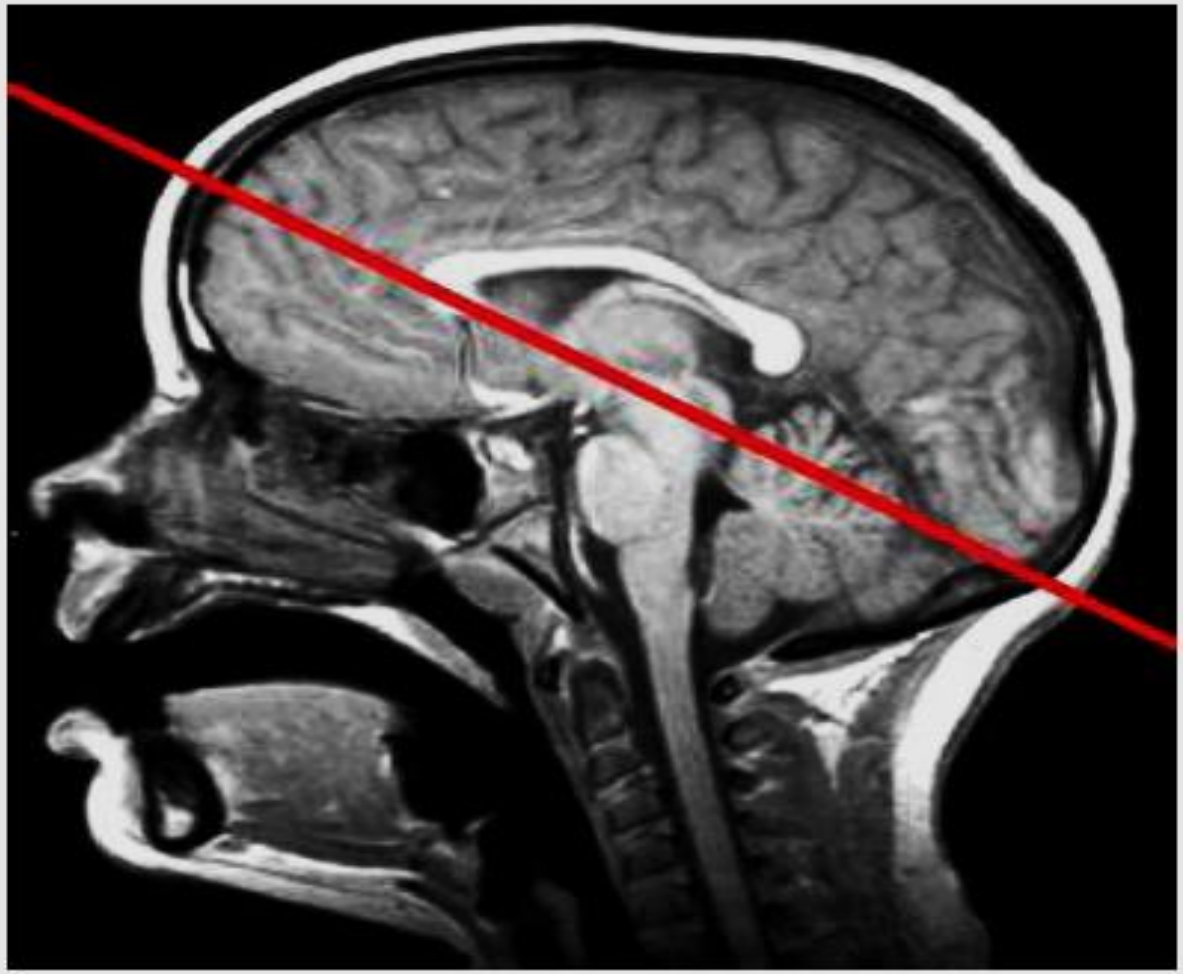
A=FRONTAL LOBE B= SYLVIAN FISSURE C=TEMPORAL LOBE

D=SUPRASELLAR CISTERN E=MIDBRAIN F=FOURTH VENTRICLE

G= CEREBELLAR HEMISPHERE

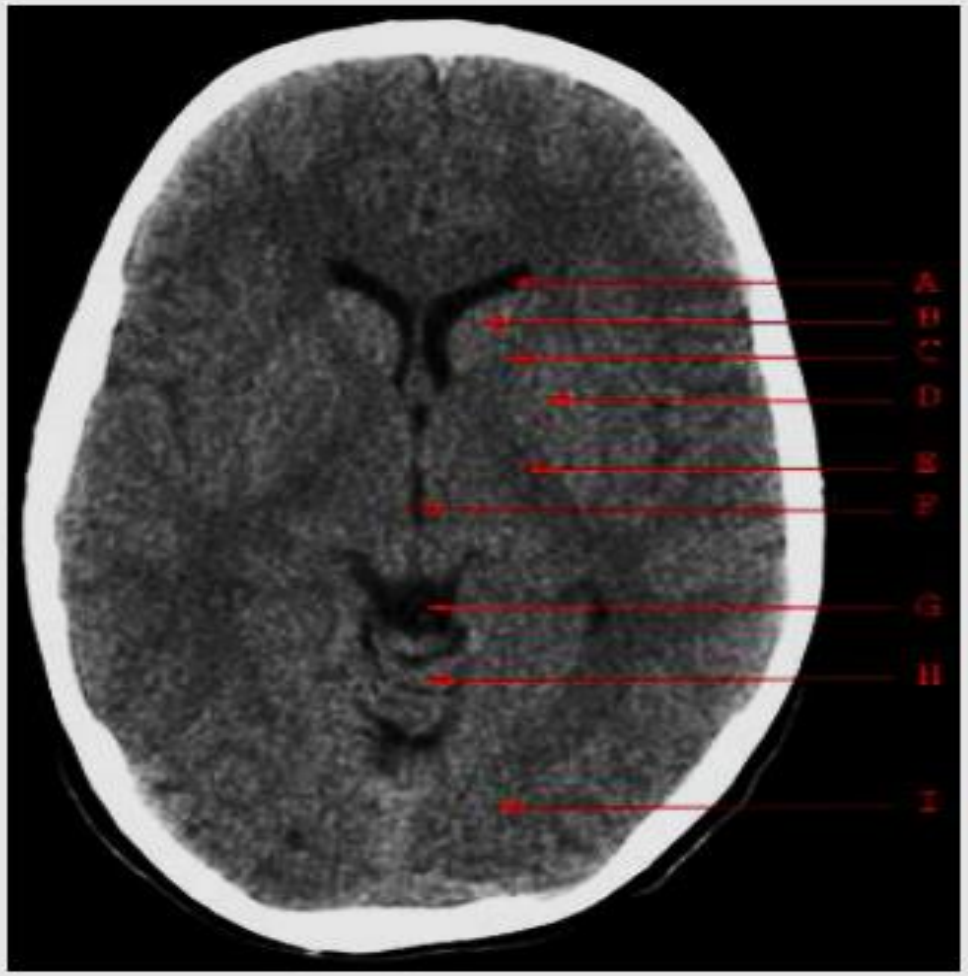
NORMAL ANATOMY.....

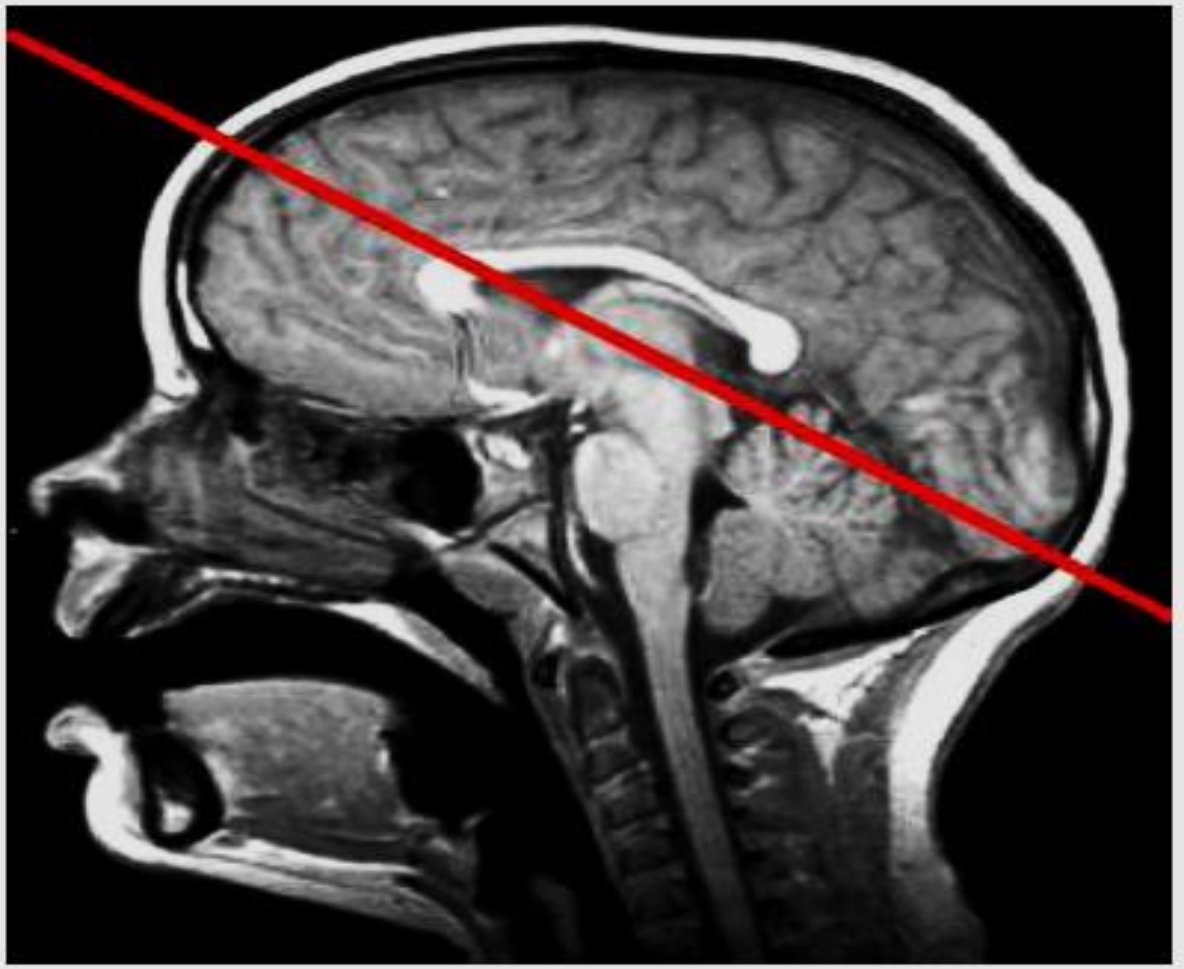




A=FALX CEREBRI B=FRONTAL LOBE C=ANTERIOR HORN LAT VENTRICLE
D=THIRD VENTRICLE E=QUADRIGEMINAL PLATE CISTERN F=CEREBELLUM

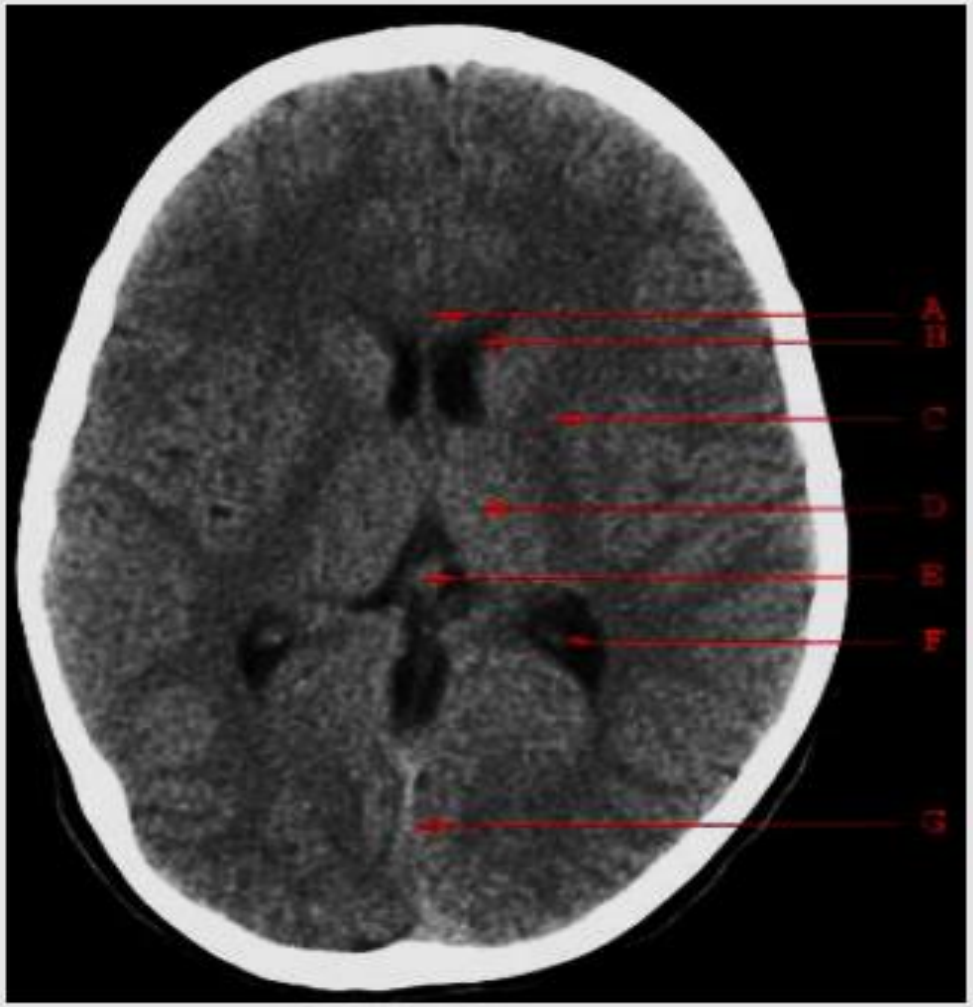
NORMAL ANATOMY.....

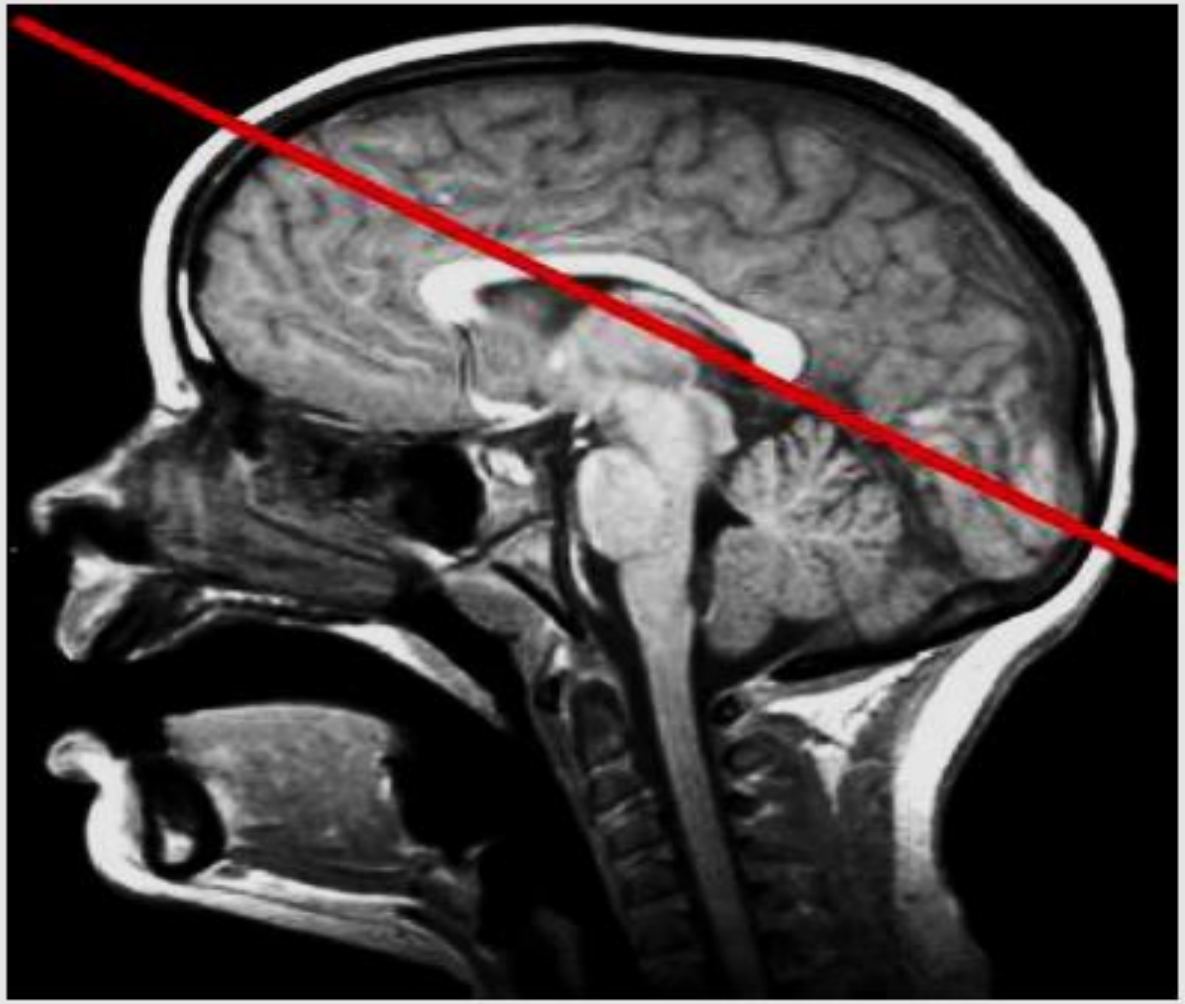




A=ANTERIOR HORN LAT VENTRICLE B=CAUDATE NUCLEUS C=ANT LIMB INT CAPSULE
D=GLOBUS PALLIDUS AND PUTAMEN E=POST LIMB INT CAPSULE F=THIRD VENTRICLE
G=QUADRIGEMINAL PLATE CISTERN H=CEREBELLAR VERMIS I=OCCIPITAL LOBE

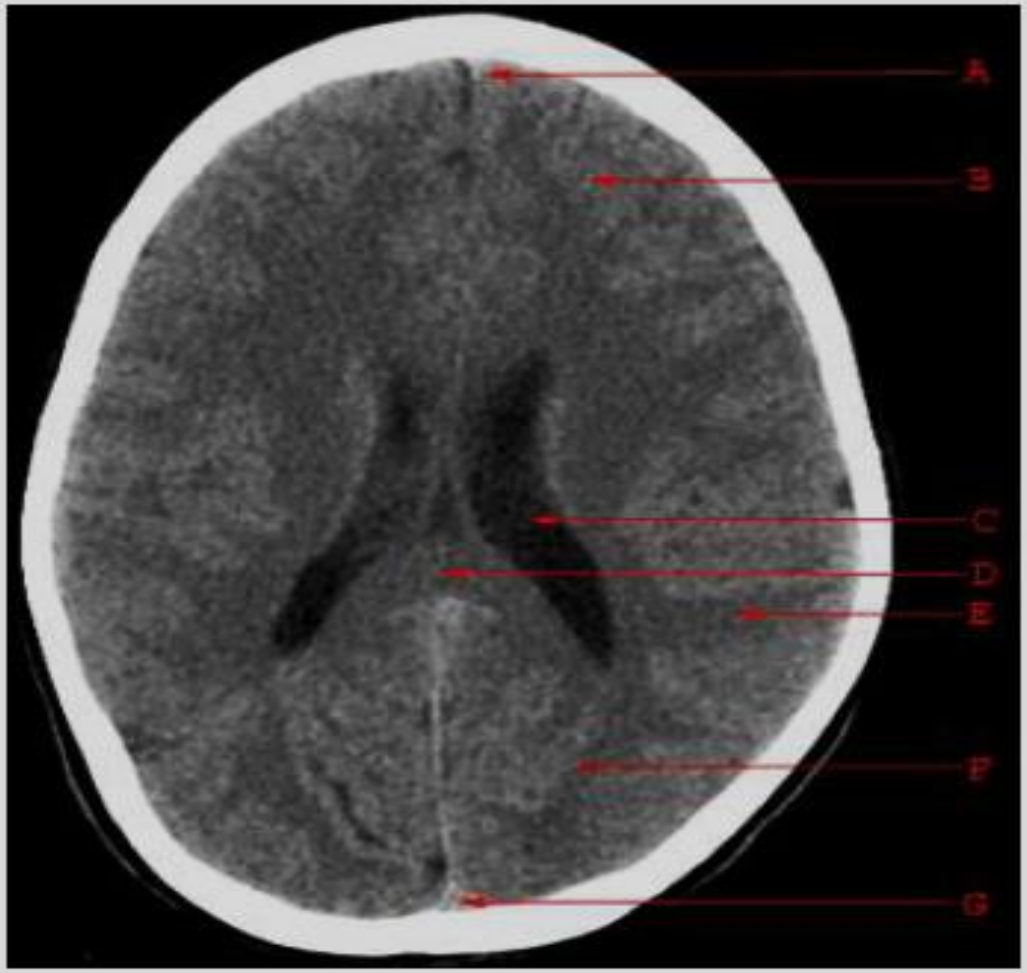
NORMAL ANATOMY.....

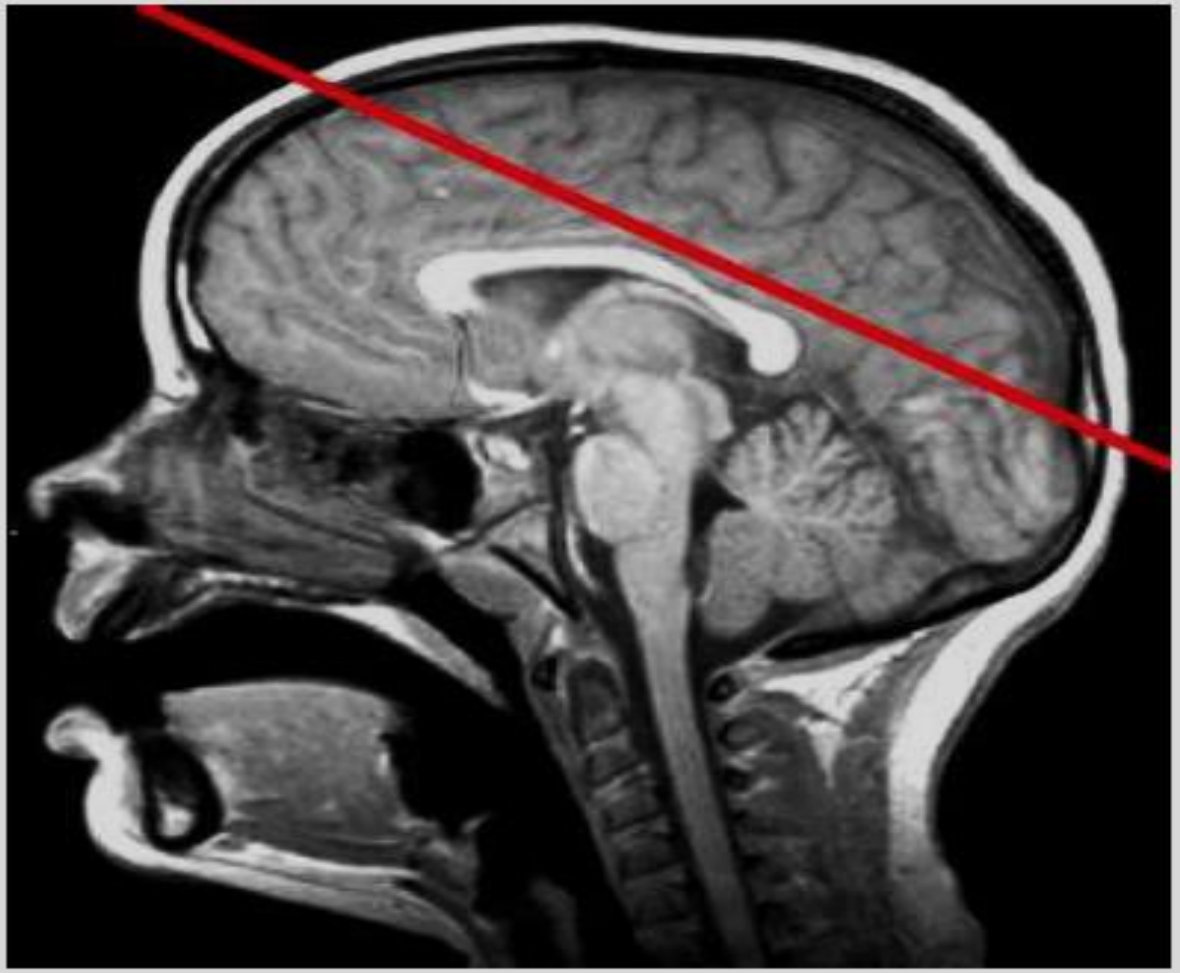




A=GENU OF CORPUS CALLOSUM B=ANT HORN OF LATERAL VENTRICLE C=INT CAPSULE
D=THALAMUS E=PINEAL GLAND F=CHOROID PLEXUS G=STARAIGHT SINUS

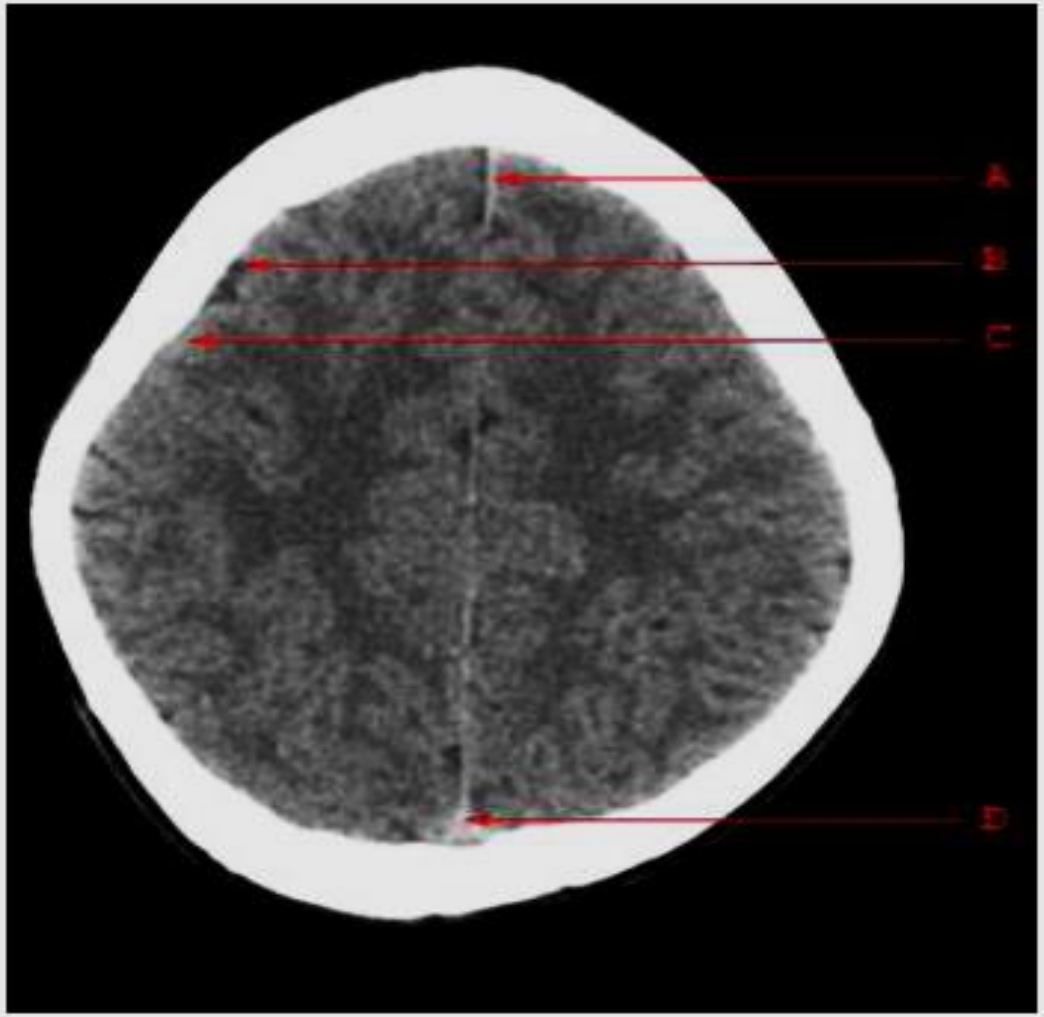
NORMAL ANATOMY.....

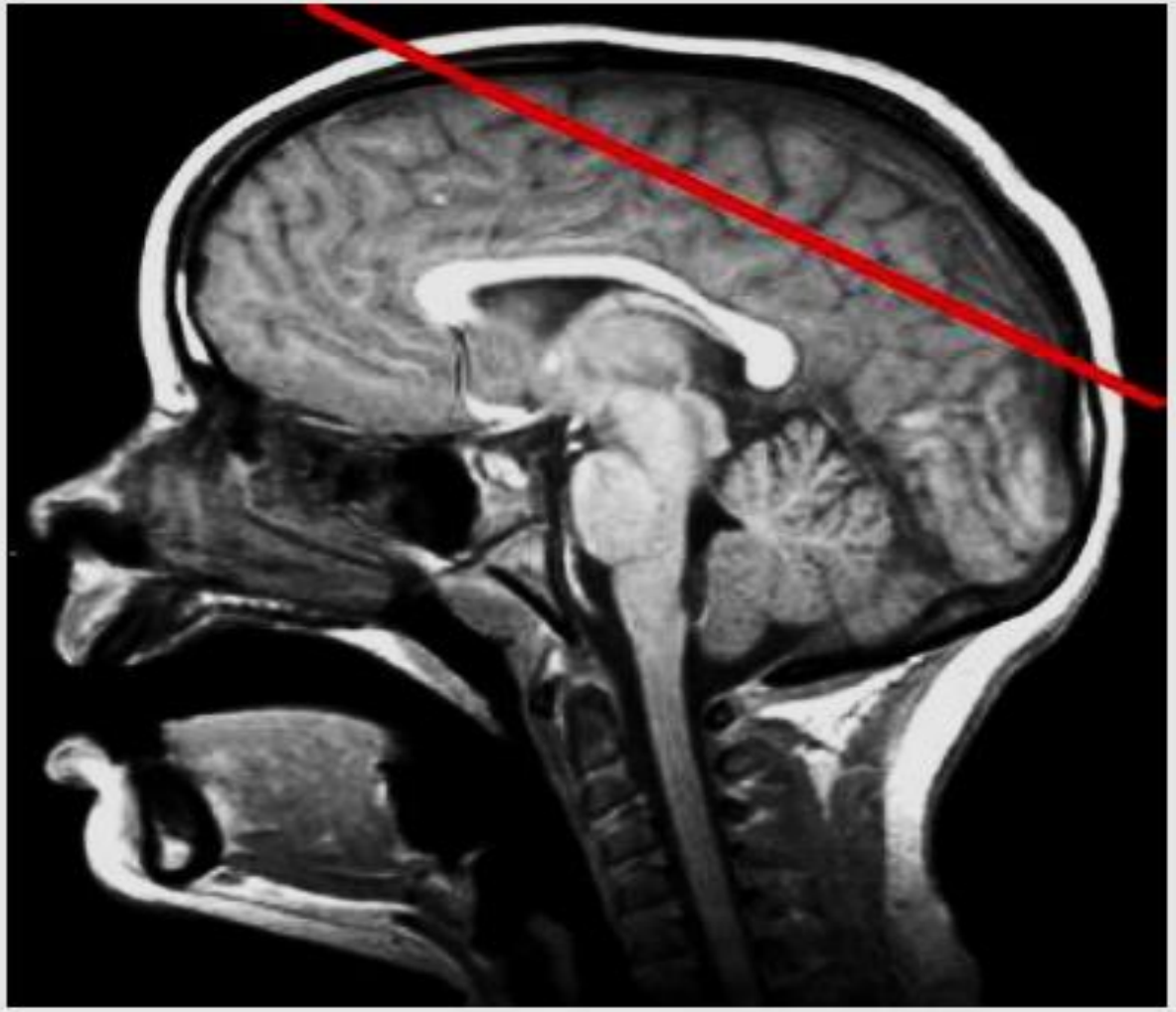




A=FALX CEREBRI B=FRONTAL LOBE C=BODY OF LATERAL VENTRICLE
D=SPLENIUM OF CORPUS CALLOSUM E=PARIETAL LOBE F=OCCIPITAL LOBE
G=SUPERIOR SAGITTAL SINUS

NORMAL ANATOMY.....

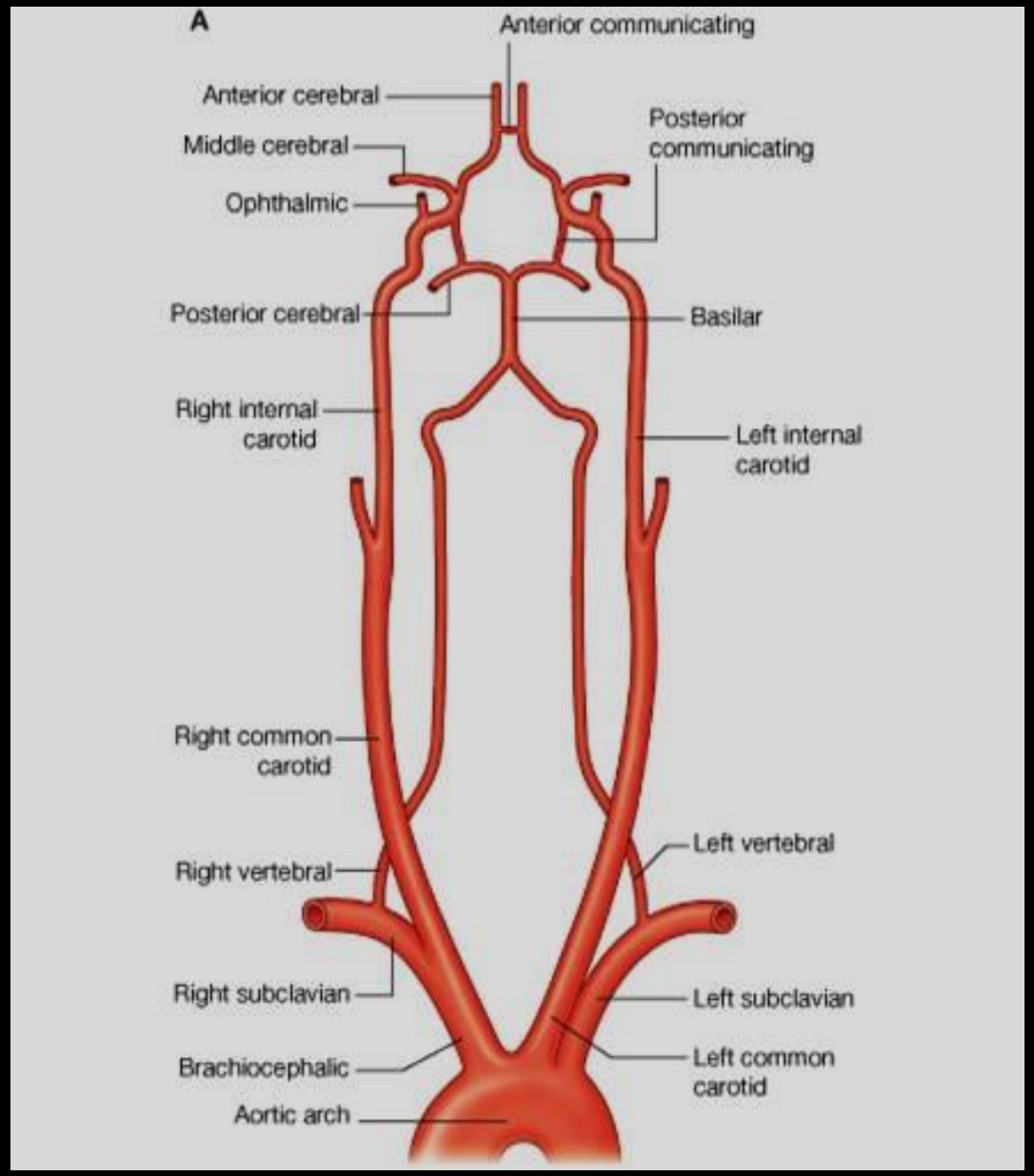




A=FALX CEREBRI B=SULCUS C=GYRUS D=SUPERIOR SAGGITAL SINUS

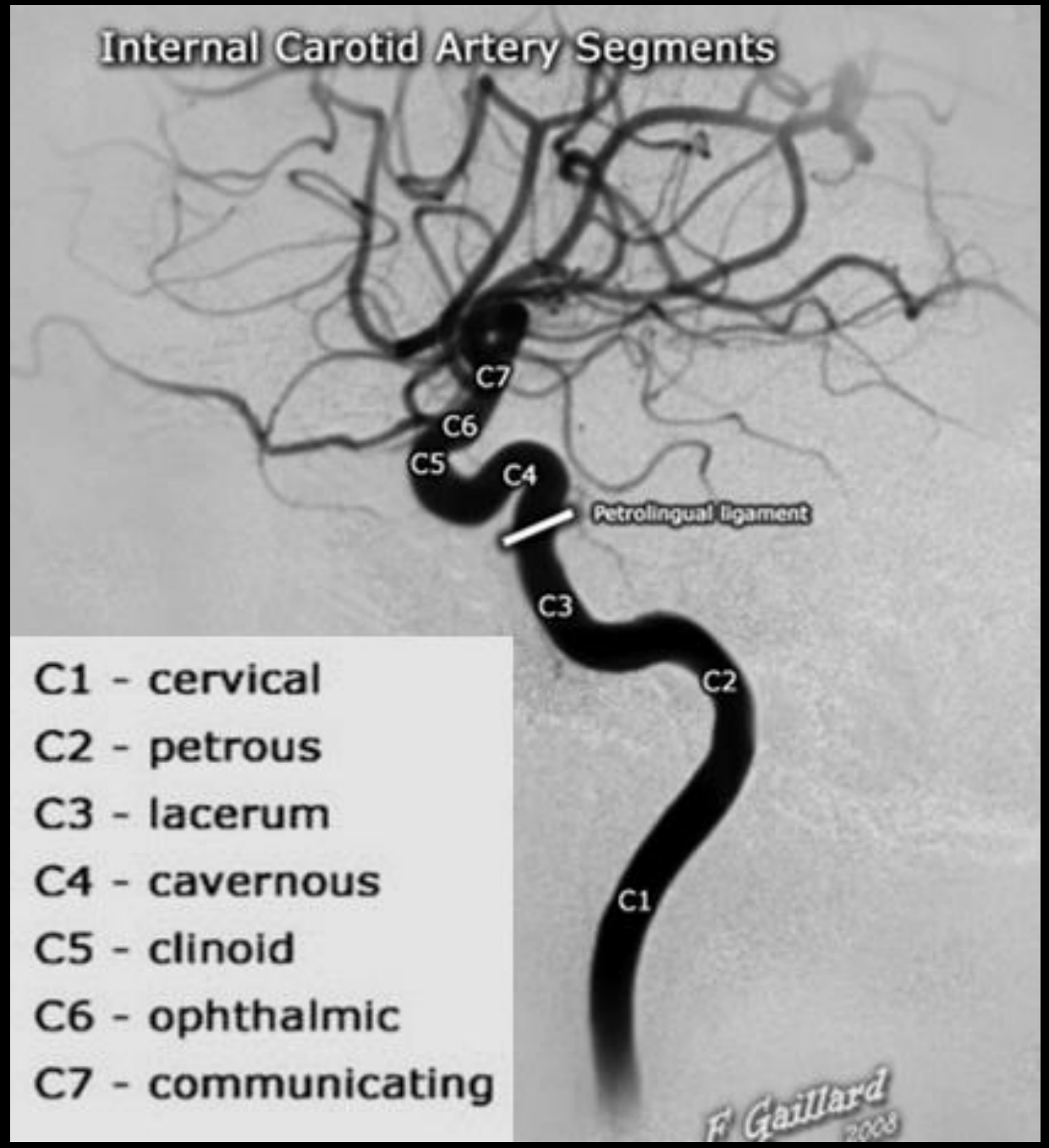
Arterial blood supply of the brain

- There are two paired arteries responsible for the blood supply to the brain:-
  - Anterior circulation:Internal carotid arteries (ICAs).
  - Posterior circulation: (vertebro-basilar system).
  - The terminal branches of these arteries form an anastomotic circle, called the Circle of Willis. The branches arise from this circle supply the majority of the cerebrum.

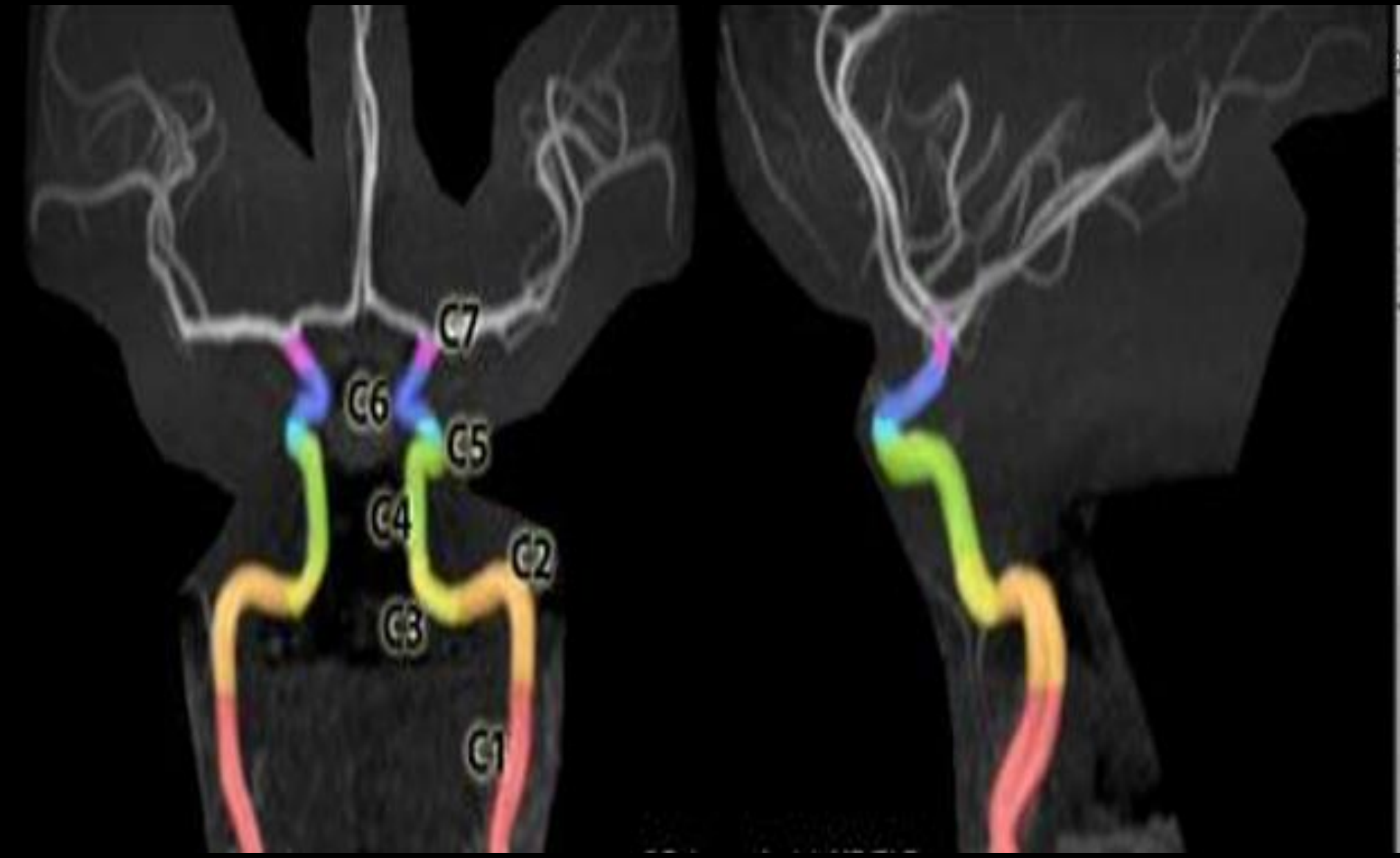


Internal carotid arteries

- The (ICAs) originate at the bifurcation of CCAs, at the level of (C4).
- They ascend within the carotid sheath, and enter the brain via the carotid canal of the temporal bone.
- No branches to the face or neck.
- It has seven segments ; cervical, petrous, lacerum, cavernous, clinoid, ophthalmic and supraclinoid.
- Supraclinoid segment gives rise to anterior cerebral artery (ACA) and middle cerebral artery (MCA).



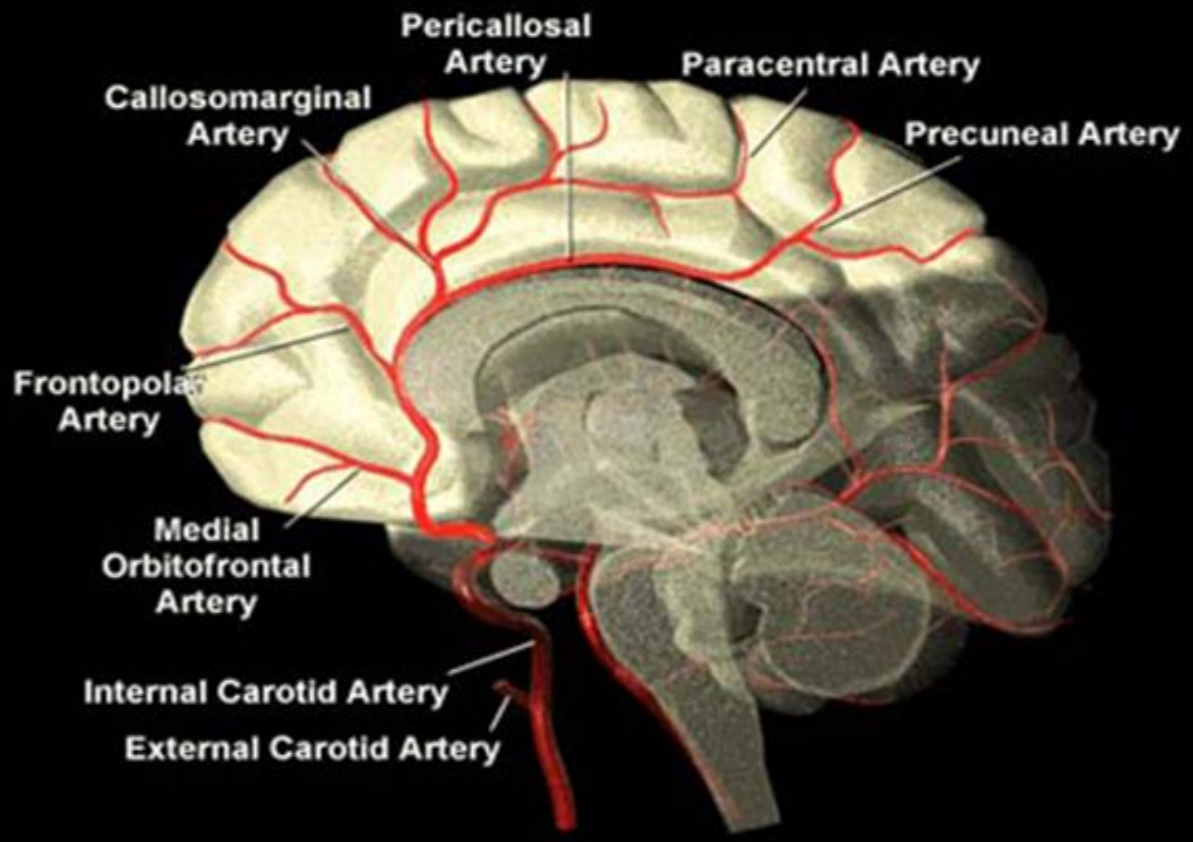
Internal carotid artery segments





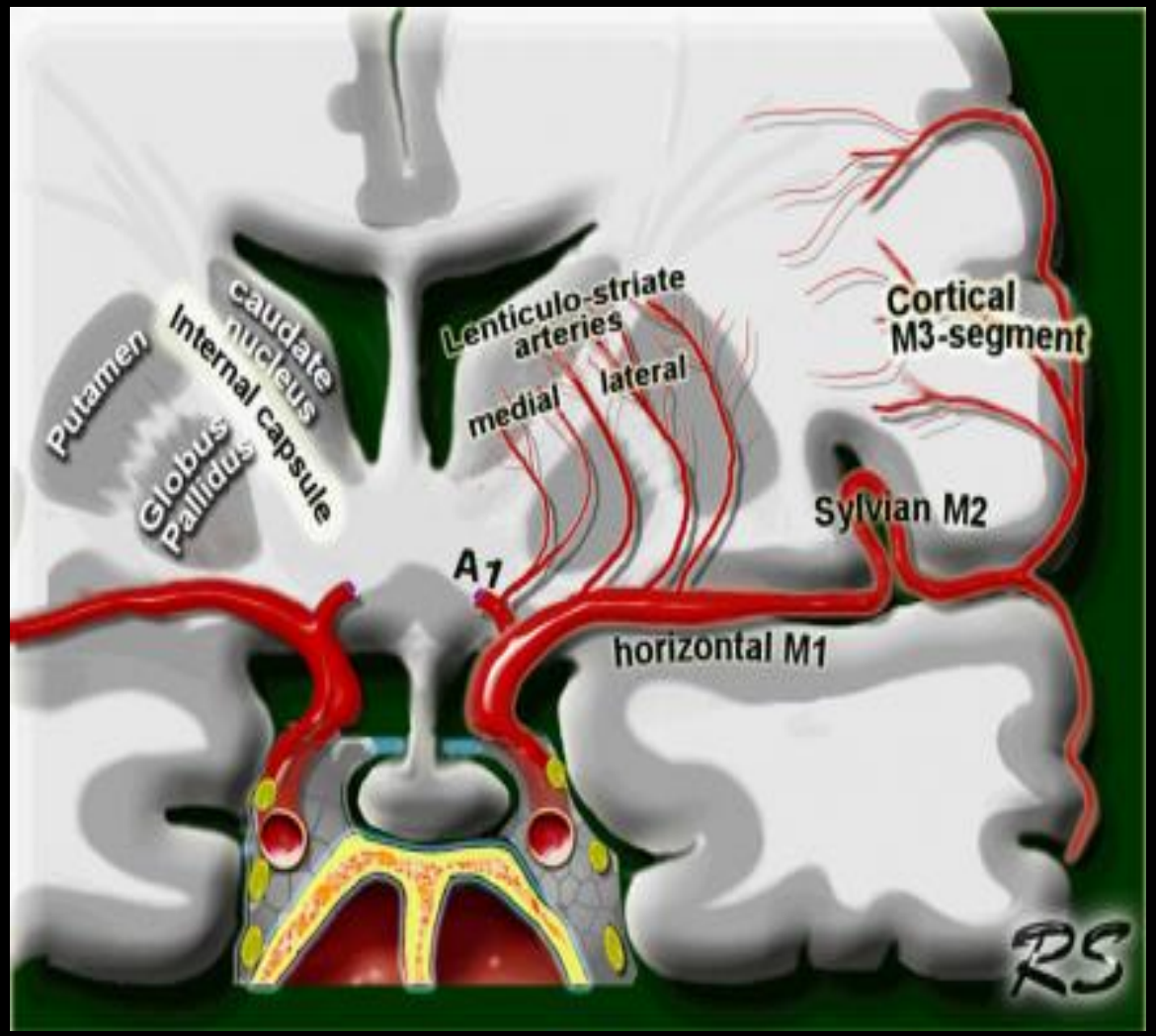
The ACA is divided into five segments

- A1(pre-communicating) : horizontal , terminating at the anterior communicating A.
- A2 (post-communicating, infracallosal):vertical from ACOM, to callosomarginal A. at the genu of the corpus callosum .
- A3 (precallosal): distal to the callosomarginal artery, turns directly posterior above the corpus callosum
- A4 (supracallosal): above the body of the corpus callosum .
- A5 (postcallosal): above the body of the corpus callosum posterior to the plane of the coronal suture.



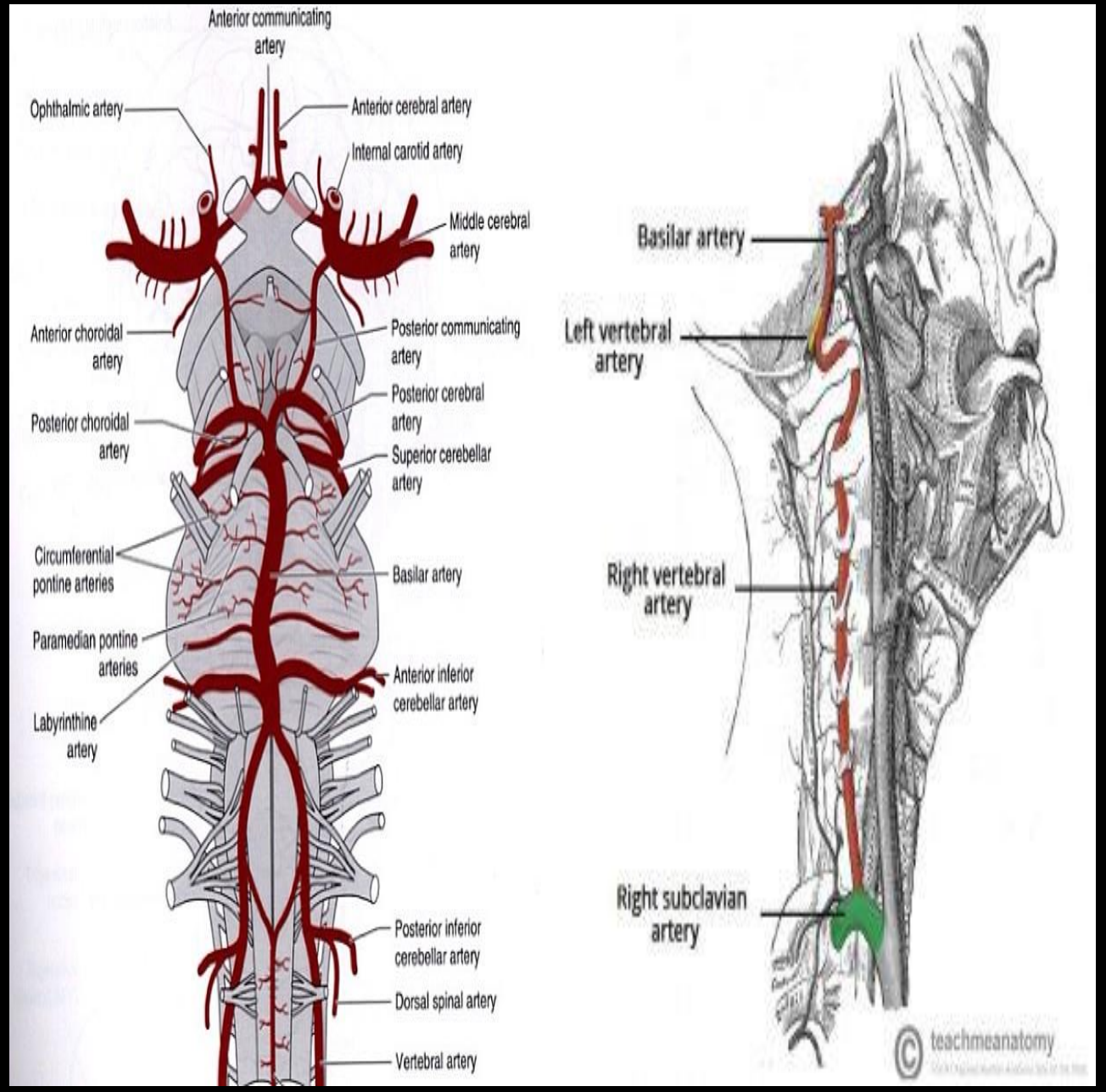
Middle cerebral artery segments

- Horizontal M1-segment supply parts of basal ganglia and internal capsule.
- Sylvian M2-segment supply the temporal lobe and insular cortex , parts of parietal and frontal lobes
- Cortical M3-segment supply the lateral cerebral cortex



Vertebral Arteries

- Arise from the **subclavian** arteries, then ascend through **foramena transversarium**.
- Enter the cranial cavity via the **foramen magnum.**
- After this, the two vertebral arteries converge to form the **basilar artery**.
- The basilar artery terminates by bifurcating into the posterior cerebral arteries (PCA).



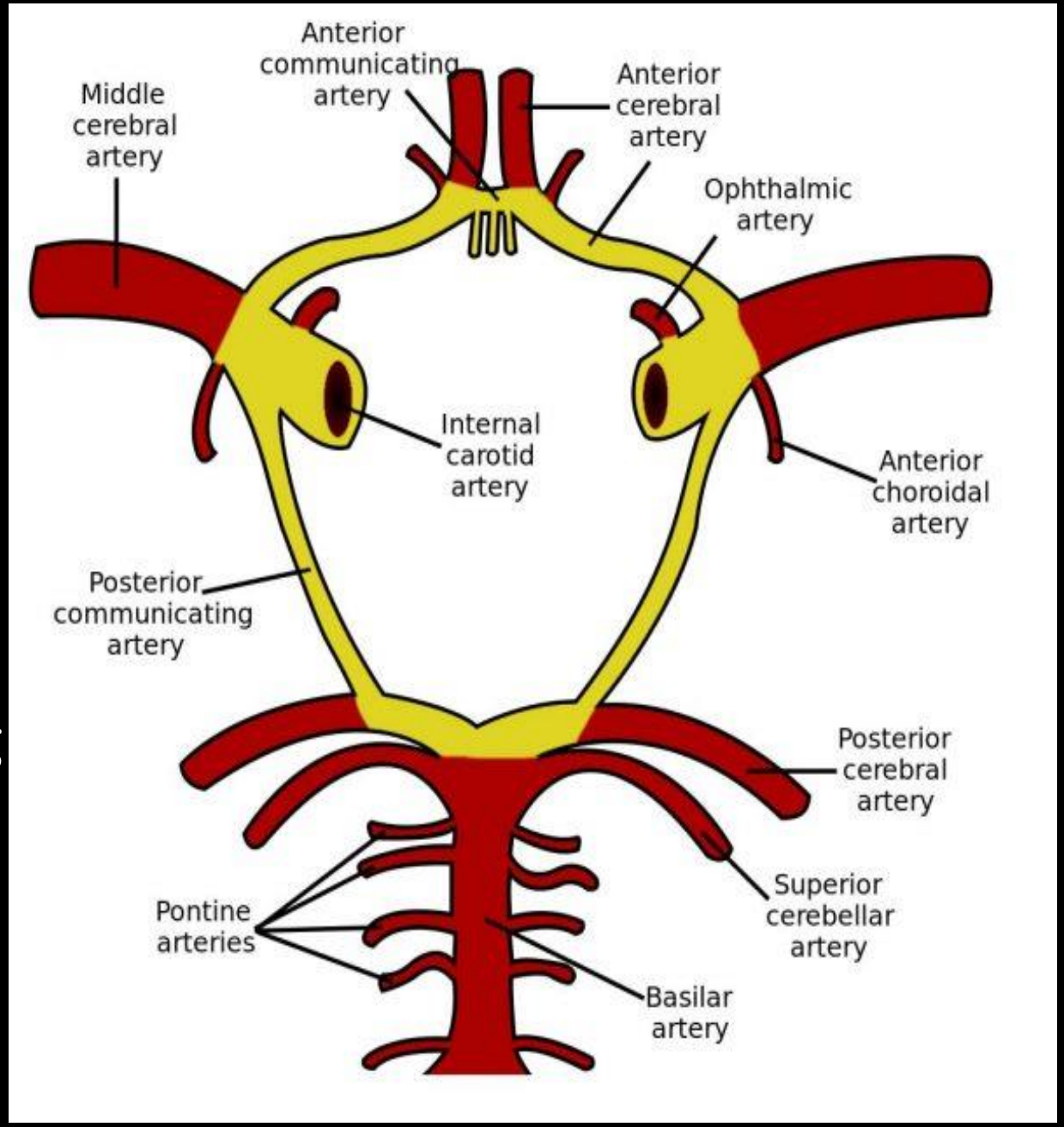
The PCA is divided into 4 segments:

- P1: from its origin at the termination of the basilar artery to posterior communicating artery (PCOM), within interpeduncular cistern
- P2: from the PCOM around the midbrain
- P3: within the quadrigeminal cistern
- P4: cortical segment.



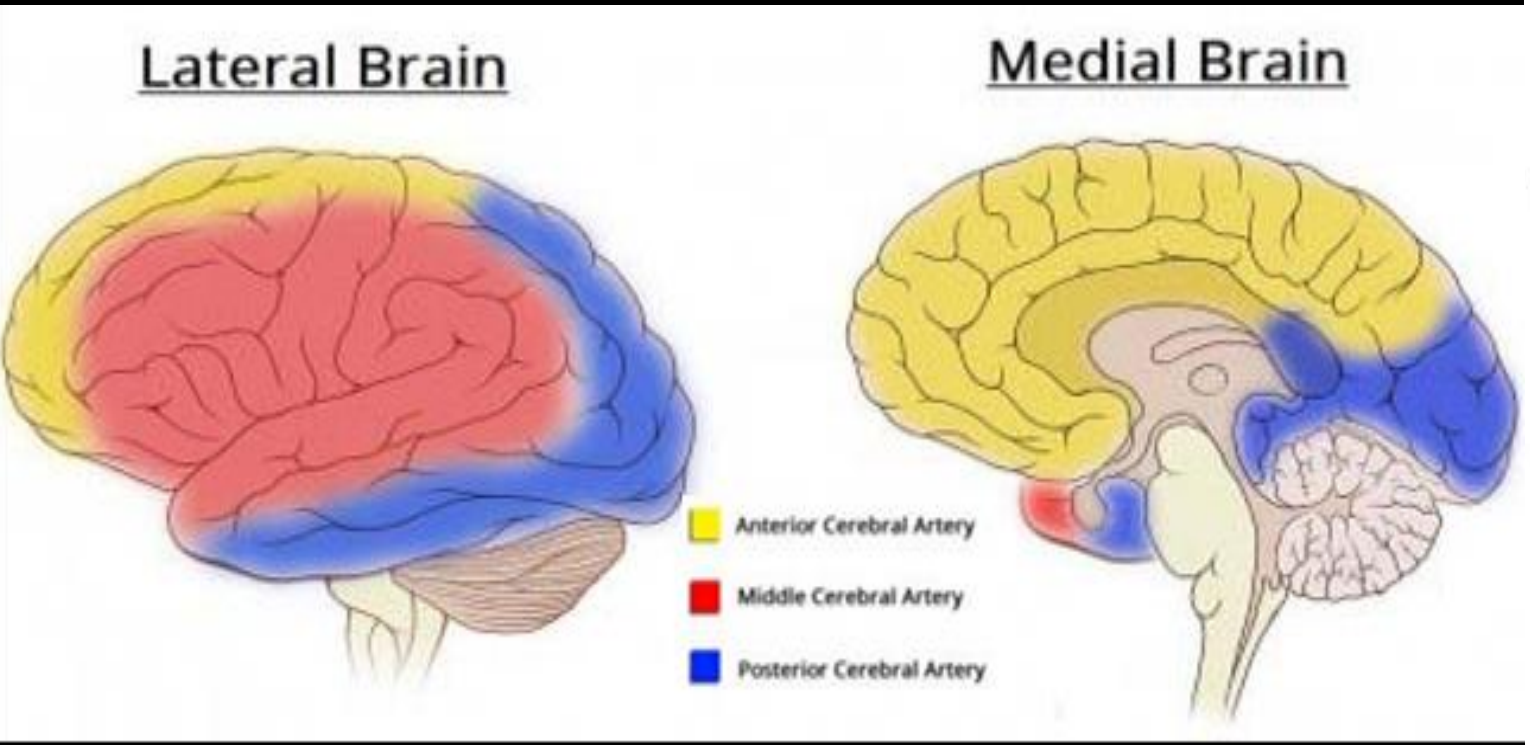
Arterial Circle of Willis

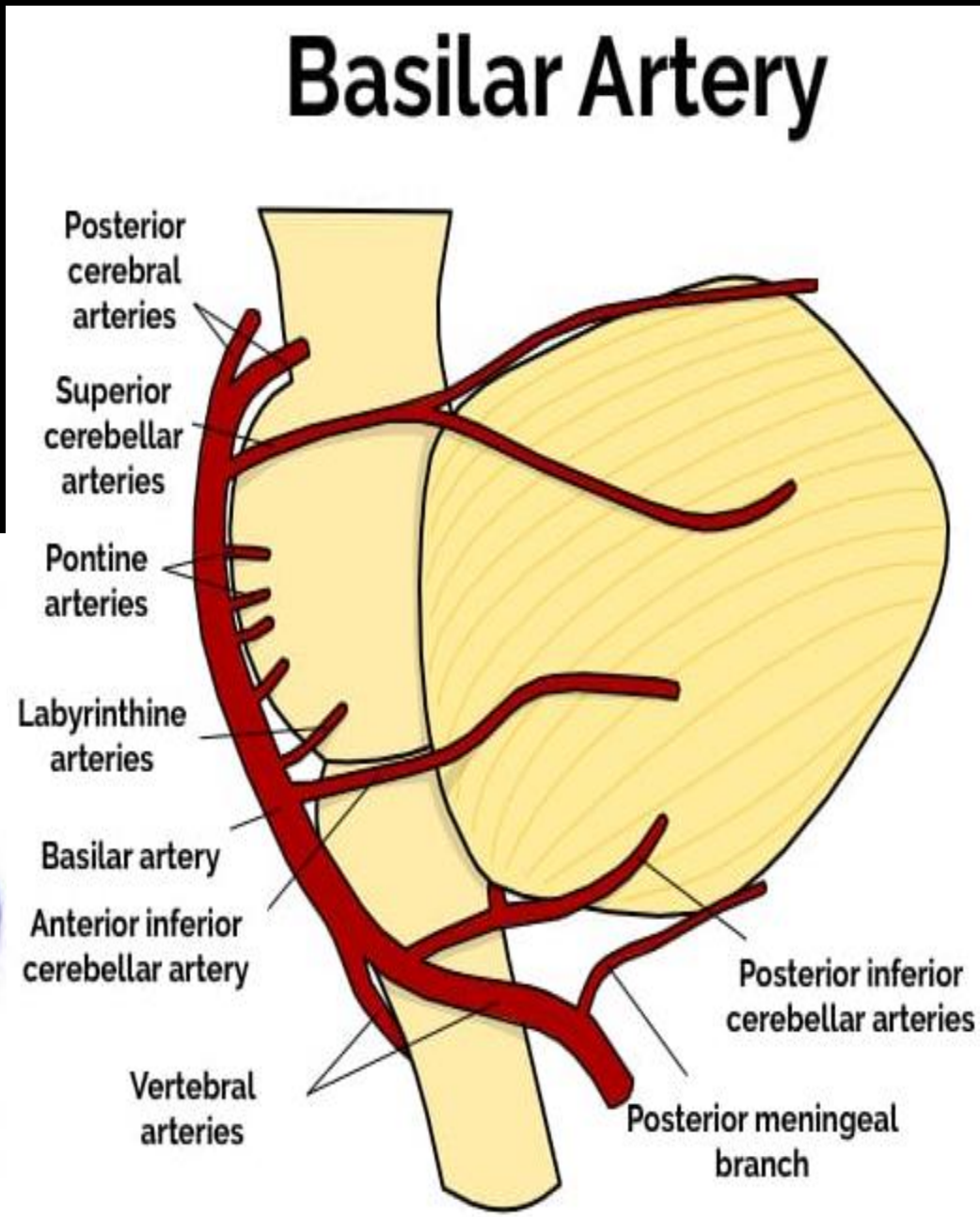
- The terminal branches of the vertebral and internal carotid arteries all anastomose to form a circular blood vessel, called the Circle of Willis.
- There are three main (paired) constituents of the Circle of Willis:
  - ACAs (A1 segment).
  - ICAs (C7 segment).
  - PCAs (P1 segment).
- To complete the circle, two 'connecting vessels' are also present:
  - Anterior communicating artery (Acom) : – connects the two ACAs.
  - Posterior communicating artery (Pcom) :– branch of the internal carotid artery connects the ICA to the PCA.



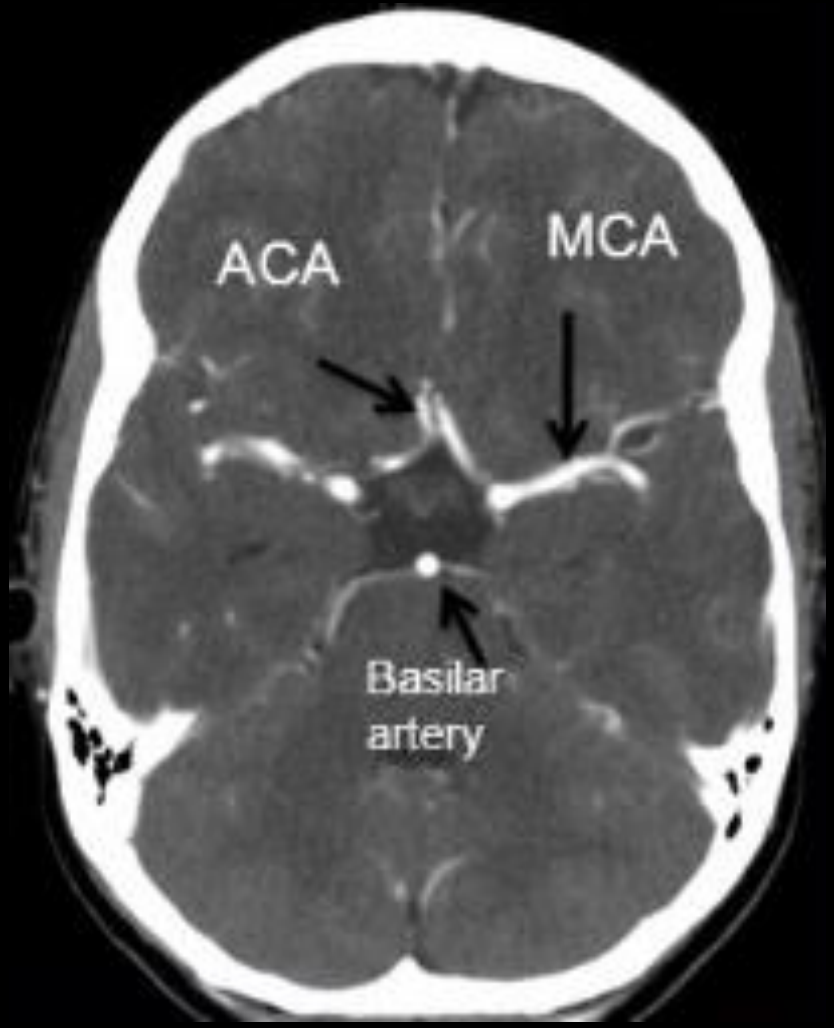
Regional Blood Supply to the brain

- Cerebrum:
  - ACA:supply the **antero-medial** portion.
  - MCA: supply the **lateral** portion.
  - PCA: supply the **posterior** cerebrum & mid brain.
- Cerebellum and brain stem are supplied by branches arise from vertebro-basilar arteries



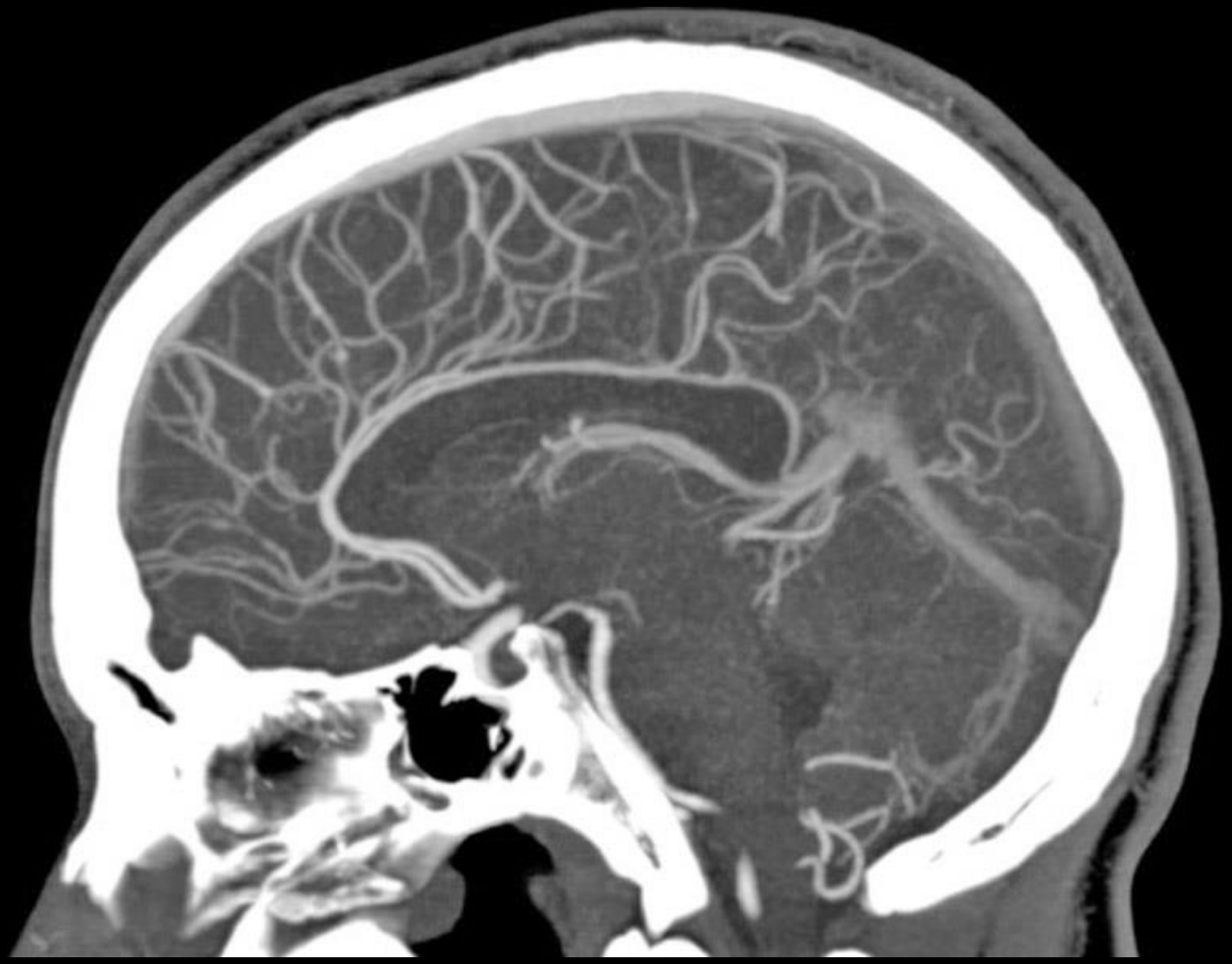


Ct angiography





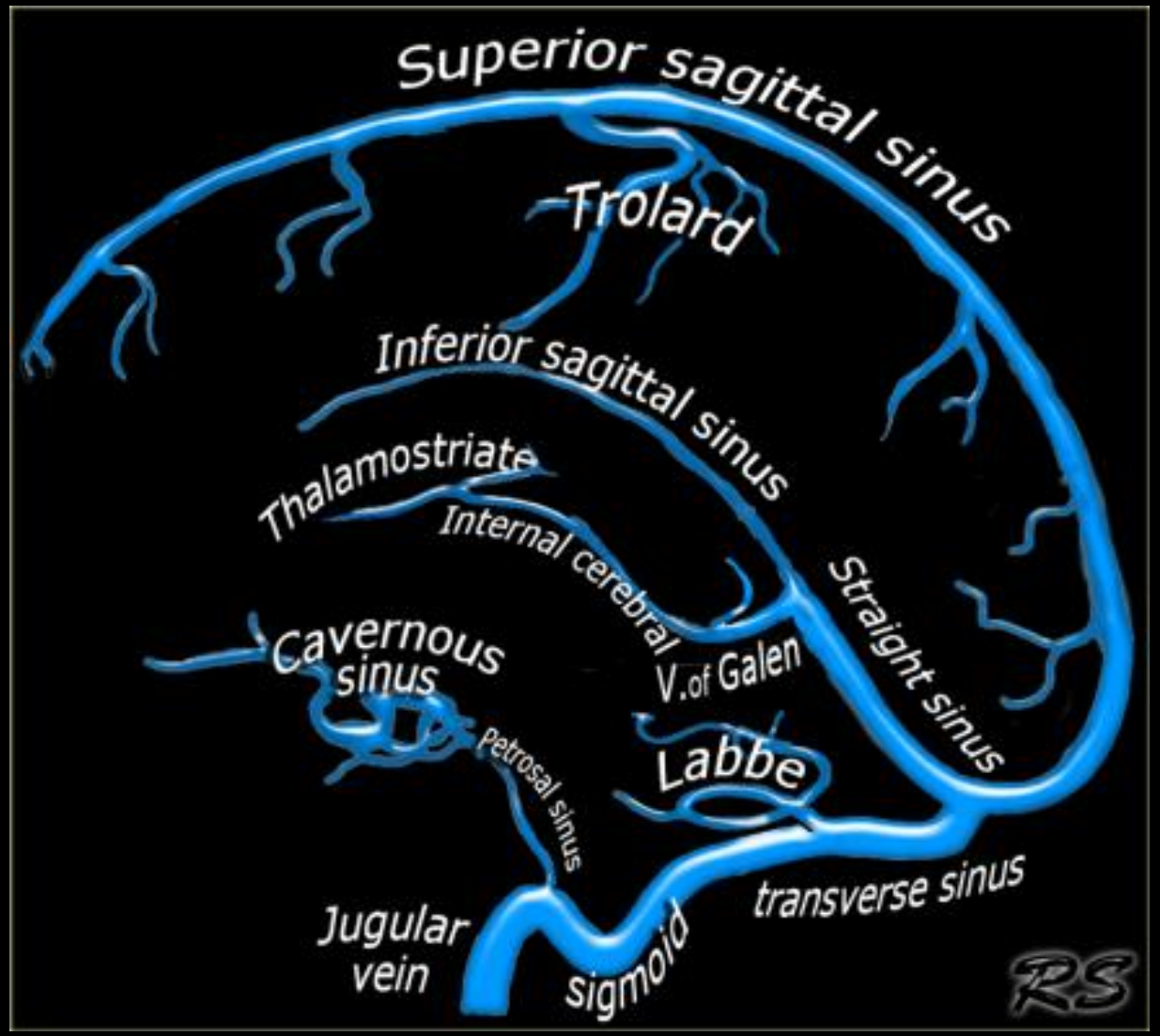
Ct angiography



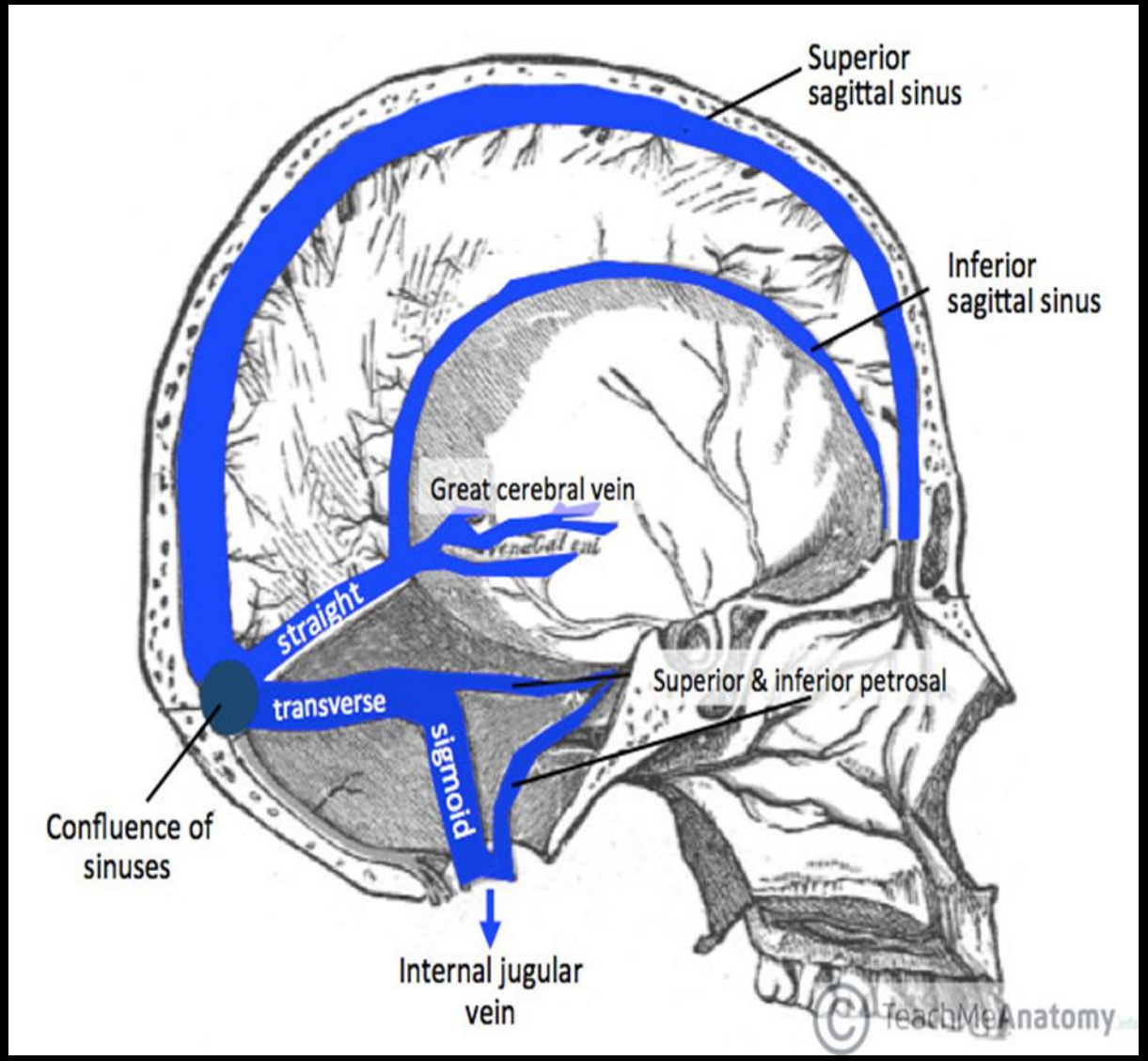


The Venous Drainage of the Central Nervous System

- The dural venous sinuses lie between the **periosteal** and **meningeal** layers of the dura mater. They are best thought of as collecting pools of blood, which drain the brain.
- Unlike most veins of the body, the dural venous sinuses do not have valves.
- All the dural venous sinuses ultimately drain into the **internal jugular vein**.

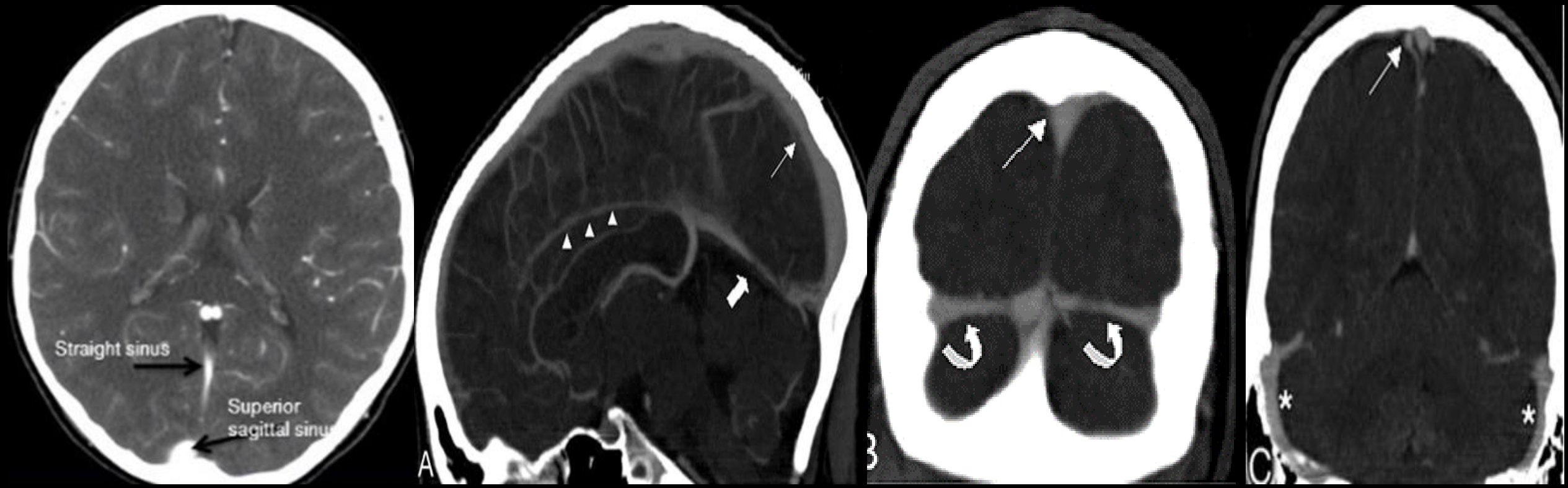


The Venous SINUSES

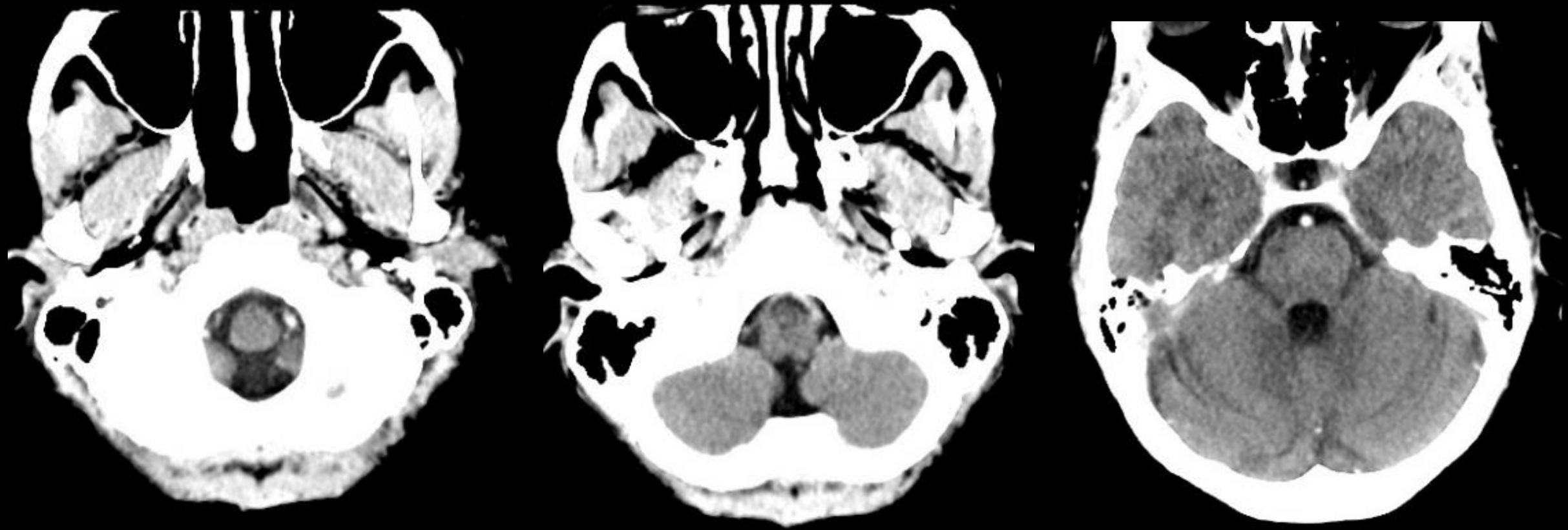


- The straight, superior, and inferior sagittal sinuses are found in the **falx cerebri** . They converge at the **confluence of sinuses** (overlying the **internal occipital protuberance**).
- The straight sinus is a continuation of the great cerebral vein and the inferior sagittal sinus.
- From the confluence, the **transverse** sinus continues bi-laterally and curves into the **sigmoid** sinus to meet the opening of the internal jugular vein.
- The **cavernous** sinus drains the ophthalmic veins and can be found on either side of the sella turcica. From here, the blood returns to the internal jugular vein via the **superior** or **inferior petrosal** sinuses.

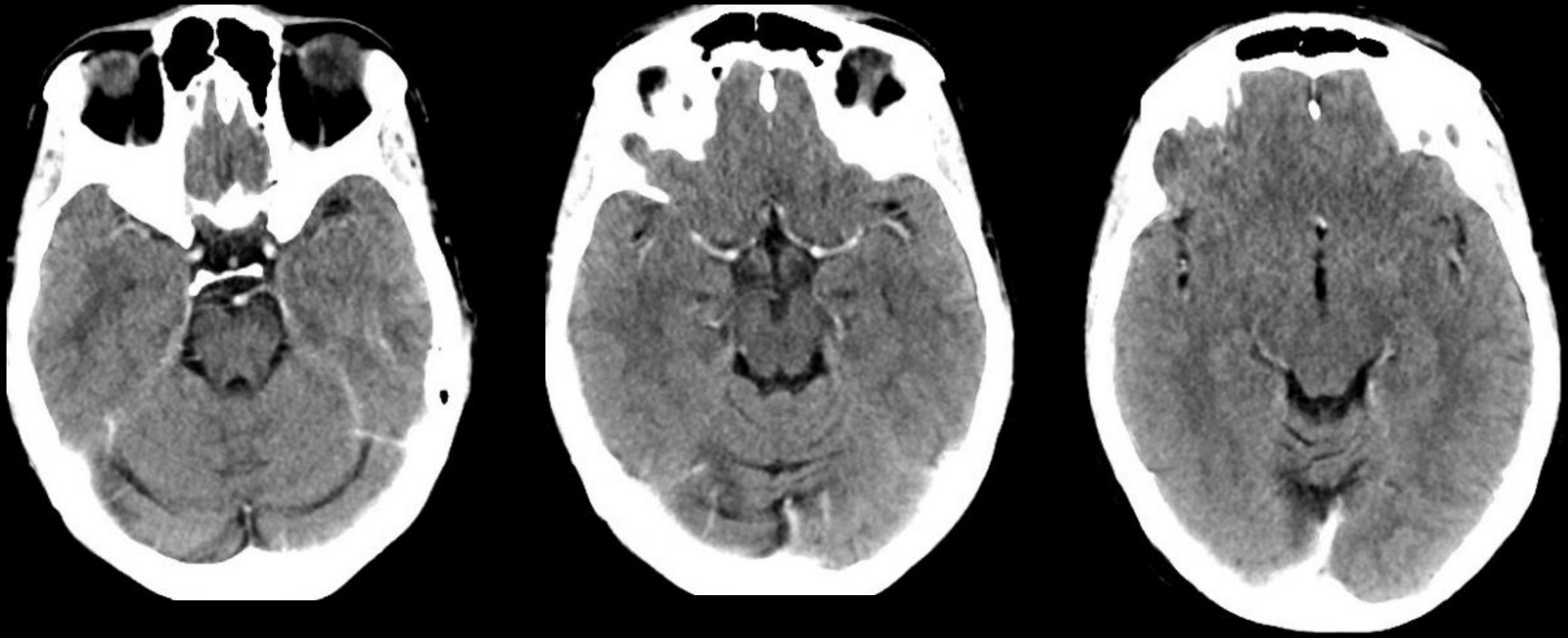
Ct venography



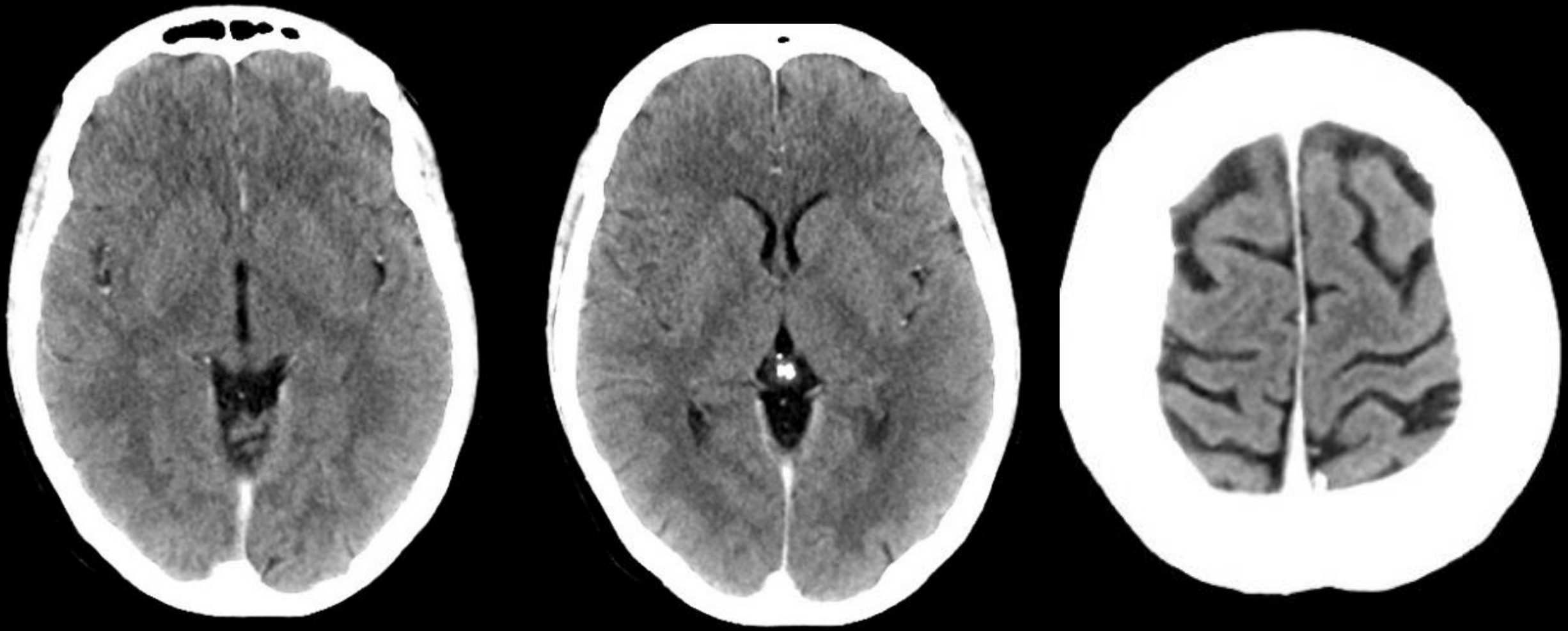
Post contrast Ct brain



Post contrast Ct brain



Post contrast Ct brain




Thank
you!!